GNU LibreDWG

for version 0.13.4, 17 March 2026

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This manual is for GNU LibreDWG (version 0.13.4, 17 March 2026).

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1 Overview

LibreDWG is a free C library to read and write *DWG files*. The DWG file format is proprietary and was created in the 1970s for one then-emerging CAD application.

This library is part of the GNU project, released under the aegis of GNU. It is made available under GPLv3+, i.e., under the terms of the GNU General Public License version 3, or (at your option) any later version.

It came out of code from the QCad Community Edition product from Ribbonsoft.

1.1 API/ABI version

LibreDWG 0.13.4 provides the API/ABI version 1. We hope that this doesn't need to change much in the future.

See `include/dwg.h` for `LIBREDWG_VERSION_MAJOR`, `LIBREDWG_VERSION_MINOR` and `LIBREDWG_SO_VERSION`.

1.2 Coverage

Because the DWG file format is not open, its specification had to be reverse-engineered. The specification is almost complete. The LibreDWG implementation of the specification is an ongoing process; as of version 0.13.4, coverage is approximately 99%.

It can read all DWG formats from r1.2 to r2018 for 99%. The internally supported DWG revisions are identified as r1.1, r1.2, r1.3, r1.4, r2.0b, r2.0, r2.10, r2.21, r2.22, r2.4, r2.5, r2.6, r9, r9c1, r10, r11b1, r11b2, r11, r13b1, r13b2, r13, r13c3, r14, r2000b, r2000, r2000i, r2002, r2004a, r2004b, r2004c, r2004, r2007a, r2007b, r2007, r2010b, r2010, r2013b, r2013, r2018b and r2018.

Here is a list of features that are still missing:

Writing r2004+ DWG
: Writing DWG formats for r2004 and later: r2007, r2010, r2013, r2018 is an ongoing effort. You need to patch the code to enable writing to the r2004+ format. See the `work/2004` branch. Writing to the specific r2007 format is not implemented. We write as default in the r2000 or older formats only.

Reading binary DXF
: DXF support is now mostly implemented. ASCII DXF's are generated fully, with much more elements and fields and other free libraries, but AutoCAD ® fails to import some of them. See the `TODO` file for a detailed coverage report.

  Reading binary DXF should be complete but is undertested.

Reading and writing pre-R13 DXF
: Reading and writing pre-R13 DXF's is work in progress. There is no support planned for the very old pre-R2.0 DXF format.

Enabled entities and objects, but unstable, undertested. Field names may change:
: For a detailed overview see the file `src/classes.inc` or `src/classes.c`.

  ACSH_BREP_CLASS ACSH_CHAMFER_CLASS ACSH_CONE_CLASS ACSH_PYRAMID_CLASS ARC_DIMENSION ASSOCACTION

ASSOCBLENDSURFACEACTIONBODY ASSOCEXTENDSURFACEACTIONBODY ASSOCEXTRUDEDSURFACEACTIONBODY ASSOCFILLETSURFACEACTIONBODY ASSOCGEOMDEPENDENCY ASSOCLOFTEDSURFACEACTIONBODY ASSOCNETWORK ASSOCDEPENDENCY ASSOCVALUEDEPENDENCY ASSOCNETWORKSURFACEACTIONBODY ASSOCOFFSETSURFACEACTIONBODY ASSOCPATCHSURFACEACTIONBODY ASSOCPLANESURFACEACTIONBODY ASSOCREVOLVEDSURFACEACTIONBODY ASSOCTRIMSURFACEACTIONBODY BACKGROUND BLOCKLINEARPARAMETER BLOCKROTATIONPARAMETER BLOCKXYPARAMETER BLOCKVISIBILITYGRIP BLOCKVISIBILITYPARAMETER EVALUATION_GRAPH HELIX LARGE_RADIAL_DIMENSION LIGHTLIST MATERIAL MENTALRAYRENDERSETTINGS OBJECT_PTR RAPIDRTRENDERSETTINGS RENDERSETTINGS SECTION_SETTINGS SPATIAL_INDEX TABLESTYLE (works only pre-2010)

Unhandled (fields spec'ed, but broken/undertested):

ACMECOMMANDHISTORY ACMESCOPE ACMESTATEMGR ACSH_EXTRUSION_CLASS ACSH_LOFT_CLASS ACSH_REVOLVE_CLASS ACSH_SWEEP_CLASS ALDIMOBJECTCONTEXTDATA ANNOTSCALEOBJECTCONTEXTDATA ASSOC2DCONSTRAINTGROUP ASSOCACTION ASSOCALIGNEDDIMACTIONBODY ASSOCEXTRUDEDSURFACEACTIONBODY ASSOCGEOMDEPENDENCY ASSOCLOFTEDSURFACEACTIONBODY ASSOCNETWORK ASSOCOSNAPPOINTREFACTIONPARAM ASSOCOSNAPPOINTREFACTIONPARAM ASSOCPERSSUBENTMANAGER ASSOCREVOLVEDSURFACEACTIONBODY ASSOCVERTEXACTIONPARAM ATEXT BLKREFOBJECTCONTEXTDATA CONTEXTDATAMANAGER CSACDOCUMENTOPTIONS CURVEPATH DATALINK DATATABLE DIMASSOC DYNAMICBLOCKPROXYNODE EXTRUDEDSURFACE FCFOBJECTCONTEXTDATA GEOMAPIMAGE GEOPOSITIONMARKER LAYOUTPRINTCONFIG LEADEROBJECTCONTEXTDATA LOFTEDSURFACE MLEADEROBJECTCONTEXTDATA MOTIONPATH MTEXTATTRIBUTEOBJECTCONTEXTDATA MTEXTOBJECTCONTEXTDATA NAVISWORKSMODEL NAVISWORKSMODELDEF NURBSURFACE PERSUBENTMGR PLANESURFACE POINTPATH RENDERENTRY RENDERGLOBAL REVOLVEDSURFACE RTEXT SUNSTUDY SWEPTSURFACE TABLE (works only pre-2010) TABLECONTENT TEXTOBJECTCONTEXTDATA TVDEVICEPROPERTIES

ASSOCDIMDEPENDENCYBODY BLOCKPARAMDEPENDENCYBODY ALIGNMENTPARAMETERENTITY BASEPOINTPARAMETERENTITY FLIPPARAMETERENTITY LINEARPARAMETERENTITY POINTPARAMETERENTITY ROTATIONPARAMETERENTITY VISIBILITYPARAMETERENTITY VISIBILITYGRIPENTITY XYPARAMETERENTITY BLOCKALIGNEDCONSTRAINTPARAMETER BLOCKANGULARCONSTRAINTPARAMETER BLOCKARRAYACTION BLOCKDIAMETRICCONSTRAINTPARAMETER BLOCKHORIZONTAL-

CONSTRAINTPARAMETER BLOCKLINEARCONSTRAINTPARAMETER BLOCKLOOKUPACTION BLOCKLOOKUPPARAMETER BLOCKPOINTPARAMETER BLOCKPOLARGRIP BLOCKPOLARPARAMETER BLOCKPOLARSTRETCHACTION BLOCKPROPERTIESTABLE BLOCKPROPERTIESTABLEGRIP BLOCKRADIALCONSTRAINTPARAMETER BLOCKREPRESENTATION BLOCKSTRETCHACTION BLOCKUSERPARAMETER BLOCKVERTICALCONSTRAINTPARAMETER BLOCKXYGRIP POINTCLOUD POINTCLOUDEX POINTCLOUDDEF POINTCLOUDDEFEX POINTCLOUDDEF_REACTOR POINTCLOUDDEF_REACTOR_EX POINTCLOUDCOLORMAP

See `src/classes.inc`.

Missing:

* PROXY subentities, PROXY_ENTITY

Halfway:

SUNSTUDY VBA_PROJECT ASSOCACTION ASSOCNETWORK ASSOCALIGNEDDIMACTIONBODY ASSOCOSNAPPOINTREFACTIONPARAM ASSOCPERSSUBENTMANAGER PERSUBENTMGR ASSOC2DCONSTRAINTGROUP EVALUATION_GRAPH ASSOCOSNAPPOINTREFACTIONPARAM ACSH_BOX_CLASS ACSH_EXTRUSION_CLASS ACSH_HISTORY_CLASS ACSH_SWEEP_CLASS NAVISWORKSMODEL (i.e. COORDINATION MODEL) NAVISWORKSMODELDEF DATATABLE TABLESTYLE ASSOCGEOMDEPENDENCY LAYOUTPRINTCONFIG RENDERENVIRONMENT RENDERGLOBAL LIGHTLIST SECTION_SETTINGS

Unhandled (i.e. passed through, no DXF and fields):

ACDSRECORD ACDSSCHEMA NPOCOLLECTION RAPIDRTRENDERENVIRONMENT XREFPANELOBJECT

no test coverage for entities:

I.e. we need an extended `example_2018.dwg` with all types, with the following missing entities:

ARCALIGNEDTEXT BODY CAMERA DIMENSION_ANG3PT DIMENSION_DIAMETER DIMENSION_RADIUS DGNUNDERLAY DWFUNDERLAY GEOPOSITIONMARKER IMAGE LEADER LONG_TRANSACTION MESH MINSERT OLE2FRAME OLEFRAME POLYLINE_2D POLYLINE_MESH PROXY_ENTITY PROXY_LWPOLYLINE SHAPE TOLERANCE VERTEX_2D VERTEX_MESH

and objects:

CSACDOCUMENTOPTIONS XREFPANELOBJECT IDBUFFER IMAGEDEF IMAGEDEF_REACTOR LAYER_INDEX LIGHTLIST NPOCOLLECTION OBJECT_PTR PLOTSETTINGS PROXY_OBJECT RASTERVARIABLES SPATIAL_INDEX UCS VBA_PROJECT

1.3 Related projects

Some projects that use DWG (and specifically LibreDWG) are:

FreeCAD `https://freecadweb.org/`

GRASS GIS
`http://grass.osgeo.org/`

Plans are to add support for SolveSpace, LibreCAD, FreeCAD, OpenSCAD and PythonCAD.

Related libraries:

libdwg The old version (documented in Esperanto) which was forked to LibreDWG in 2009. But in the meantime it got a DXF reader.

libdxfrw Read the DWG format for all versions r13+ but with much less elements, only those needed for DXF. Written in C++, under the GPLv2 license.

libopencad
Read the r2000 DWG format. Written in C++, under the GPLv2 license.

ACadSharp
Read and write the basic objects in most DWG and DXF formats. Written in C#, under the MIT license.

OpenDWG
The OpenDWG's license does not allow the usage in free software projects.

Compared to libdwg, libdxfrw and libopencad, LibreDWG can read and write much more DWG versions and details. Which is especially important for attached links and data from third party applications: BIM, MAP, GIS, AEC, MECH, ..., for 3D solids and dynamic parametric constraints.

2 Usage

This chapter describes how to compile and link a program against LibreDWG. To access LibreDWG interface elements (see Chapter 3 [Types], page 6, see Chapter 7 [Functions], page 273), include its header file in the C code.

```
#include <dwg.h>
```

Optionally you can also use the

```
#include <dwg_api.h>
```

API.

Make sure you specify '`-lredwg`' when linking, such as in this `Makefile.am` fragment:

```
AM_LDFLAGS += -lredwg
```

Note that the shared object library is named `libredwg` (with some system-specific extension, e.g., `.so`), so you do **not** want to specify `-llibredwg`, as that would (try to) link against `liblibredwg` and fail.

3 Types

LibreDWG types map closely to the type system of the DWG file format. This chapter describes the enums and structs used to define the single DWG structure, which is passed around the functions (see Chapter 7 [Functions], page 273).

`BITCODE_RC char` [define]
: 1 raw unsigned char, uint8_t

`BITCODE_RS short` [define]
: 1 raw unsigned short int, uint16_t

`BITCODE_RL long` [define]
: 1 raw unsigned long int, uint32_t

`BITCODE_RD double` [define]
: 1 raw IEEE-754 double

`BITCODE_B byte` [define]
: 1 bit

`BITCODE_BB byte` [define]
: 2 bits

`BITCODE_3B byte` [define]
: 1-3 bits

`BITCODE_4BITS byte` [define]
: 4 bits (for VIEW view_mode)

`BITCODE_BS short` [define]
: 1 bit-encoded unsigned short

`BITCODE_BL long` [define]
: 1 bit-encoded unsigned long (max 32bit)

`BITCODE_BLL uint64_t` [define]
: 1 bit-encoded unsigned 64bit long

`BITCODE_BD double` [define]
: 1 bit-encoded double

`BITCODE_DD double` [define]
: 1 bit-encoded double with default

`BITCODE_MC long` *int* [define]
: 1-4 modular chars

`BITCODE_UMC long` *unsigned int* [define]
: 1-4 unsigned modular chars

`BITCODE_MS long` *unsigned int* [define]
: 1 modular short, max 2 words

`BITCODE_BE double[3]` [define]
: 1 bitencoded extrusion vector.

  Note that this specifies an OCS (Object Coordinate System) for each entity, with the default (0, 0, 1). An extrusion of (0, 0, -1) is typically caused by exploding a block inserted with a negative x scale, i.e. the sign of each X point needs to be flipped. For more see the vendor DXF documentation on OCS and `programs/geom.c`.

`BITCODE_BT double` [define]
: 1 bitencoded thickness value

`BITCODE_TV char*` [define]
: length + ASCIIZ string The default text type until r2004.

`BITCODE_TU wchar*` [define]
: length + windows 2-byte wchar string (UCS-2). The default text type since r2007.

`BITCODE_TF char*` [define]
: Fixed length buffer, which can include NUL characters.

`BITCODE_TFF char*` [define]
: Embedded fixed length string, which can include NUL characters.

`BITCODE_H void*` [define]
: handle-references

`BITCODE_CMC struct` *Dwg_Color* [define]
: Dwg_Color struct with index or rgb, alpha and optional DBCOLOR handle, name, book_name.

[and some more]

Two types that do not derive from the type system of the DWG file format are the enums for return codes and error codes.

On non-C99 systems ensure that stdint.h and inttypes.h are available to use the proper C99 `int32_t`,... types, and not just the native fallback types int/long, which are different across platforms.

4 Objects

4.1 HEADER

All header variables.

`ACADMAINTVER`
: RC, DXF 90

`ANGBASE`
: BD, DXF 50

`ANGDIR`
: B, DXF 70

`APPID_CONTROL_OBJECT`
: H

`ATTDIA`
: B, DXF 70

`ATTMODE`
: BS, DXF 70

`ATTREQ`
: B, DXF 70

`AUNITS`
: BS, DXF 70

`AUPREC`
: BS, DXF 70

`AXISMODE`
: BS, DXF 70

`AXISUNIT`
: 2RD, DXF 20

`BACKZ`
: BD

`BLIPMODE`
: B, DXF 70

`BLOCK_CONTROL_OBJECT`
: H

`BLOCK_RECORD_MSPACE`
: H

`BLOCK_RECORD_PSPACE`
: H

`CAMERADISPLAY`
: B, DXF 290

`CAMERAHEIGHT`
: BD, DXF 40

`CECOLOR`
: CMC, DXF 62

`CELTSCALE`
: BD, DXF 40

`CELTYPE`
: H, DXF 6

`CELWEIGHT`
: BSd, DXF 370

`CEPSNTYPE`
: BS, DXF 380

`CHAMFERA` BD, DXF 40

`CHAMFERB` BD, DXF 40

`CHAMFERC` BD, DXF 40

`CHAMFERD` BD, DXF 40

`CLAYER` H, DXF 8

`CMATERIAL`
: H, DXF 347

`CMLJUST` BS, DXF 70

`CMLSCALE` BD, DXF 40

`CMLSTYLE` H, DXF 2

`COORDS` BS, DXF 70

`CPSNID` H, DXF 390

`CSHADOW` RC, DXF 280

`DELOBJ` B, DXF 70

`DGNFRAME` RC, DXF 280

`DICTIONARY_ACAD_GROUP`
: H

`DICTIONARY_ACAD_MLINESTYLE`
: H

`DICTIONARY_COLOR`
: H

`DICTIONARY_LAYOUT`
: H

`DICTIONARY_LIGHTLIST`
: H

`DICTIONARY_MATERIAL`
: H

`DICTIONARY_NAMED_OBJECT`
: H

`DICTIONARY_PLOTSETTINGS`
: H

`DICTIONARY_PLOTSTYLENAME`
: H

`DICTIONARY_VISUALSTYLE`
: H

`DIMADEC` BS, DXF 70

`DIMALT` B, DXF 70

`DIMALTD` BS, DXF 70

`DIMALTF` BD, DXF 40

`DIMALTMZF`
BD

`DIMALTMZS`
T, DXF 1

`DIMALTRND`
BD, DXF 40

`DIMALTTD` BS, DXF 70

`DIMALTTZ` BS, DXF 70

`DIMALTU` BS, DXF 70

`DIMALTZ` BS, DXF 70

`DIMAPOST` TV, DXF 1

`DIMAPOST_T`
T

`DIMARCSYM`
BS, DXF 70

`DIMARROW` BD

`DIMASO` B, DXF 70

`DIMASSOC` RC, DXF 280

`DIMASZ` BD, DXF 40

`DIMATFIT` BS, DXF 70

`DIMAUNIT` BS, DXF 70

`DIMAZIN` BS, DXF 70

`DIMBLK` H, DXF 1

`DIMBLK1` H, DXF 1

`DIMBLK1_T`
T

`DIMBLK2` H, DXF 1

`DIMBLK2_T`
T

`DIMBLK_T` T

`DIMCEN` BD, DXF 40

`DIMCLRD` CMC, DXF 70

`DIMCLRD_C` RS

`DIMCLRE` CMC, DXF 70

`DIMCLRE_C` RS

`DIMCLRT` CMC, DXF 70

`DIMCLRT_C` RS

`DIMDEC` BS, DXF 70

`DIMDLE` BD, DXF 40

`DIMDLI` BD, DXF 40

`DIMDSEP` BS, DXF 70

`DIMEXE` BD, DXF 40

`DIMEXO` BD, DXF 40

`DIMFIT` BS, DXF 70

`DIMFRAC` BS, DXF 70

`DIMFXL` BD, DXF 40

`DIMFXLON` B, DXF 70

`DIMGAP` BD, DXF 40

`DIMJOGANG` BD, DXF 40

`DIMJUST` BS, DXF 70

`DIMLDRBLK` H, DXF 1

`DIMLFAC` BD, DXF 40

`DIMLIM` B, DXF 70

`DIMLTEX1` H, DXF 6

`DIMLTEX2` H, DXF 6

`DIMLTYPE` H, DXF 6

`DIMLUNIT` BS, DXF 70

`DIMLWD` BSd, DXF 70

`DIMLWE` BSd, DXF 70

`DIMMZF` BD

`DIMMZS` T, DXF 1

`DIMPOST` TV, DXF 1

`DIMPOST_T`
T

`DIMRND` BD, DXF 40

`DIMSAH` B, DXF 70

`DIMSAV` B

`DIMSCALE` BD, DXF 40

`DIMSD1` B, DXF 70

`DIMSD2` B, DXF 70

`DIMSE1` B, DXF 70

`DIMSE2` B, DXF 70

`DIMSHO` B, DXF 70

`DIMSOXD` B, DXF 70

`DIMSTYLE` H, DXF 2

`DIMSTYLE_CONTROL_OBJECT`
H

`DIMTAD` BS, DXF 70

`DIMTDEC` BS, DXF 70

`DIMTFAC` BD, DXF 40

`DIMTFILL` BS, DXF 70

`DIMTFILLCLR`
CMC, DXF 70

`DIMTIH` B, DXF 70

`DIMTIX` B, DXF 70

`DIMTM` BD, DXF 40

`DIMTMOVE` BS, DXF 70

`DIMTOFL` B, DXF 70

`DIMTOH` B, DXF 70

`DIMTOL` B, DXF 70

`DIMTOLJ` BS, DXF 70

`DIMTP` BD, DXF 40

`DIMTSZ` BD, DXF 40

`DIMTVP` BD, DXF 40

`DIMTXSTY` H, DXF 7

`DIMTXT` BD, DXF 40

`DIMTXTDIRECTION`
B, DXF 70

`DIMTZIN` BS, DXF 70

`DIMUNIT` BS, DXF 70

`DIMUPT` B, DXF 70

`DIMZIN` BS, DXF 70

`DISPSILH` B, DXF 70

`DRAGMODE` BS, DXF 70

`DRAGVS` H, DXF 349

`DWFFRAME` RC, DXF 280

`DWGCODEPAGE`
TV, DXF 3

`ELEVATION`
BD, DXF 40

`ENDCAPS` B, DXF 280

`EXTMAX` 3BD, DXF 30

`EXTMIN` 3BD, DXF 30

`EXTNAMES` B, DXF 290

`FACETRES` BD

`FASTZOOM` B, DXF 70

`FILLETRAD`
BD, DXF 40

`FILLMODE` B, DXF 70

`FINGERPRINTGUID`
TV, DXF 2

`FLAGS` BL

`FLATLAND` B, DXF 70

`FRONTZ` BD

`GRIDMODE` RS, DXF 70

`GRIDUNIT` 2RD, DXF 20

`HALOGAP` RC, DXF 280

`HANDLING` BS, DXF 70

HANDSEED H, DXF 5

HIDETEXT RC, DXF 280

HYPERLINKBASE
T, DXF 1

INDEXCTL RC, DXF 280

INSBASE 3BD, DXF 30

INSUNITS BS, DXF 70

INTERFERECOLOR
CMC, DXF 62

INTERFEREOBJVS
H, DXF 345

INTERFEREVPVS
H, DXF 346

INTERSECTIONCOLOR
BS, DXF 70

INTERSECTIONDISPLAY
RC, DXF 280

ISOLINES BS

JOINSTYLE
B, DXF 280

LATITUDE BD, DXF 40

LAYER_CONTROL_OBJECT
H

LENSLENGTH
BD, DXF 40

LIGHTGLYPHDISPLAY
RC, DXF 280

LIMCHECK B, DXF 70

LIMMAX 2DPOINT, DXF 20

LIMMIN 2DPOINT, DXF 20

LOFTANG1 BD, DXF 40

LOFTANG2 BD, DXF 40

LOFTMAG1 BD, DXF 40

LOFTMAG2 BD, DXF 40

LOFTNORMALS
RC, DXF 280

`LOFTPARAM`
: BS, DXF 70

`LONGITUDE`
: BD, DXF 40

`LTSCALE` BD, DXF 40

`LTYPE_BYBLOCK`
: H

`LTYPE_BYLAYER`
: H

`LTYPE_CONTINUOUS`
: H

`LTYPE_CONTROL_OBJECT`
: H

`LUNITS` BS, DXF 70

`LUPREC` BS, DXF 70

`LWDISPLAY`
: B, DXF 290

`MAXACTVP` BS, DXF 70

`MEASUREMENT`
: BS, DXF 70

`MENU` TV, DXF 1

`MENUEXT` RC

`MIRRTEXT` B, DXF 70

`NORTHDIRECTION`
: BD, DXF 40

`OBSCOLOR` BS, DXF 70

`OBSLTYPE` RC, DXF 280

`OLESTARTUP`
: B, DXF 290

`ORTHOMODE`
: B, DXF 70

`OSMODE` BS, DXF 70

`PDMODE` BS, DXF 70

`PDSIZE` BD, DXF 40

`PELEVATION`
: BD, DXF 40

`PELLIPSE` B

`PEXTMAX` 3BD, DXF 30

`PEXTMIN` 3BD, DXF 30

`PICKSTYLE`
BS, DXF 70

`PINSBASE` 3BD, DXF 30

`PLIMCHECK`
B, DXF 70

`PLIMMAX` 2DPOINT, DXF 20

`PLIMMIN` 2DPOINT, DXF 20

`PLINEGEN` B, DXF 70

`PLINEWID` BD, DXF 40

`PROJECTNAME`
TV, DXF 1

`PROXYGRAPHICS`
BS, DXF 70

`PSLTSCALE`
B, DXF 70

`PSOLHEIGHT`
BD, DXF 40

`PSOLWIDTH`
BD, DXF 40

`PSTYLEMODE`
B, DXF 290

`PSVPSCALE`
BD, DXF 40

`PUCSBASE` H, DXF 2

`PUCSNAME` H, DXF 2

`PUCSORG` 3BD, DXF 30

`PUCSORGBACK`
3BD, DXF 30

`PUCSORGBOTTOM`
3BD, DXF 30

`PUCSORGFRONT`
3BD, DXF 30

`PUCSORGLEFT`
3BD, DXF 30

PUCSORGRIGHT
3BD, DXF 30

PUCSORGTOP
3BD, DXF 30

PUCSORTHOREF
H, DXF 2

PUCSORTHOVIEW
BS, DXF 70

PUCSXDIR 3BD, DXF 30

PUCSYDIR 3BD, DXF 30

QTEXTMODE
B, DXF 70

REALWORLDSCALE
B, DXF 290

REGENMODE
B, DXF 70

REQUIREDVERSIONS
BLL, DXF 160

SAVEIMAGES
BS

SHADEDGE BS, DXF 70

SHADEDIF BS, DXF 70

SHADOWPLANELOCATION
BD, DXF 40

SHOWHIST RC, DXF 280

SKETCHINC
BD, DXF 40

SKPOLY B, DXF 70

SNAPANG RD, DXF 50

SNAPBASE 2RD, DXF 20

SNAPISOPAIR
RS, DXF 70

SNAPMODE RS, DXF 70

SNAPSTYLE
RS, DXF 70

SNAPUNIT 2RD, DXF 20

`SOLIDHIST`
: RC, DXF 280

`SORTENTS` RC, DXF 280

`SPLFRAME` B, DXF 70

`SPLINESEGS`
: BS, DXF 70

`SPLINETYPE`
: BS, DXF 70

`STEPSIZE` BD, DXF 40

`STEPSPERSEC`
: BD, DXF 40

`STYLESHEET`
: TV, DXF 1

`STYLE_CONTROL_OBJECT`
: H

`SURFTAB1` BS, DXF 70

`SURFTAB2` BS, DXF 70

`SURFTYPE` BS, DXF 70

`SURFU` BS, DXF 70

`SURFV` BS, DXF 70

`TARGET` 3BD

`TDCREATE` TIMEBLL, DXF 40

`TDINDWG` TIMEBLL, DXF 40

`TDUCREATE`
: TIMEBLL, DXF 40

`TDUPDATE` TIMEBLL, DXF 40

`TDUSRTIMER`
: TIMEBLL, DXF 40

`TDUUPDATE`
: TIMEBLL, DXF 40

`TEXTQLTY` BS

`TEXTSIZE` BD, DXF 40

`TEXTSTYLE`
: H, DXF 7

`THICKNESS`
: BD, DXF 40

`TILEMODE` B, DXF 70

`TILEMODELIGHTSYNCH`
RC, DXF 280

`TIMEZONE` BL, DXF 70

`TRACEWID` BD, DXF 40

`TREEDEPTH`
BSd, DXF 70

`TSTACKALIGN`
BS

`TSTACKSIZE`
BS

`UCSBASE` H, DXF 2

`UCSICON` RC

`UCSNAME` H, DXF 2

`UCSORG` 3BD, DXF 30

`UCSORGBACK`
3BD, DXF 30

`UCSORGBOTTOM`
3BD, DXF 30

`UCSORGFRONT`
3BD, DXF 30

`UCSORGLEFT`
3BD, DXF 30

`UCSORGRIGHT`
3BD, DXF 30

`UCSORGTOP`
3BD, DXF 30

`UCSORTHOREF`
H, DXF 2

`UCSORTHOVIEW`
BS, DXF 70

`UCSXDIR` 3BD, DXF 30

`UCSYDIR` 3BD, DXF 30

`UCS_CONTROL_OBJECT`
H

`UNITMODE` BS, DXF 70

`USERI1` BSd, DXF 70

`USERI2` BSd, DXF 70

`USERI3` BSd, DXF 70

`USERI4` BSd, DXF 70

`USERI5` BSd, DXF 70

`USERR1` BD, DXF 40

`USERR2` BD, DXF 40

`USERR3` BD, DXF 40

`USERR4` BD, DXF 40

`USERR5` BD, DXF 40

`USRTIMER` B, DXF 70

`VERSIONGUID`
TV, DXF 2

`VIEWCTR` 3RD, DXF 20

`VIEWDIR` 3BD, DXF 30

`VIEWMODE` B

`VIEWSIZE` RD, DXF 40

`VIEWTWIST`
BD

`VIEW_CONTROL_OBJECT`
H

`VISRETAIN`
B, DXF 70

`VPOINTX` 3RD

`VPOINTXALT`
3RD

`VPOINTY` 3RD

`VPOINTYALT`
3RD

`VPOINTZ` 3RD

`VPOINTZALT`
3RD

`VPORT_CONTROL_OBJECT`
H

`VX_CONTROL_OBJECT`
H

`VX_TABLE_RECORD`
: H

`WIREFRAME`
: B

`WORLDVIEW`
: B, DXF 70

`XCLIPFRAME`
: RC, DXF 290

`XEDIT` B, DXF 290

`_3DDWFPREC`
: BD, DXF 40

`aspect_ratio`
: RD

`bitsize` RL

`bitsize_hi`
: RL

`circle_zoom_percent`
: RS

`codepage` RS

`dwg_size` RL

`flag_3d` RS

`layer_colors`
: RS

`numentities`
: RS

`oldCECOLOR_hi`
: RL

`oldCECOLOR_lo`
: RL

`size` RL

`unit1_name`
: TV

`unit1_ratio`
: BD

`unit2_name`
: TV

`unit2_ratio`
: BD

`unit3_name`
: TV

`unit3_ratio`
: BD

`unit4_name`
: TV

`unit4_ratio`
: BD

`unknown_10`
: BS

`unknown_11`
: B

`unknown_12`
: BL

`unknown_13`
: BL

`unknown_14`
: BL

`unknown_14b`
: BL

`unknown_15`
: BL

`unknown_16`
: BL

`unknown_17`
: BL

`unknown_20`
: H

`unknown_21`
: BL

`unknown_22`
: BL

`unknown_23`
: BD

`unknown_4f2`
: RL

`unknown_5`
: RS

`unknown_51e`
: RS

`unknown_520`
: RS

`unknown_52c`
: RSd

`unknown_52e`
: RS

`unknown_530`
: RC

`unknown_54`
: BS

`unknown_55`
: BS

`unknown_56`
: BS

`unknown_57`
: BS

`unknown_59`
: RS

`unknown_6`
: RS

`unknown_6a`
: RS

`unknown_6b`
: RS

`unknown_6c`
: RS

`unknown_8`
: BL

`unknown_9`
: BL

`unknown_day`
: RS

`unknown_hour`
: RS

`unknown_min`
: RS

`unknown_mon`
RS

`unknown_ms`
RS

`unknown_sec`
RS

`unknown_string`
T

`unknown_year`
RS

4.2 ENTITIES

All graphical objects with its fields. See [Common Entity fields], page 262,

3DFACE

`parent` struct _dwg_object_entity*

`has_no_flags`
B

`z_is_zero`
B

`corner1` 3BD, DXF 10

`corner2` 3BD, DXF 11

`corner3` 3BD, DXF 12

`corner4` 3BD, DXF 13

`invis_flags`
BS, DXF 70

3DLINE

`parent` struct _dwg_object_entity*

`start` 3RD, DXF 10

`end` 3RD, DXF 11

`extrusion`
3RD, DXF 210

`thickness`
RD, DXF 39

3DSOLID

`parent` struct _dwg_object_entity*

`acis_empty`
B, DXF 290

`unknown` B

`version` BS, DXF 70

`num_blocks`
BL

`block_size`
BL*

`encr_sat_data`
char **, DXF 1

`sab_size` BL

`acis_data`
RC*

`wireframe_data_present`
B

`point_present`
B

`point` 3BD

`isolines` BL

`isoline_present`
B

`num_wires`
BL

`wires` Dwg_3DSOLID_wire*

`num_silhouettes`
BL

`silhouettes`
Dwg_3DSOLID_silhouette*

`_dxf_sab_converted`
B

`acis_empty2`
B

`extra_acis_data`
struct _dwg_entity_3DSOLID*

`num_materials`
BL

`materials`
Dwg_3DSOLID_material*

`revision_guid`
RC, DXF 2

`revision_major`
BL

`revision_minor1`
BS

`revision_minor2`
BS

`revision_bytes`
RC

`end_marker`
BL

`history_id`
H, DXF 350

`has_revision_guid`
B

`acis_empty_bit`
B

ALIGNMENTPARAMETERENTITY

`parent` struct _dwg_object_entity*

ARC

`parent` struct _dwg_object_entity*

`center` 3BD, DXF 10

`radius` BD, DXF 40

`thickness`
BT, DXF 39

`extrusion`
BE, DXF 210

`start_angle`
BD, DXF 50

`end_angle`
BD, DXF 51

ARCALIGNEDTEXT

`parent` struct _dwg_object_entity*

`text_size`
D2T, DXF 42

`xscale` D2T, DXF 41

`char_spacing`
D2T, DXF 43

`style` T, DXF 7

`text_value`
T, DXF 1

`font_name`
T, DXF 2

`bigfont_name`
T, DXF 3

`offset_from_arc`
D2T, DXF 44

`right_offset`
D2T, DXF 45

`left_offset`
D2T, DXF 46

`center` 3BD, DXF 10

`radius` BD, DXF 40

`start_angle`
BD, DXF 50

`end_angle`
BD, DXF 51

`extrusion`
3BD, DXF 210

`color` BL, DXF 90

`is_reverse`
BS, DXF 70

`text_direction`
BS, DXF 71

`alignment`
BS, DXF 72

`text_position`
BS, DXF 73

`is_bold` BS, DXF 74

`is_italic`
BS, DXF 75

`is_underlined`
BS, DXF 76

`character_set`
BS, DXF 77

`pitch_and_family`
BS, DXF 78

`is_shx` BS, DXF 79

`wizard_flag`
BS, DXF 280

`arc_handle`
H, DXF 330

ARC_DIMENSION

`parent` struct _dwg_object_entity*

`class_version`
RC, DXF 280

`extrusion`
BE, DXF 210

`def_pt` 3BD, DXF 10

`text_midpt`
2RD, DXF 11

`elevation`
BD, DXF 31

`flag` RC, DXF 70

`flag1` RC

`user_text`
T, DXF 1

`text_rotation`
BD, DXF 53

`horiz_dir`
BD, DXF 51

`ins_scale`
3BD_1

`ins_rotation`
BD, DXF 54

`attachment`
BS, DXF 71

`lspace_style`
BS, DXF 72

`lspace_factor`
BD, DXF 41

`act_measurement`
BD, DXF 42

`unknown` B, DXF 73

`flip_arrow1`
B, DXF 74

`flip_arrow2`
B, DXF 75

`clone_ins_pt`
2RD, DXF 12

`dimstyle` H, DXF 3

`block` H, DXF 2

`xline1_pt`
3BD, DXF 13

`xline2_pt`
3BD, DXF 14

`center_pt`
3BD, DXF 15

`is_partial`
B, DXF 70

`arc_start_param`
BD, DXF 40

`arc_end_param`
BD, DXF 41

`has_leader`
B, DXF 71

`leader1_pt`
3BD, DXF 16

`leader2_pt`
3BD, DXF 17

ATTDEF

`parent` struct _dwg_object_entity*

`elevation`
BD, DXF 30

`ins_pt` 2DPOINT, DXF 10

`alignment_pt`
2DPOINT, DXF 11

`extrusion`
BE, DXF 210

`thickness`
RD, DXF 39

`oblique_angle`
: RD, DXF 51

`rotation`
: RD, DXF 50

`height`
: RD, DXF 40

`width_factor`
: RD, DXF 41

`default_value`
: T, DXF 1

`generation`
: BS, DXF 71

`horiz_alignment`
: BS, DXF 72

`vert_alignment`
: BS, DXF 74

`dataflags`
: RC

`is_locked_in_block`
: RC, DXF 280

`keep_duplicate_records`
: RC

`tag`
: T, DXF 2

`field_length`
: BS, DXF 73

`flags`
: RC, DXF 70

`lock_position_flag`
: B

`style`
: H, DXF 7

`mtext_type`
: RC, DXF 70

`mtext`
: Dwg_AcDbMTextObjectEmbedded

`is_really_locked`
: B

`annotative_data_size`
: BS

`annotative_data`
: RC*

`annotative_flag`
: BS

`annotative_style`
H

`prompt` T, DXF 3

ATTRIB

`parent` struct _dwg_object_entity*

`elevation`
BD, DXF 30

`ins_pt` 2DPOINT, DXF 10

`alignment_pt`
2DPOINT, DXF 11

`extrusion`
BE, DXF 210

`thickness`
RD, DXF 39

`oblique_angle`
RD, DXF 51

`rotation` RD, DXF 50

`height` RD, DXF 40

`width_factor`
RD, DXF 41

`text_value`
T, DXF 1

`generation`
BS, DXF 71

`horiz_alignment`
BS, DXF 72

`vert_alignment`
BS, DXF 74

`dataflags`
RC

`is_locked_in_block`
RC, DXF 280

`keep_duplicate_records`
RC, DXF 280

`tag` T, DXF 2

`field_length`
BS, DXF 73

`flags` RC, DXF 70

`lock_position_flag`
B

`style` H, DXF 7

`mtext_type`
RC, DXF 70

`mtext` Dwg_AcDbMTextObjectEmbedded

`is_really_locked`
B

`annotative_data_size`
BS

`annotative_data`
RC*

`annotative_flag`
BS

`annotative_style`
H

BASEPOINTPARAMETERENTITY

`parent` struct _dwg_object_entity*

BLOCK

`parent` struct _dwg_object_entity*

`name` TV, DXF 2

`xref_pname`
TV, DXF 1

`base_pt` 2RD, DXF 10

BODY

See [3DSOLID], page 24,

CAMERA

`parent` struct _dwg_object_entity*

`view` H

CIRCLE

`parent` struct _dwg_object_entity*

`center` 3BD, DXF 10

`radius` BD, DXF 40

`thickness`
BT, DXF 39

`extrusion`
: BE, DXF 210

DGNUNDERLAY

`parent` struct _dwg_object_entity*

`extrusion`
: BE, DXF 210

`ins_pt` 3BD, DXF 10

`scale` 3BD_1, DXF 41

`angle` BD, DXF 50

`flag` RC, DXF 280

`contrast` RC, DXF 281

`fade` RC, DXF 282

`num_clip_verts`
: BL

`clip_verts`
: 2RD*, DXF 11

`num_clip_inverts`
: BS, DXF 170

`clip_inverts`
: 2RD*, DXF 12

`definition_id`
: H, DXF 340

DIMENSION_ALIGNED

`parent` struct _dwg_object_entity*

`class_version`
: RC, DXF 280

`extrusion`
: BE, DXF 210

`def_pt` 3BD, DXF 10

`text_midpt`
: 2RD, DXF 11

`elevation`
: BD, DXF 31

`flag` RC, DXF 70

`flag1` RC

`user_text`
: T, DXF 1

`text_rotation`
BD, DXF 53

`horiz_dir`
BD, DXF 51

`ins_scale`
3BD_1

`ins_rotation`
BD, DXF 54

`attachment`
BS, DXF 71

`lspace_style`
BS, DXF 72

`lspace_factor`
BD, DXF 41

`act_measurement`
BD, DXF 42

`unknown` B, DXF 73

`flip_arrow1`
B, DXF 74

`flip_arrow2`
B, DXF 75

`clone_ins_pt`
2RD, DXF 12

`dimstyle` H, DXF 3

`block` H, DXF 2

`xline1_pt`
3BD, DXF 13

`xline2_pt`
3BD, DXF 14

`oblique_angle`
BD, DXF 50

DIMENSION_ANG2LN

`parent` struct _dwg_object_entity*

`class_version`
RC, DXF 280

`extrusion`
BE, DXF 210

`def_pt` 3BD, DXF 10

`text_midpt`
: 2RD, DXF 11

`elevation`
: BD, DXF 31

`flag` RC, DXF 70

`flag1` RC

`user_text`
: T, DXF 1

`text_rotation`
: BD, DXF 53

`horiz_dir`
: BD, DXF 51

`ins_scale`
: 3BD_1

`ins_rotation`
: BD, DXF 54

`attachment`
: BS, DXF 71

`lspace_style`
: BS, DXF 72

`lspace_factor`
: BD, DXF 41

`act_measurement`
: BD, DXF 42

`unknown` B, DXF 73

`flip_arrow1`
: B, DXF 74

`flip_arrow2`
: B, DXF 75

`clone_ins_pt`
: 2RD, DXF 12

`dimstyle` H, DXF 3

`block` H, DXF 2

`xline1start_pt`
: 3BD, DXF 13

`xline1end_pt`
: 3BD, DXF 14

`xline2start_pt`
3BD, DXF 15

`xline2end_pt`
3BD, DXF 16

DIMENSION_ANG3PT

`parent` struct _dwg_object_entity*

`class_version`
RC, DXF 280

`extrusion`
BE, DXF 210

`def_pt` 3BD, DXF 10

`text_midpt`
2RD, DXF 11

`elevation`
BD, DXF 31

`flag` RC, DXF 70

`flag1` RC

`user_text`
T, DXF 1

`text_rotation`
BD, DXF 53

`horiz_dir`
BD, DXF 51

`ins_scale`
3BD_1

`ins_rotation`
BD, DXF 54

`attachment`
BS, DXF 71

`lspace_style`
BS, DXF 72

`lspace_factor`
BD, DXF 41

`act_measurement`
BD, DXF 42

`unknown` B, DXF 73

`flip_arrow1`
B, DXF 74

`flip_arrow2`
B, DXF 75

`clone_ins_pt`
2RD, DXF 12

`dimstyle` H, DXF 3

`block` H, DXF 2

`xline1_pt`
3BD, DXF 13

`xline2_pt`
3BD, DXF 14

`center_pt`
3BD, DXF 15

`xline2end_pt`
3RD

DIMENSION_DIAMETER

`parent` struct _dwg_object_entity*

`class_version`
RC, DXF 280

`extrusion`
BE, DXF 210

`def_pt` 3BD, DXF 10

`text_midpt`
2RD, DXF 11

`elevation`
BD, DXF 31

`flag` RC, DXF 70

`flag1` RC

`user_text`
T, DXF 1

`text_rotation`
BD, DXF 53

`horiz_dir`
BD, DXF 51

`ins_scale`
3BD_1

`ins_rotation`
BD, DXF 54

`attachment`
BS, DXF 71

`lspace_style`
BS, DXF 72

`lspace_factor`
BD, DXF 41

`act_measurement`
BD, DXF 42

`unknown` B, DXF 73

`flip_arrow1`
B, DXF 74

`flip_arrow2`
B, DXF 75

`clone_ins_pt`
2RD, DXF 12

`dimstyle` H, DXF 3

`block` H, DXF 2

`first_arc_pt`
3BD, DXF 15

`leader_len`
BD, DXF 40

DIMENSION_LINEAR

`parent` struct _dwg_object_entity*

`class_version`
RC, DXF 280

`extrusion`
BE, DXF 210

`def_pt` 3BD, DXF 10

`text_midpt`
2RD, DXF 11

`elevation`
BD, DXF 31

`flag` RC, DXF 70

`flag1` RC

`user_text`
T, DXF 1

`text_rotation`
BD, DXF 53

`horiz_dir`
: BD, DXF 51

`ins_scale`
: 3BD_1

`ins_rotation`
: BD, DXF 54

`attachment`
: BS, DXF 71

`lspace_style`
: BS, DXF 72

`lspace_factor`
: BD, DXF 41

`act_measurement`
: BD, DXF 42

`unknown` B, DXF 73

`flip_arrow1`
: B, DXF 74

`flip_arrow2`
: B, DXF 75

`clone_ins_pt`
: 2RD, DXF 12

`dimstyle` H, DXF 3

`block` H, DXF 2

`xline1_pt`
: 3BD, DXF 13

`xline2_pt`
: 3BD, DXF 14

`oblique_angle`
: BD, DXF 52

`dim_rotation`
: BD, DXF 50

DIMENSION_ORDINATE

`parent` struct _dwg_object_entity*

`class_version`
: RC, DXF 280

`extrusion`
: BE, DXF 210

`def_pt` 3BD, DXF 10

`text_midpt`
: 2RD, DXF 11

`elevation`
: BD, DXF 31

`flag` RC, DXF 70

`flag1` RC

`user_text`
: T, DXF 1

`text_rotation`
: BD, DXF 53

`horiz_dir`
: BD, DXF 51

`ins_scale`
: 3BD_1

`ins_rotation`
: BD, DXF 54

`attachment`
: BS, DXF 71

`lspace_style`
: BS, DXF 72

`lspace_factor`
: BD, DXF 41

`act_measurement`
: BD, DXF 42

`unknown` B, DXF 73

`flip_arrow1`
: B, DXF 74

`flip_arrow2`
: B, DXF 75

`clone_ins_pt`
: 2RD, DXF 12

`dimstyle` H, DXF 3

`block` H, DXF 2

`feature_location_pt`
: 3BD, DXF 13

`leader_endpt`
: 3BD, DXF 14

`flag2` RC

DIMENSION_RADIUS

`parent` struct _dwg_object_entity*

`class_version`
RC, DXF 280

`extrusion`
BE, DXF 210

`def_pt` 3BD, DXF 10

`text_midpt`
2RD, DXF 11

`elevation`
BD, DXF 31

`flag` RC, DXF 70

`flag1` RC

`user_text`
T, DXF 1

`text_rotation`
BD, DXF 53

`horiz_dir`
BD, DXF 51

`ins_scale`
3BD_1

`ins_rotation`
BD, DXF 54

`attachment`
BS, DXF 71

`lspace_style`
BS, DXF 72

`lspace_factor`
BD, DXF 41

`act_measurement`
BD, DXF 42

`unknown` B, DXF 73

`flip_arrow1`
B, DXF 74

`flip_arrow2`
B, DXF 75

`clone_ins_pt`
2RD, DXF 12

`dimstyle` H, DXF 3

`block` H, DXF 2

`first_arc_pt`
3BD, DXF 15

`leader_len`
BD, DXF 40

DWFUNDERLAY

See [UNDERLAY], page 33,

ELLIPSE

`parent` struct _dwg_object_entity*

`center` 3BD, DXF 10

`sm_axis` 3BD, DXF 11

`extrusion`
BE, DXF 210

`axis_ratio`
BD, DXF 40

`start_angle`
BD, DXF 41

`end_angle`
BD, DXF 42

ENDBLK

`parent` struct _dwg_object_entity*

ENDREP

`parent` struct _dwg_object_entity*

`numcols` RS, DXF 70

`numrows` RS, DXF 71

`colspacing`
RD, DXF 40

`rowspacing`
RD, DXF 41

EXTRUDEDSURFACE

`parent` struct _dwg_object_entity*

`acis_empty`
B, DXF 290

`unknown` B

`version` BS, DXF 70

`num_blocks`
: BL

`block_size`
: BL*

`encr_sat_data`
: char **, DXF 1

`sab_size` BL

`acis_data`
: RC*

`wireframe_data_present`
: B

`point_present`
: B

`point` 3BD

`isolines` BL

`isoline_present`
: B

`num_wires`
: BL

`wires` Dwg_3DSOLID_wire*

`num_silhouettes`
: BL

`silhouettes`
: Dwg_3DSOLID_silhouette*

`_dxf_sab_converted`
: B

`acis_empty2`
: B

`extra_acis_data`
: struct _dwg_entity_3DSOLID*

`num_materials`
: BL

`materials`
: Dwg_3DSOLID_material*

`revision_guid`
: RC, DXF 2

`revision_major`
: BL

`revision_minor1`
BS

`revision_minor2`
BS

`revision_bytes`
RC

`end_marker`
BL

`history_id`
H, DXF 350

`has_revision_guid`
B

`acis_empty_bit`
B

`modeler_format_version`
BS

`bindata_size`
BL

`bindata` TF

`u_isolines`
BS, DXF 71

`v_isolines`
BS, DXF 72

`class_version`
BL

`draft_angle`
BD, DXF 42

`draft_start_distance`
BD, DXF 43

`draft_end_distance`
BD, DXF 44

`twist_angle`
BD, DXF 45

`scale_factor`
BD, DXF 48

`align_angle`
BD, DXF 49

`sweep_entity_transmatrix`
BD*, DXF 46

`path_entity_transmatrix`
BD*, DXF 47

`is_solid` B, DXF 290

`sweep_alignment_flags`
BS, DXF 70

`path_flags`
BS, DXF 71

`align_start`
B, DXF 292

`bank` B, DXF 293

`base_point_set`
B, DXF 294

`sweep_entity_transform_computed`
B, DXF 295

`path_entity_transform_computed`
B, DXF 296

`reference_vector_for_controlling_twist`
3BD, DXF 11

`sweep_entity`
H

`path_entity`
H

`sweep_vector`
3BD, DXF 10

`sweep_transmatrix`
BD*, DXF 40

FLIPGRIPENTITY

`parent` struct _dwg_object_entity*

FLIPPARAMETERENTITY

`parent` struct _dwg_object_entity*

GEOPOSITIONMARKER

`parent` struct _dwg_object_entity*

`class_version`
BS, DXF 90

`position` 3BD, DXF 10

`radius` BD, DXF 40

`landing_gap`
BD, DXF 40

`notes` T, DXF 1

`text_alignment`
RC, DXF 280

`mtext_visible`
B, DXF 290

`enable_frame_text`
B, DXF 290

`ins_pt` 3BD

`alignment_pt`
2RD

`mtext` Dwg_AcDbMTextObjectEmbedded

`is_really_locked`
B

`annotative_data_size`
BS

`annotative_data`
RC*

`annotative_flag`
BS

`annotative_style`
H

HATCH

`parent` struct _dwg_object_entity*

`is_gradient_fill`
BL, DXF 450

`reserved` BL, DXF 451

`gradient_angle`
BD, DXF 460

`gradient_shift`
BD, DXF 461

`single_color_gradient`
BL, DXF 452

`gradient_tint`
BD, DXF 462

`num_colors`
BL, DXF 453

`colors` Dwg_HATCH_Color*

`gradient_name`
T, DXF 470

`elevation`
BD, DXF 30

`extrusion`
BE, DXF 210

`name` T, DXF 2

`is_solid_fill`
B, DXF 70

`is_associative`
B, DXF 71

`num_paths`
BL, DXF 91

`paths` Dwg_HATCH_Path*

`style` BS, DXF 75

`pattern_type`
BS, DXF 76

`angle` BD, DXF 52

`scale_spacing`
BD, DXF 41

`double_flag`
B, DXF 77

`num_deflines`
BS, DXF 78

`deflines` Dwg_HATCH_DefLine*

`has_derived`
B

`pixel_size`
BD, DXF 47

`num_seeds`
BL, DXF 98

`seeds` 2RD*, DXF 10

HELIX

`parent` struct _dwg_object_entity*

`flag` BS, DXF 70

`scenario` BS

`degree` BS, DXF 71

`splineflags`
BL

`knotparam`
BL

`fit_tol` BD, DXF 44

`beg_tan_vec`
3BD, DXF 12

`end_tan_vec`
3BD, DXF 13

`rational` B

`closed_b` B

`periodic` B

`weighted` B

`knot_tol` BD, DXF 42

`ctrl_tol` BD, DXF 43

`num_fit_pts`
BS, DXF 74

`fit_pts` 3DPOINT*, DXF 11

`num_knots`
BL, DXF 72

`knots` BD*, DXF 40

`num_ctrl_pts`
BL, DXF 73

`ctrl_pts` Dwg_SPLINE_control_point*

`major_version`
BL, DXF 90

`maint_version`
BL, DXF 91

`axis_base_pt`
3BD, DXF 10

`start_pt` 3BD, DXF 11

`axis_vector`
3BD, DXF 12

`radius` BD, DXF 40

`turns` BD, DXF 41

`turn_height`
BD, DXF 42

`handedness`
B, DXF 290

`constraint_type`
RC, DXF 280

IMAGE

`parent` struct _dwg_object_entity*

`class_version`
BL, DXF 90

`pt0` 3BD, DXF 10

`uvec` 3BD, DXF 11

`vvec` 3BD, DXF 12

`image_size`
2RD, DXF 13

`display_props`
BS, DXF 70

`clipping` B, DXF 280

`brightness`
RC, DXF 281

`contrast` RC, DXF 282

`fade` RC, DXF 283

`clip_mode`
B, DXF 290

`clip_boundary_type`
BS, DXF 71

`num_clip_verts`
BL, DXF 91

`clip_verts`
2RD*, DXF 14

`imagedef` H, DXF 340

`imagedefreactor`
H, DXF 360

INSERT

`parent` struct _dwg_object_entity*

`ins_pt` 3DPOINT, DXF 10

`scale_flag`
: BB

`scale` 3BD_1, DXF 41

`rotation` BD, DXF 50

`extrusion`
: BE, DXF 210

`has_attribs`
: B, DXF 66

`num_owned`
: BL

`block_header`
: H, DXF 2

`first_attrib`
: H

`last_attrib`
: H

`attribs` H*

`seqend` H

`num_cols` RS, DXF 70

`num_rows` RS, DXF 71

`col_spacing`
: RD, DXF 44

`row_spacing`
: RD, DXF 45

`block_name`
: TV, DXF 2

JUMP

`parent` struct _dwg_object_entity*

`jump_address_raw`
: RL

`jump_address`
: RL

`jump_entity_section`
: Dwg_Entity_Sections

LARGE_RADIAL_DIMENSION

`parent` struct _dwg_object_entity*

`class_version`
: RC, DXF 280

`extrusion`
: BE, DXF 210

`def_pt`
: 3BD, DXF 10

`text_midpt`
: 2RD, DXF 11

`elevation`
: BD, DXF 31

`flag`
: RC, DXF 70

`flag1`
: RC

`user_text`
: T, DXF 1

`text_rotation`
: BD, DXF 53

`horiz_dir`
: BD, DXF 51

`ins_scale`
: 3BD_1

`ins_rotation`
: BD, DXF 54

`attachment`
: BS, DXF 71

`lspace_style`
: BS, DXF 72

`lspace_factor`
: BD, DXF 41

`act_measurement`
: BD, DXF 42

`unknown`
: B, DXF 73

`flip_arrow1`
: B, DXF 74

`flip_arrow2`
: B, DXF 75

`clone_ins_pt`
: 2RD, DXF 12

`dimstyle`
: H, DXF 3

`block`
: H, DXF 2

`jog_angle`
: BD, DXF 40

`jog_pt` 3BD, DXF 13

`ovr_center`
: 3BD, DXF 14

`chord_pt` 3BD, DXF 15

LAYOUTPRINTCONFIG

`parent` struct _dwg_object_entity*

`class_version`
: BS

`flag` BS, DXF 93

LEADER

`parent` struct _dwg_object_entity*

`unknown_bit_1`
: B

`path_type`
: BS, DXF 72

`annot_type`
: BS, DXF 73

`num_points`
: BL, DXF 76

`points` 3DPOINT*, DXF 10

`origin` 3DPOINT

`extrusion`
: BE, DXF 210

`x_direction`
: 3DPOINT, DXF 211

`inspt_offset`
: 3DPOINT, DXF 212

`endptproj`
: 3DPOINT, DXF 213

`dimgap` BD

`box_height`
: BD, DXF 40

`box_width`
: BD, DXF 41

`hookline_dir`
: B, DXF 74

`hookline_on`
: B, DXF 75

`arrowhead_on`
: B, DXF 71

`arrowhead_type`
: BS

`dimasz` BD

`unknown_bit_2`
: B

`unknown_bit_3`
: B

`unknown_short_1`
: BS

`byblock_color`
: BS, DXF 77

`unknown_bit_4`
: B

`unknown_bit_5`
: B

`associated_annotation`
: H, DXF 340

`dimstyle` H, DXF 3

LIGHT

`parent` struct _dwg_object_entity*

`class_version`
: BL, DXF 90

`name` T, DXF 1

`type` BL, DXF 70

`status` B, DXF 290

`light_color`
: CMC, DXF 63

`plot_glyph`
: B, DXF 291

`intensity`
: BD, DXF 40

`position` 3BD, DXF 10

`target` 3BD, DXF 11

`attenuation_type`
BL, DXF 72

`use_attenuation_limits`
B, DXF 292

`attenuation_start_limit`
BD, DXF 41

`attenuation_end_limit`
BD, DXF 42

`hotspot_angle`
BD, DXF 50

`falloff_angle`
BD, DXF 51

`cast_shadows`
B, DXF 293

`shadow_type`
BL, DXF 73

`shadow_map_size`
BS, DXF 91

`shadow_map_softness`
RC, DXF 280

`is_photometric`
B

`has_photometric_data`
B, DXF 1

`has_webfile`
B, DXF 290

`webfile` T, DXF 300

`physical_intensity_method`
BS, DXF 70

`physical_intensity`
BD, DXF 40

`illuminance_dist`
BD, DXF 41

`lamp_color_type`
BS, DXF 71

`lamp_color_temp`
BD, DXF 42

`lamp_color_preset`
BS, DXF 72

`lamp_color_rgb`
BL

`web_rotation`
3BD_1, DXF 43

`extlight_shape`
BS, DXF 73

`extlight_length`
BD, DXF 46

`extlight_width`
BD, DXF 47

`extlight_radius`
BD, DXF 48

`webfile_type`
BS, DXF 74

`web_symetry`
BS, DXF 75

`has_target_grip`
BS, DXF 76

`web_flux` BD, DXF 49

`web_angle1`
BD, DXF 50

`web_angle2`
BD, DXF 51

`web_angle3`
BD, DXF 52

`web_angle4`
BD, DXF 53

`web_angle5`
BD, DXF 54

`glyph_display_type`
BS, DXF 77

LINE

`parent` struct _dwg_object_entity*

`z_is_zero`
RC

`start` 3BD, DXF 10

`end` 3BD, DXF 11

`thickness`
BT, DXF 39

`extrusion`
BE, DXF 210

`unknown_r11`
2RD

LINEARGRIPENTITY

`parent` struct _dwg_object_entity*

LINEARPARAMETERENTITY

`parent` struct _dwg_object_entity*

LOAD

`parent` struct _dwg_object_entity*

`file_name`
TV, DXF 1

LOFTEDSURFACE

`parent` struct _dwg_object_entity*

`acis_empty`
B, DXF 290

`unknown` B

`version` BS, DXF 70

`num_blocks`
BL

`block_size`
BL*

`encr_sat_data`
char **, DXF 1

`sab_size` BL

`acis_data`
RC*

`wireframe_data_present`
B

`point_present`
B

`point` 3BD

`isolines` BL

`isoline_present`
: B

`num_wires`
: BL

`wires` Dwg_3DSOLID_wire*

`num_silhouettes`
: BL

`silhouettes`
: Dwg_3DSOLID_silhouette*

`_dxf_sab_converted`
: B

`acis_empty2`
: B

`extra_acis_data`
: struct _dwg_entity_3DSOLID*

`num_materials`
: BL

`materials`
: Dwg_3DSOLID_material*

`revision_guid`
: RC, DXF 2

`revision_major`
: BL

`revision_minor1`
: BS

`revision_minor2`
: BS

`revision_bytes`
: RC

`end_marker`
: BL

`history_id`
: H, DXF 350

`has_revision_guid`
: B

`acis_empty_bit`
: B

`modeler_format_version`
: BS, DXF 70

`u_isolines`
: BS, DXF 71

`v_isolines`
: BS, DXF 72

`loft_entity_transmatrix`
: BD*, DXF 40

`plane_normal_lofting_type`
: BL, DXF 70

`start_draft_angle`
: BD, DXF 41

`end_draft_angle`
: BD, DXF 42

`start_draft_magnitude`
: BD, DXF 43

`end_draft_magnitude`
: BD, DXF 44

`arc_length_parameterization`
: B, DXF 290

`no_twist` B, DXF 291

`align_direction`
: B, DXF 292

`simple_surfaces`
: B, DXF 293

`closed_surfaces`
: B, DXF 294

`solid` B, DXF 295

`ruled_surface`
: B, DXF 296

`virtual_guide`
: B, DXF 297

`num_cross_sections`
: BS

`num_guide_curves`
: BS

`cross_sections`
: H*, DXF 310

`guide_curves`
: H*, DXF 310

`path_curve`
H

LWPOLYLINE

`parent` struct _dwg_object_entity*

`flag` BS, DXF 70

`const_width`
BD, DXF 43

`elevation`
BD, DXF 38

`thickness`
BD, DXF 39

`extrusion`
BE, DXF 210

`num_points`
BL, DXF 90

`points` 2RD*, DXF 10

`num_bulges`
BL

`bulges` BD*, DXF 42

`num_vertexids`
BL

`vertexids`
BL*, DXF 91

`num_widths`
BL

`widths` Dwg_LWPOLYLINE_width*

MESH

`parent` struct _dwg_object_entity*

`dlevel` BS, DXF 71

`is_watertight`
B, DXF 72

`num_subdiv_vertex`
BL, DXF 91

`subdiv_vertex`
3DPOINT*, DXF 10

`num_vertex`
BL, DXF 92

`vertex` 3DPOINT*, DXF 10

`num_faces`
BL, DXF 93

`faces` BL*, DXF 90

`num_edges`
BL, DXF 94

`edges` Dwg_MESH_edge*

`num_crease`
BL, DXF 95

`crease` BD*, DXF 140

`unknown_b1`
B

`unknown_b2`
B

MINSERT

`parent` struct _dwg_object_entity*

`ins_pt` 3DPOINT, DXF 10

`scale_flag`
BB

`scale` 3BD_1, DXF 41

`rotation` BD, DXF 50

`extrusion`
BE, DXF 210

`has_attribs`
B, DXF 66

`num_owned`
BL

`num_cols` BS, DXF 70

`num_rows` BS, DXF 71

`col_spacing`
BD, DXF 44

`row_spacing`
BD, DXF 45

`block_header`
H, DXF 2

`first_attrib`
H

`last_attrib`
H

`attribs` H*

`seqend` H

MLINE

`parent` struct _dwg_object_entity*

`scale` BD, DXF 40

`justification`
RC, DXF 70

`base_point`
3BD, DXF 10

`extrusion`
BE, DXF 210

`flags` BS, DXF 71

`num_lines`
RC, DXF 73

`num_verts`
BS, DXF 72

`verts` Dwg_MLINE_vertex*

`mlinestyle`
H, DXF 340

MPOLYGON

`parent` struct _dwg_object_entity*

`is_gradient_fill`
BL

`reserved` BL

`gradient_angle`
BD

`gradient_shift`
BD

`single_color_gradient`
BL

`gradient_tint`
BD

`num_colors`
BL

`colors` Dwg_HATCH_Color*

`gradient_name`
T

`elevation`
BD, DXF 30

`extrusion`
BE, DXF 210

`name` T, DXF 2

`is_solid_fill`
B, DXF 70

`is_associative`
B, DXF 71

`num_paths`
BL, DXF 91

`paths` Dwg_HATCH_Path*

`style` BS, DXF 75

`pattern_type`
BS, DXF 76

`angle` BD, DXF 52

`scale_spacing`
BD, DXF 41

`double_flag`
B, DXF 77

`num_deflines`
BS, DXF 78

`deflines` Dwg_HATCH_DefLine*

`hatch_color`
CMC, DXF 62

`x_dir` 2RD, DXF 11

`num_boundary_handles`
BL, DXF 99

MTEXT

`parent` struct _dwg_object_entity*

`ins_pt` 3BD, DXF 10

`extrusion`
BE, DXF 210

`x_axis_dir`
3BD, DXF 11

`rect_height`
: BD, DXF 41

`rect_width`
: BD, DXF 40

`text_height`
: BD, DXF 40

`attachment`
: BS, DXF 71

`flow_dir` BS, DXF 72

`extents_width`
: BD, DXF 42

`extents_height`
: BD, DXF 43

`text` T, DXF 1

`style` H, DXF 7

`linespace_style`
: BS, DXF 73

`linespace_factor`
: BD, DXF 44

`unknown_b0`
: B

`bg_fill_flag`
: BL, DXF 90

`bg_fill_scale`
: BL, DXF 45

`bg_fill_color`
: CMC, DXF 63

`bg_fill_trans`
: BL, DXF 441

`is_not_annotative`
: B

`class_version`
: BS

`default_flag`
: B, DXF 70

`appid` H

`ignore_attachment`
: BL

`column_type`
BS, DXF 71

`numfragments`
BL, DXF 72

`column_width`
BD, DXF 44

`gutter` BD, DXF 45

`auto_height`
B, DXF 73

`flow_reversed`
B, DXF 74

`num_column_heights`
BL, DXF 72

`column_heights`
BD*, DXF 46

MULTILEADER

`parent` struct _dwg_object_entity*

`class_version`
BS, DXF 270

`ctx` Dwg_MLEADER_AnnotContext

`mleaderstyle`
H, DXF 340

`flags` BL, DXF 90

`type` BS, DXF 170

`line_color`
CMC, DXF 91

`line_ltype`
H, DXF 341

`line_linewt`
BLd, DXF 171

`has_landing`
B, DXF 290

`has_dogleg`
B, DXF 291

`landing_dist`
BD, DXF 41

`arrow_handle`
H, DXF 342

`arrow_size`
: BD, DXF 42

`style_content`
: BS, DXF 172

`text_style`
: H, DXF 343

`text_left`
: BS, DXF 173

`text_right`
: BS, DXF 95

`text_angletype`
: BS, DXF 174

`text_alignment`
: BS, DXF 175

`text_color`
: CMC, DXF 92

`has_text_frame`
: B, DXF 292

`block_style`
: H, DXF 344

`block_color`
: CMC, DXF 93

`block_scale`
: 3BD, DXF 10

`block_rotation`
: BD, DXF 43

`style_attachment`
: BS, DXF 176

`is_annotative`
: B, DXF 293

`num_arrowheads`
: BL

`arrowheads`
: Dwg_LEADER_ArrowHead*

`num_blocklabels`
: BL

`blocklabels`
: Dwg_LEADER_BlockLabel*

`is_neg_textdir`
: B, DXF 294

`ipe_alignment`
: BS, DXF 178

`justification`
: BS, DXF 179

`scale_factor`
: BD, DXF 45

`attach_dir`
: BS, DXF 271

`attach_top`
: BS, DXF 273

`attach_bottom`
: BS, DXF 272

`is_text_extended`
: B, DXF 295

NAVISWORKSMODEL

`parent` struct _dwg_object_entity*

`flags` BS, DXF 70

`definition`
: H, DXF 340

`transmatrix`
: BD*, DXF 40

`unitfactor`
: BD, DXF 40

NURBSURFACE

`parent` struct _dwg_object_entity*

`acis_empty`
: B, DXF 290

`unknown` B

`version` BS, DXF 70

`num_blocks`
: BL

`block_size`
: BL*

`encr_sat_data`
: char **, DXF 1

`sab_size` BL

`acis_data`
RC*

`wireframe_data_present`
B

`point_present`
B

`point` 3BD

`isolines` BL

`isoline_present`
B

`num_wires`
BL

`wires` Dwg_3DSOLID_wire*

`num_silhouettes`
BL

`silhouettes`
Dwg_3DSOLID_silhouette*

`_dxf_sab_converted`
B

`acis_empty2`
B

`extra_acis_data`
struct _dwg_entity_3DSOLID*

`num_materials`
BL

`materials`
Dwg_3DSOLID_material*

`revision_guid`
RC, DXF 2

`revision_major`
BL

`revision_minor1`
BS

`revision_minor2`
BS

`revision_bytes`
RC

`end_marker`
BL

`history_id`
H, DXF 350

`has_revision_guid`
B

`acis_empty_bit`
B

`u_isolines`
BS, DXF 71

`v_isolines`
BS, DXF 72

`short170` BS, DXF 170

`cv_hull_display`
B, DXF 290

`uvec1` 3BD, DXF 10

`vvec1` 3BD, DXF 11

`uvec2` 3BD, DXF 12

`vvec2` 3BD, DXF 13

OLE2FRAME

`parent` struct _dwg_object_entity*

`type` BS, DXF 71

`mode` BS, DXF 72

`lock_aspect`
RC, DXF 73

`data_size`
BL, DXF 90

`data` TF, DXF 310

`oleversion`
BS, DXF 70

`oleclient`
TF, DXF 3

`pt1` 3BD, DXF 10

`pt2` 3BD, DXF 11

OLEFRAME

`parent` struct _dwg_object_entity*

`flag` BS, DXF 70

`mode` BS

`data_size`
BL, DXF 90

`data` TF, DXF 310

PDFUNDERLAY

See [UNDERLAY], page 33,

PLANESURFACE

`parent` struct _dwg_object_entity*

`acis_empty`
B, DXF 290

`unknown` B

`version` BS, DXF 70

`num_blocks`
BL

`block_size`
BL*

`encr_sat_data`
char **, DXF 1

`sab_size` BL

`acis_data`
RC*

`wireframe_data_present`
B

`point_present`
B

`point` 3BD

`isolines` BL

`isoline_present`
B

`num_wires`
BL

`wires` Dwg_3DSOLID_wire*

`num_silhouettes`
BL

`silhouettes`
Dwg_3DSOLID_silhouette*

`_dxf_sab_converted`
: B

`acis_empty2`
: B

`extra_acis_data`
: struct _dwg_entity_3DSOLID*

`num_materials`
: BL

`materials`
: Dwg_3DSOLID_material*

`revision_guid`
: RC, DXF 2

`revision_major`
: BL

`revision_minor1`
: BS

`revision_minor2`
: BS

`revision_bytes`
: RC

`end_marker`
: BL

`history_id`
: H, DXF 350

`has_revision_guid`
: B

`acis_empty_bit`
: B

`modeler_format_version`
: BS, DXF 70

`u_isolines`
: BS, DXF 71

`v_isolines`
: BS, DXF 72

`class_version`
: BL

POINT

`parent` struct _dwg_object_entity*

`x` BD, DXF 10

`y` BD, DXF 20

`z` BD, DXF 30

`thickness`
BT, DXF 39

`extrusion`
BE, DXF 210

`x_ang` BD, DXF 50

POINTCLOUD

`parent` struct _dwg_object_entity*

`class_version`
BS, DXF 70

`origin` 3BD, DXF 10

`saved_filename`
T, DXF 1

`num_source_files`
BL, DXF 90

`source_files`
TV*, DXF 2

`extents_min`
3BD, DXF 11

`extents_max`
3BD, DXF 12

`numpoints`
RLL, DXF 92

`ucs_name` T, DXF 3

`ucs_origin`
3BD, DXF 13

`ucs_x_dir`
3BD, DXF 210

`ucs_y_dir`
3BD, DXF 211

`ucs_z_dir`
3BD, DXF 212

`pointclouddef`
H, DXF 330

`reactor` H, DXF 360

`show_intensity`
B

`intensity_scheme`
BS, DXF 71

`intensity_style`
Dwg_POINTCLOUD_IntensityStyle

`show_clipping`
B

`num_clippings`
BL

`clippings`
Dwg_POINTCLOUD_Clippings*

POINTCLOUDEX

`parent` struct _dwg_object_entity*

`class_version`
BS, DXF 70

`extents_min`
3BD, DXF 10

`extents_max`
3BD, DXF 11

`ucs_origin`
3BD, DXF 12

`ucs_x_dir`
3BD, DXF 210

`ucs_y_dir`
3BD, DXF 211

`ucs_z_dir`
3BD, DXF 212

`is_locked`
B, DXF 290

`pointclouddefex`
H, DXF 330

`reactor` H, DXF 360

`name` T, DXF 1

`show_intensity`
B, DXF 291

`stylization_type`
BS, DXF 71

`intensity_colorscheme`
: T, DXF 1

`cur_colorscheme`
: T, DXF 1

`classification_colorscheme`
: T, DXF 1

`elevation_min`
: BD, DXF 40

`elevation_max`
: BD, DXF 41

`intensity_min`
: BL, DXF 90

`intensity_max`
: BL, DXF 91

`intensity_out_of_range_behavior`
: BS, DXF 71

`elevation_out_of_range_behavior`
: BS, DXF 72

`elevation_apply_to_fixed_range`
: B, DXF 292

`intensity_as_gradient`
: B, DXF 293

`elevation_as_gradient`
: B, DXF 294

`show_cropping`
: B, DXF 295

`unknown_bl0`
: BL, DXF 93

`unknown_bl1`
: BL, DXF 93

`num_croppings`
: BL, DXF 92

`croppings`
: Dwg_POINTCLOUDEX_Croppings*

POINTPARAMETERENTITY

`parent` struct _dwg_object_entity*

POLARGRIPENTITY

`parent` struct _dwg_object_entity*

POLYLINE_2D

`parent` struct _dwg_object_entity*

`has_vertex`
B, DXF 66

`num_owned`
BL

`first_vertex`
H

`last_vertex`
H

`vertex` H*

`seqend` H

`flag` BS, DXF 70

`curve_type`
BS, DXF 75

`start_width`
BD, DXF 40

`end_width`
BD, DXF 41

`thickness`
BT, DXF 39

`elevation`
BD

`extrusion`
BE, DXF 210

`extra_r11_size`
BL

`extra_r11_text`
TV

`num_m_verts`
BS, DXF 71

`num_n_verts`
BS, DXF 72

POLYLINE_3D

`parent` struct _dwg_object_entity*

`has_vertex`
B, DXF 66

`num_owned`
BL

`first_vertex`
H

`last_vertex`
H

`vertex` H*

`seqend` H

`curve_type`
RC, DXF 75

`start_width`
BD, DXF 40

`end_width`
BD, DXF 41

`flag` RC, DXF 70

`extrusion`
BE, DXF 210

POLYLINE_MESH

`parent` struct _dwg_object_entity*

`has_vertex`
B, DXF 66

`num_owned`
BL

`first_vertex`
H

`last_vertex`
H

`vertex` H*

`seqend` H

`flag` BS, DXF 70

`curve_type`
BS, DXF 75

`num_m_verts`
BS, DXF 71

`num_n_verts`
BS, DXF 72

`m_density`
BS, DXF 73

`n_density`
: BS, DXF 74

POLYLINE_PFACE

`parent` struct _dwg_object_entity*

`has_vertex`
: B, DXF 66

`num_owned`
: BL

`first_vertex`
: H

`last_vertex`
: H

`vertex` H*

`seqend` H

`flag` BS, DXF 70

`numverts` BS, DXF 71

`numfaces` BS, DXF 72

PROXY_ENTITY

`parent` struct _dwg_object_entity*

`proxy_id` BL, DXF 90

`class_id` BL, DXF 91

`dwg_versions`
: BL, DXF 95

`maint_version`
: BL, DXF 95

`dwg_version`
: BL, DXF 95

`from_dxf` B, DXF 70

`proxy_data_size`
: BL, DXF 92

`proxy_data`
: TF, DXF 310

`data_numbits`
: BL, DXF 93

`data_size`
: BL

`data` TF, DXF 310

`num_objids`
BL

`objids` H*

RAY

`parent` struct _dwg_object_entity*

`point` 3BD, DXF 10

`vector` 3BD, DXF 11

REGION

See [3DSOLID], page 24,

REPEAT

`parent` struct _dwg_object_entity*

REVOLVEDSURFACE

`parent` struct _dwg_object_entity*

`acis_empty`
B, DXF 290

`unknown` B

`version` BS, DXF 70

`num_blocks`
BL

`block_size`
BL*

`encr_sat_data`
char **, DXF 1

`sab_size` BL

`acis_data`
RC*

`wireframe_data_present`
B

`point_present`
B

`point` 3BD

`isolines` BL

`isoline_present`
B

`num_wires`
BL

`wires` Dwg_3DSOLID_wire*

`num_silhouettes`
BL

`silhouettes`
Dwg_3DSOLID_silhouette*

`_dxf_sab_converted`
B

`acis_empty2`
B

`extra_acis_data`
struct _dwg_entity_3DSOLID*

`num_materials`
BL

`materials`
Dwg_3DSOLID_material*

`revision_guid`
RC, DXF 2

`revision_major`
BL

`revision_minor1`
BS

`revision_minor2`
BS

`revision_bytes`
RC

`end_marker`
BL

`history_id`
H, DXF 350

`has_revision_guid`
B

`acis_empty_bit`
B

`modeler_format_version`
BS, DXF 70

`u_isolines`
BS, DXF 71

`v_isolines`
: BS, DXF 72

`class_version`
: BL, DXF 90

`id` BL, DXF 90

`axis_point`
: 3BD, DXF 10

`axis_vector`
: 3BD, DXF 11

`revolve_angle`
: BD, DXF 40

`start_angle`
: BD, DXF 41

`revolved_entity_transmatrix`
: BD*, DXF 42

`draft_angle`
: BD, DXF 43

`draft_start_distance`
: BD, DXF 44

`draft_end_distance`
: BD, DXF 45

`twist_angle`
: BD, DXF 46

`solid` B, DXF 290

`close_to_axis`
: B, DXF 291

ROTATIONGRIPENTITY

`parent` struct _dwg_object_entity*

ROTATIONPARAMETERENTITY

`parent` struct _dwg_object_entity*

RTEXT

`parent` struct _dwg_object_entity*

`pt` 3BD, DXF 10

`extrusion`
: BE, DXF 210

`rotation` BD, DXF 50

`height` BD, DXF 50

`flags` BS, DXF 70

`text_value`
T, DXF 1

`style` H, DXF 7

SECTIONOBJECT

`parent` struct _dwg_object_entity*

`state` BL, DXF 90

`flags` BL, DXF 91

`name` T, DXF 1

`vert_dir` 3BD, DXF 10

`top_height`
BD, DXF 40

`bottom_height`
BD, DXF 41

`indicator_alpha`
BS, DXF 70

`indicator_color`
CMC, DXF 62

`num_verts`
BL, DXF 92

`verts` 3BD*, DXF 11

`num_blverts`
BL, DXF 93

`blverts` 3BD*, DXF 12

`section_settings`
H, DXF 360

SEQEND

`parent` struct _dwg_object_entity*

`begin_addr_r11`
RL

SHAPE

`parent` struct _dwg_object_entity*

`ins_pt` 3BD, DXF 10

`scale` BD, DXF 40

`rotation` BD, DXF 50

`width_factor`
BD, DXF 41

`oblique_angle`
BD, DXF 51

`thickness`
BD, DXF 39

`style_id` BS, DXF 2

`extrusion`
BE, DXF 210

`style` H, DXF 7

SOLID

`parent` struct _dwg_object_entity*

`thickness`
BT, DXF 39

`elevation`
BD, DXF 38

`corner1` 2RD, DXF 10

`corner2` 2RD, DXF 11

`corner3` 2RD, DXF 12

`corner4` 2RD, DXF 13

`extrusion`
BE, DXF 210

SPLINE

`parent` struct _dwg_object_entity*

`flag` RS

`scenario` BS

`degree` BS, DXF 71

`splineflags`
BL

`knotparam`
BL

`fit_tol` BD, DXF 44

`beg_tan_vec`
3BD, DXF 12

`end_tan_vec`
3BD, DXF 13

`closed_b` B

`periodic` B

`rational` B

`weighted` B

`knot_tol` BD, DXF 42

`ctrl_tol` BD, DXF 43

`num_fit_pts`
BS, DXF 74

`fit_pts` 3DPOINT*, DXF 11

`num_knots`
BL, DXF 72

`knots` BD*, DXF 40

`num_ctrl_pts`
BL, DXF 73

`ctrl_pts` Dwg_SPLINE_control_point*

SWEPTSURFACE

`parent` struct _dwg_object_entity*

`acis_empty`
B, DXF 290

`unknown` B

`version` BS, DXF 70

`num_blocks`
BL

`block_size`
BL*

`encr_sat_data`
char **, DXF 1

`sab_size` BL

`acis_data`
RC*

`wireframe_data_present`
B

`point_present`
B

`point` 3BD

`isolines` BL

`isoline_present`
B

`num_wires`
BL

`wires` Dwg_3DSOLID_wire*

`num_silhouettes`
BL

`silhouettes`
Dwg_3DSOLID_silhouette*

`_dxf_sab_converted`
B

`acis_empty2`
B

`extra_acis_data`
struct _dwg_entity_3DSOLID*

`num_materials`
BL

`materials`
Dwg_3DSOLID_material*

`revision_guid`
RC, DXF 2

`revision_major`
BL

`revision_minor1`
BS

`revision_minor2`
BS

`revision_bytes`
RC

`end_marker`
BL

`history_id`
H, DXF 350

`has_revision_guid`
B

`acis_empty_bit`
B

`modeler_format_version`
BS, DXF 70

`u_isolines`
: BS, DXF 71

`v_isolines`
: BS, DXF 72

`class_version`
: BL, DXF 90

`sweep_entity_id`
: BL, DXF 90

`sweepdata_size`
: BL, DXF 90

`sweepdata`
: TF, DXF 310

`path_entity_id`
: BL, DXF 90

`pathdata_size`
: BL, DXF 90

`pathdata` TF, DXF 310

`draft_angle`
: BD, DXF 42

`draft_start_distance`
: BD, DXF 43

`draft_end_distance`
: BD, DXF 44

`twist_angle`
: BD, DXF 45

`scale_factor`
: BD, DXF 48

`align_angle`
: BD, DXF 49

`sweep_entity_transmatrix`
: BD*, DXF 46

`path_entity_transmatrix`
: BD*, DXF 47

`is_solid` B, DXF 290

`sweep_alignment_flags`
: BS, DXF 70

`path_flags`
: BS, DXF 71

`align_start`
B, DXF 292

`bank` B, DXF 293

`base_point_set`
B, DXF 294

`sweep_entity_transform_computed`
B, DXF 295

`path_entity_transform_computed`
B, DXF 296

`reference_vector_for_controlling_twist`
3BD, DXF 11

`sweep_entity`
H

`path_entity`
H

TABLE

`parent` struct _dwg_object_entity*

`ldata` Dwg_LinkedData

`tdata` Dwg_LinkedTableData

`fdata` Dwg_FormattedTableData

`tablestyle`
H, DXF 342

`unknown_rc`
RC

`unknown_h`
H

`unknown_bl`
BL

`unknown_b`
B

`unknown_bl1`
BL

`ins_pt` 3BD, DXF 10

`scale` 3BD_1, DXF 41

`scale_flag`
BB

`rotation` BD, DXF 50

`extrusion`
: BE, DXF 210

`has_attribs`
: B, DXF 66

`num_owned`
: BL

`flag_for_table_value`
: BS, DXF 90

`horiz_direction`
: 3BD, DXF 11

`num_cols` BL, DXF 92

`num_rows` BL, DXF 91

`num_cells`
: unsigned long

`col_widths`
: BD*, DXF 142

`row_heights`
: BD*, DXF 141

`cells` Dwg_TABLE_Cell*

`has_table_overrides`
: B

`table_flag_override`
: BL, DXF 93

`title_suppressed`
: B, DXF 280

`header_suppressed`
: B, DXF 281

`flow_direction`
: BS, DXF 70

`horiz_cell_margin`
: BD, DXF 40

`vert_cell_margin`
: BD, DXF 41

`title_row_color`
: CMC, DXF 64

`header_row_color`
: CMC, DXF 64

`data_row_color`
: CMC, DXF 64

`title_row_fill_none`
: B, DXF 283

`header_row_fill_none`
: B, DXF 283

`data_row_fill_none`
: B, DXF 283

`title_row_fill_color`
: CMC, DXF 63

`header_row_fill_color`
: CMC, DXF 63

`data_row_fill_color`
: CMC, DXF 63

`title_row_alignment`
: BS, DXF 170

`header_row_alignment`
: BS, DXF 170

`data_row_alignment`
: BS, DXF 170

`title_text_style`
: H, DXF 7

`header_text_style`
: H, DXF 7

`data_text_style`
: H, DXF 7

`title_row_height`
: BD, DXF 140

`header_row_height`
: BD, DXF 140

`data_row_height`
: BD, DXF 140

`has_border_color_overrides`
: B

`border_color_overrides_flag`
: BL, DXF 94

`title_horiz_top_color`
: CMC, DXF 64

`title_horiz_ins_color`
: CMC, DXF 65

`title_horiz_bottom_color`
: CMC, DXF 66

`title_vert_left_color`
: CMC, DXF 63

`title_vert_ins_color`
: CMC, DXF 68

`title_vert_right_color`
: CMC, DXF 69

`header_horiz_top_color`
: CMC, DXF 64

`header_horiz_ins_color`
: CMC, DXF 65

`header_horiz_bottom_color`
: CMC, DXF 66

`header_vert_left_color`
: CMC, DXF 63

`header_vert_ins_color`
: CMC, DXF 68

`header_vert_right_color`
: CMC, DXF 69

`data_horiz_top_color`
: CMC, DXF 64

`data_horiz_ins_color`
: CMC, DXF 65

`data_horiz_bottom_color`
: CMC, DXF 66

`data_vert_left_color`
: CMC, DXF 63

`data_vert_ins_color`
: CMC, DXF 68

`data_vert_right_color`
: CMC, DXF 69

`has_border_lineweight_overrides`
: B

`border_lineweight_overrides_flag`
: BL, DXF 95

`title_horiz_top_linewt`
: BS

`title_horiz_ins_linewt`
: BS

`title_horiz_bottom_linewt`
: BS

`title_vert_left_linewt`
: BS

`title_vert_ins_linewt`
: BS

`title_vert_right_linewt`
: BS

`header_horiz_top_linewt`
: BS

`header_horiz_ins_linewt`
: BS

`header_horiz_bottom_linewt`
: BS

`header_vert_left_linewt`
: BS

`header_vert_ins_linewt`
: BS

`header_vert_right_linewt`
: BS

`data_horiz_top_linewt`
: BS

`data_horiz_ins_linewt`
: BS

`data_horiz_bottom_linewt`
: BS

`data_vert_left_linewt`
: BS

`data_vert_ins_linewt`
: BS

`data_vert_right_linewt`
: BS

`has_border_visibility_overrides`
: B

`border_visibility_overrides_flag`
: BL, DXF 96

`title_horiz_top_visibility`
: BS

`title_horiz_ins_visibility`
: BS

`title_horiz_bottom_visibility`
: BS

`title_vert_left_visibility`
: BS

`title_vert_ins_visibility`
: BS

`title_vert_right_visibility`
: BS

`header_horiz_top_visibility`
: BS

`header_horiz_ins_visibility`
: BS

`header_horiz_bottom_visibility`
: BS

`header_vert_left_visibility`
: BS

`header_vert_ins_visibility`
: BS

`header_vert_right_visibility`
: BS

`data_horiz_top_visibility`
: BS

`data_horiz_ins_visibility`
: BS

`data_horiz_bottom_visibility`
: BS

`data_vert_left_visibility`
: BS

`data_vert_ins_visibility`
: BS

`data_vert_right_visibility`
: BS

`block_header`
: H, DXF 2

`first_attrib`
H

`last_attrib`
H

`attribs` H*

`seqend` H

`title_row_style_override`
H, DXF 7

`header_row_style_override`
H

`data_row_style_override`
H

`unknown_bs`
BS

`hor_dir` 3BD, DXF 11

`has_break_data`
BL

`break_flag`
BL

`break_flow_direction`
BL

`break_spacing`
BD

`break_unknown1`
BL

`break_unknown2`
BL

`num_break_heights`
BL

`break_heights`
Dwg_TABLE_BreakHeight*

`num_break_rows`
BL

`break_rows`
Dwg_TABLE_BreakRow*

TEXT

`parent` struct _dwg_object_entity*

`dataflags`
RC

`elevation`
RD, DXF 30

`ins_pt` 2DPOINT, DXF 10

`alignment_pt`
2DPOINT, DXF 11

`extrusion`
BE, DXF 210

`thickness`
RD, DXF 39

`oblique_angle`
RD, DXF 51

`rotation` RD, DXF 50

`height` RD, DXF 40

`width_factor`
RD, DXF 41

`text_value`
T, DXF 1

`generation`
BS, DXF 71

`horiz_alignment`
BS, DXF 72

`vert_alignment`
BS, DXF 73

`style` H, DXF 7

TOLERANCE

`parent` struct _dwg_object_entity*

`unknown_short`
BS

`height` BD

`dimgap` BD

`ins_pt` 3BD, DXF 10

`x_direction`
3BD, DXF 11

`extrusion`
BE

`text_value`
T, DXF 1

`dimstyle` H, DXF 3

TRACE

`parent` struct _dwg_object_entity*

`thickness`
BT, DXF 39

`elevation`
BD, DXF 38

`corner1` 2RD, DXF 10

`corner2` 2RD, DXF 11

`corner3` 2RD, DXF 12

`corner4` 2RD, DXF 13

`extrusion`
BE, DXF 210

UNKNOWN_ENT

`parent` struct _dwg_object_entity*

VERTEX_2D

`parent` struct _dwg_object_entity*

`flag` RC, DXF 70

`point` 3BD, DXF 10

`start_width`
BD, DXF 40

`end_width`
BD, DXF 41

`id` BL, DXF 91

`bulge` BD, DXF 42

`tangent_dir`
BD, DXF 50

VERTEX_3D

`parent` struct _dwg_object_entity*

`flag` RC, DXF 70

`point` 3BD, DXF 10

VERTEX_MESH

See [VERTEX_3D], page 93,

VERTEX_PFACE

See [VERTEX_3D], page 93,

VERTEX_PFACE_FACE

`parent` struct _dwg_object_entity*

`flag` RC, DXF 70

`vertind` BSd, DXF 71

VIEWPORT

`parent` struct _dwg_object_entity*

`center` 3BD, DXF 10

`width` BD, DXF 40

`height` BD, DXF 41

`on_off` RS, DXF 68

`id` RS, DXF 69

`view_target`
3BD, DXF 17

`VIEWDIR` 3BD, DXF 16

`twist_angle`
BD, DXF 51

`VIEWSIZE` BD, DXF 45

`lens_length`
BD, DXF 42

`front_clip_z`
BD, DXF 43

`back_clip_z`
BD, DXF 44

`SNAPANG` BD, DXF 50

`VIEWCTR` 2RD, DXF 12

`SNAPBASE` 2RD, DXF 13

`SNAPUNIT` 2RD, DXF 14

`GRIDUNIT` 2RD, DXF 15

`circle_zoom`
BS, DXF 72

`grid_major`
BS, DXF 61

`num_frozen_layers`
: BL

`status_flag`
: BL, DXF 90

`style_sheet`
: T, DXF 1

`render_mode`
: RC, DXF 281

`ucs_at_origin`
: B, DXF 74

`UCSVP` B, DXF 71

`ucsorg` 3BD, DXF 110

`ucsxdir` 3BD, DXF 111

`ucsydir` 3BD, DXF 112

`ucs_elevation`
: BD, DXF 146

`UCSORTHOVIEW`
: BS, DXF 79

`shadeplot_mode`
: BS, DXF 170

`use_default_lights`
: B, DXF 292

`default_lighting_type`
: RC, DXF 282

`brightness`
: BD, DXF 141

`contrast` BD, DXF 142

`ambient_color`
: CMC, DXF 63

`vport_entity_header`
: H

`frozen_layers`
: H*, DXF 341

`clip_boundary`
: H, DXF 340

`named_ucs`
: H, DXF 345

`base_ucs` H, DXF 346

`background`
: H, DXF 332

`visualstyle`
: H, DXF 348

`shadeplot`
: H, DXF 333

`sun` H, DXF 361

VISIBILITYGRIPENTITY

`parent` struct _dwg_object_entity*

VISIBILITYPARAMETERENTITY

`parent` struct _dwg_object_entity*

WIPEOUT

`parent` struct _dwg_object_entity*

`class_version`
: BL, DXF 90

`pt0` 3BD, DXF 10

`uvec` 3BD, DXF 11

`vvec` 3BD, DXF 12

`image_size`
: 2RD, DXF 13

`display_props`
: BS, DXF 70

`clipping` B, DXF 280

`brightness`
: RC, DXF 281

`contrast` RC, DXF 282

`fade` RC, DXF 283

`clip_mode`
: B, DXF 290

`clip_boundary_type`
: BS, DXF 71

`num_clip_verts`
: BL, DXF 91

`clip_verts`
: 2RD*, DXF 14

`imagedef` H, DXF 340

`imagedefreactor`
H, DXF 360

XLINE
See [RAY], page 77,

XYGRIPENTITY

`parent` struct _dwg_object_entity*

XYPARAMETERENTITY

`parent` struct _dwg_object_entity*

4.3 OBJECTS

All non-graphical objects with its fields. See [Common Object fields], page 264,

ACMECOMMANDHISTORY

`parent` struct _dwg_object_object*

`class_version`
BS

ACMESCOPE

`parent` struct _dwg_object_object*

`class_version`
BS

ACMESTATEMGR

`parent` struct _dwg_object_object*

`class_version`
BS

ACSH_BOOLEAN_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`operation`
RCd, DXF 280

`operand1` BL, DXF 92

`operand2` BL, DXF 93

ACSH_BOX_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`length` BD, DXF 40

`width` BD, DXF 41

`height` BD, DXF 42

ACSH_BREP_CLASS

`parent` struct _dwg_object_object*

`acis_empty`
B, DXF 290

`unknown` B

`version` BS, DXF 70

`num_blocks`
BL

`block_size`
BL*

`encr_sat_data`
char **, DXF 1

`sab_size` BL

`acis_data`
RC*

`wireframe_data_present`
B

`point_present`
B

`point` 3BD

`isolines` BL

`isoline_present`
B

`num_wires`
BL

`wires` Dwg_3DSOLID_wire*

`num_silhouettes`
BL

`silhouettes`
: Dwg_3DSOLID_silhouette*

`_dxf_sab_converted`
: B

`acis_empty2`
: B

`extra_acis_data`
: struct _dwg_entity_3DSOLID*

`num_materials`
: BL

`materials`
: Dwg_3DSOLID_material*

`revision_guid`
: RC, DXF 2

`revision_major`
: BL

`revision_minor1`
: BS

`revision_minor2`
: BS

`revision_bytes`
: RC

`end_marker`
: BL

`history_id`
: H, DXF 350

`has_revision_guid`
: B

`acis_empty_bit`
: B

`evalexpr` Dwg_EvalExpr

`history_node`
: Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

ACSH_CHAMFER_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
: Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`bl92` BL, DXF 92

`base_dist`
: BD, DXF 41

`other_dist`
: BD, DXF 42

`num_edges`
: BL, DXF 93

`edges` BL*, DXF 94

`bl95` BL, DXF 95

ACSH_CONE_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
: Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`height` BD, DXF 40

`major_radius`
: BD, DXF 41

`minor_radius`
: BD, DXF 42

`x_radius` BD, DXF 43

ACSH_CYLINDER_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
: Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`height` BD, DXF 40

`major_radius`
: BD, DXF 41

`minor_radius`
BD, DXF 42

`x_radius` BD, DXF 43

ACSH_EXTRUSION_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`direction`
3BD, DXF 10

`bl92` BL, DXF 92

`shsw_text_size`
BL, DXF 90

`shsw_text`
TF, DXF 310

`shsw_bl93`
BL, DXF 93

`shsw_text2_size`
BL, DXF 90

`shsw_text2`
TF, DXF 310

`draft_angle`
BD, DXF 42

`start_draft_dist`
BD, DXF 43

`end_draft_dist`
BD, DXF 44

`scale_factor`
BD, DXF 45

`twist_angle`
BD, DXF 48

`align_angle`
BD, DXF 49

`sweepentity_transform`
BD*, DXF 46

`pathentity_transform`
BD*, DXF 47

`align_option`
RC, DXF 70

`miter_option`
RC, DXF 71

`has_align_start`
B, DXF 290

`bank` B, DXF 292

`check_intersections`
B, DXF 293

`shsw_b294`
B, DXF 294

`shsw_b295`
B, DXF 295

`shsw_b296`
B, DXF 296

`pt2` 3BD, DXF 11

ACSH_FILLET_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`bl92` BL, DXF 92

`num_edges`
BL, DXF 93

`edges` BL*, DXF 94

`num_radiuses`
BL, DXF 95

`num_startsetbacks`
BL, DXF 96

`num_endsetbacks`
BL, DXF 97

`radiuses` BD*, DXF 41

`startsetbacks`
BD*, DXF 42

`endsetbacks`
BD*, DXF 43

ACSH_HISTORY_CLASS

`parent` struct _dwg_object_object*

`major` BL, DXF 90

`minor` BL, DXF 91

`owner` H, DXF 360

`h_nodeid` BL, DXF 92

`show_history`
B, DXF 280

`record_history`
B, DXF 281

ACSH_LOFT_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`num_crosssects`
BL, DXF 92

`crosssects`
H*

`num_guides`
BL, DXF 95

`guides` H*

ACSH_PYRAMID_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`height` BD, DXF 40

`sides` BL, DXF 92

`radius` BD, DXF 41

`topradius`
BD, DXF 42

ACSH_REVOLVE_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`axis_pt` 3BD, DXF 10

`direction`
2RD, DXF 11

`revolve_angle`
BD, DXF 40

`start_angle`
BD, DXF 41

`draft_angle`
BD, DXF 43

`bd44` BD, DXF 44

`bd45` BD, DXF 45

`twist_angle`
BD, DXF 46

`b290` B, DXF 290

`is_close_to_axis`
B, DXF 291

`sweep_entity`
H

ACSH_SPHERE_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`radius` BD, DXF 40

ACSH_SWEEP_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`direction`
3BD, DXF 10

`bl92` BL, DXF 92

`shsw_text_size`
BL, DXF 90

`shsw_text`
TF, DXF 310

`shsw_bl93`
BL, DXF 93

`shsw_text2_size`
BL, DXF 90

`shsw_text2`
TF, DXF 310

`draft_angle`
BD, DXF 42

`start_draft_dist`
BD, DXF 43

`end_draft_dist`
BD, DXF 44

`scale_factor`
BD, DXF 45

`twist_angle`
BD, DXF 48

`align_angle`
BD, DXF 49

`sweepentity_transform`
BD*, DXF 46

`pathentity_transform`
BD*, DXF 47

`align_option`
: RC, DXF 70

`miter_option`
: RC, DXF 71

`has_align_start`
: B, DXF 290

`bank` B, DXF 292

`check_intersections`
: B, DXF 293

`shsw_b294`
: B, DXF 294

`shsw_b295`
: B, DXF 295

`shsw_b296`
: B, DXF 296

`pt2` 3BD, DXF 11

ACSH_TORUS_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
: Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`major_radius`
: BD, DXF 40

`minor_radius`
: BD, DXF 41

ACSH_WEDGE_CLASS

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`history_node`
: Dwg_ACSH_HistoryNode

`major` BL, DXF 90

`minor` BL, DXF 91

`length` BD, DXF 40

`width` BD, DXF 41

`height` BD, DXF 42

ALDIMOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`dimension`
Dwg_OCD_Dimension

`dimline_pt`
3BD, DXF 11

ANGDIMOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`dimension`
Dwg_OCD_Dimension

`arc_pt` 3BD, DXF 11

ANNOTSCALEOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

APPID

APPID is a table object.

`parent` struct _dwg_object_object*

`flag` RC

`name` TV

`used` RSd

`is_xref_ref`
: B

`is_xref_resolved`
: BS

`is_xref_dep`
: B

`xref` H

`unknown` RC, DXF 71

APPID_CONTROL

APPID_CONTROL is a table_control object.

`parent` struct _dwg_object_object*

`num_entries`
: BS, DXF 70

`entries` H*

`flags_r11`
: RS

ASSOC2DCONSTRAINTGROUP

`parent` struct _dwg_object_object*

`class_version`
: BS, DXF 90

`geometry_status`
: BL, DXF 90

`owningnetwork`
: H, DXF 330

`actionbody`
: H, DXF 360

`action_index`
: BL, DXF 90

`max_assoc_dep_index`
: BL, DXF 90

`num_deps` BL, DXF 90

`deps` Dwg_ASSOCACTION_Deps*

`num_owned_params`
: BL

`owned_params`
: H*

`num_values`
BL

`values` Dwg_VALUEPARAM*

`version` BL, DXF 90

`b1` B, DXF 70

`workplane`
3BD

`h1` H, DXF 360

`num_actions`
BL, DXF 90

`actions` H*, DXF 360

`num_nodes`
BL, DXF 90

`nodes` Dwg_CONSTRAINTGROUPNODE*

ASSOC3POINTANGULARDIMACTIONBODY

`parent` struct _dwg_object_object*

`aaab_version`
BS, DXF 90

`assoc_dep`
H, DXF 330

`aab_version`
BS, DXF 90

`actionbody`
H, DXF 360

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`class_version`
BS, DXF 90

`r_node` H, DXF 330

`d_node` H, DXF 330

`assocdep` H, DXF 330

ASSOCACTION

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 90

`geometry_status`
BL, DXF 90

`owningnetwork`
: H, DXF 330

`actionbody`
: H, DXF 360

`action_index`
: BL, DXF 90

`max_assoc_dep_index`
: BL, DXF 90

`num_deps` BL, DXF 90

`deps` Dwg_ASSOCACTION_Deps*

`num_owned_params`
: BL

`owned_params`
: H*

`num_values`
: BL

`values` Dwg_VALUEPARAM*

ASSOCACTIONPARAM

`parent` struct _dwg_object_object*

`is_r2013` BS, DXF 90

`aap_version`
: BL, DXF 90

`name` T, DXF 1

ASSOCALIGNEDDIMACTIONBODY

`parent` struct _dwg_object_object*

`aaab_version`
: BS, DXF 90

`assoc_dep`
: H, DXF 330

`aab_version`
: BS, DXF 90

`actionbody`
: H, DXF 360

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`class_version`
: BL, DXF 90

`r_node` H, DXF 330

`d_node` H, DXF 330

ASSOCARRAYACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`aaab_version`
BL, DXF 90

`paramblock`
T, DXF 1

`transmatrix`
BD*, DXF 40

ASSOCARRAYMODIFYACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`aaab_version`
BL, DXF 90

`paramblock`
T, DXF 1

`transmatrix`
BD*, DXF 40

`status` BS, DXF 70

`num_items`
BL, DXF 90

`items` Dwg_ARRAYITEMLOCATOR*

ASSOCASMBODYACTIONPARAM

`parent` struct _dwg_object_object*

`is_r2013` BS, DXF 90

`aap_version`
BL, DXF 90

`name` T, DXF 1

`asdap_class_version`
BL, DXF 90

`dep` H, DXF 330

`class_version`
: BL, DXF 90

`acis_empty`
: B

`unknown` B

`version` BS

`num_blocks`
: BL

`block_size`
: BL*

`encr_sat_data`
: char **

`sab_size` BL

`acis_data`
: RC*

`wireframe_data_present`
: B

`point_present`
: B

`point` 3BD

`isolines` BL

`isoline_present`
: B

`num_wires`
: BL

`wires` Dwg_3DSOLID_wire*

`num_silhouettes`
: BL

`silhouettes`
: Dwg_3DSOLID_silhouette*

`_dxf_sab_converted`
: B

`acis_empty2`
: B

`extra_acis_data`
: struct _dwg_entity_3DSOLID*

`num_materials`
: BL

`materials`
: Dwg_3DSOLID_material*

`revision_guid`
: RC

`revision_major`
: BL

`revision_minor1`
: BS

`revision_minor2`
: BS

`revision_bytes`
: RC

`end_marker`
: BL

`history_id`
: H

`has_revision_guid`
: B

`acis_empty_bit`
: B

ASSOCBLENDSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
: BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
: BL, DXF 90

`class_version`
: BL, DXF 90

`b1` B, DXF 290

`b2` B, DXF 291

`b3` B, DXF 292

`b4` B, DXF 293

`b5` B, DXF 294

`blend_options`
: BS, DXF 72

`bs2` BS, DXF 73

ASSOCCOMPOUNDACTIONPARAM

`parent` struct _dwg_object_object*

`is_r2013` BS, DXF 90

`aap_version`
BL, DXF 90

`name` T, DXF 1

`class_version`
BS, DXF 90

`bs1` BS, DXF 90

`num_params`
BL, DXF 90

`params` H*, DXF 360

`has_child_param`
B

`child_status`
BS, DXF 90

`child_id` BL, DXF 90

`child_param`
H, DXF 330

`h330_2` H, DXF 330

`bl2` BL, DXF 90

`h330_3` H, DXF 330

ASSOCDEPENDENCY

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 90

`status` BL, DXF 90

`is_read_dep`
B, DXF 290

`is_write_dep`
B, DXF 290

`is_attached_to_object`
B, DXF 290

`is_delegating_to_owning_action`
B, DXF 290

`order` BLd, DXF 90

`dep_on` H, DXF 330

`has_name` B, DXF 290

`name` T, DXF 1

`depbodyid`
BLd, DXF 90

`readdep` H, DXF 330

`dep_body` H, DXF 360

`node` H, DXF 330

ASSOCDIMDEPENDENCYBODY

`parent` struct _dwg_object_object*

`adb_version`
BS, DXF 90

`dimbase_version`
BS, DXF 90

`name` T, DXF 1

`class_version`
BS, DXF 90

ASSOCEDGEACTIONPARAM

`parent` struct _dwg_object_object*

`is_r2013` BS, DXF 90

`aap_version`
BL, DXF 90

`name` T, DXF 1

`asdap_class_version`
BL, DXF 90

`dep` H, DXF 330

`class_version`
BL, DXF 90

`param` H, DXF 330

`has_action`
B, DXF 290

`action_type`
BL, DXF 90

`subent` H

ASSOCEDGECHAMFERACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

ASSOCEDGEFILLETACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

ASSOCEXTENDSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

`class_version`
BL, DXF 90

`option` RC, DXF 280

ASSOCEXTRUDEDSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

`class_version`
BL, DXF 90

ASSOCFACEACTIONPARAM

`parent` struct _dwg_object_object*

`is_r2013` BS, DXF 90

`aap_version`
BL, DXF 90

`name` T, DXF 1

`asdap_class_version`
BL, DXF 90

`dep` H, DXF 330

`class_version`
BL, DXF 90

`index` BL, DXF 90

ASSOCFILLETSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

`class_version`
BL, DXF 90

`status` BS, DXF 70

`pt1` 2RD, DXF 10

`pt2` 2RD, DXF 10

ASSOCGEOMDEPENDENCY

`parent` struct _dwg_object_object*

`assocdep` Dwg_Object_ASSOCDEPENDENCY

`class_version`
BS, DXF 90

`enabled` B, DXF 290

`classname`
T, DXF 1

`dependent_on_compound_object`
B, DXF 290

ASSOCLOFTEDSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

`class_version`
BL, DXF 90

ASSOCMLEADERACTIONBODY

`parent` struct _dwg_object_object*

`aaab_version`
BS, DXF 90

`assoc_dep`
H, DXF 330

`aab_version`
BS, DXF 90

`actionbody`
H, DXF 360

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`class_version`
BL, DXF 90

`num_actions`
BL, DXF 90

`actions` Dwg_ASSOCACTIONBODY_action*

ASSOCNETWORK

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 90

`geometry_status`
BL, DXF 90

`owningnetwork`
H, DXF 330

`actionbody`
: H, DXF 360

`action_index`
: BL, DXF 90

`max_assoc_dep_index`
: BL, DXF 90

`num_deps` BL, DXF 90

`deps` Dwg_ASSOCACTION_Deps*

`num_owned_params`
: BL

`owned_params`
: H*

`num_values`
: BL

`values` Dwg_VALUEPARAM*

`network_version`
: BS, DXF 90

`network_action_index`
: BL, DXF 90

`num_actions`
: BL, DXF 90

`actions` Dwg_ASSOCACTION_Deps*

`num_owned_actions`
: BL, DXF 90

`owned_actions`
: H*, DXF 330

ASSOCNETWORKSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
: BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
: BL, DXF 90

`class_version`
: BL, DXF 90

ASSOCOBJECTACTIONPARAM

`parent` struct _dwg_object_object*

`is_r2013` BS, DXF 90

`aap_version`
BL, DXF 90

`name` T, DXF 1

`asdap_class_version`
BL, DXF 90

`dep` H, DXF 330

`class_version`
BS, DXF 90

ASSOCOFFSETSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

`class_version`
BL, DXF 90

`b1` B, DXF 290

ASSOCORDINATEDIMACTIONBODY

`parent` struct _dwg_object_object*

`aaab_version`
BS, DXF 90

`assoc_dep`
H, DXF 330

`aab_version`
BS, DXF 90

`actionbody`
H, DXF 360

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`class_version`
BL, DXF 90

`r_node` H, DXF 330

`d_node` H, DXF 330

ASSOCOSNAPPOINTREFACTIONPARAM

`parent` struct _dwg_object_object*

`is_r2013` BS, DXF 90

`aap_version`
BL, DXF 90

`name` T, DXF 1

`class_version`
BS, DXF 90

`bs1` BS, DXF 90

`num_params`
BL, DXF 90

`params` H*, DXF 360

`has_child_param`
B

`child_status`
BS, DXF 90

`child_id` BL, DXF 90

`child_param`
H, DXF 330

`h330_2` H, DXF 330

`bl2` BL, DXF 90

`h330_3` H, DXF 330

`status` BS, DXF 90

`osnap_mode`
RC, DXF 90

`param` BD, DXF 40

ASSOCPATCHSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

`class_version`
BL, DXF 90

ASSOCPATHACTIONPARAM

`parent` struct _dwg_object_object*

`is_r2013` BS, DXF 90

`aap_version`
BL, DXF 90

`name` T, DXF 1

`class_version`
BS, DXF 90

`bs1` BS, DXF 90

`num_params`
BL, DXF 90

`params` H*, DXF 360

`has_child_param`
B

`child_status`
BS, DXF 90

`child_id` BL, DXF 90

`child_param`
H, DXF 330

`h330_2` H, DXF 330

`bl2` BL, DXF 90

`h330_3` H, DXF 330

`version` BL, DXF 90

ASSOCPERSSUBENTMANAGER

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`unknown_3`
BL, DXF 90

`unknown_0`
BL, DXF 90

`unknown_2`
BL, DXF 90

`num_steps`
BL, DXF 90

`num_subents`
BL, DXF 90

`steps` BL*, DXF 90

`subents` BL*

`unknown_bl6`
BL, DXF 90

`unknown_bl6a`
BL, DXF 90

`unknown_bl7a`
BL, DXF 90

`unknown_bl7`
BL, DXF 90

`unknown_bl8`
BL, DXF 90

`unknown_bl9`
BL, DXF 90

`unknown_bl10`
BL, DXF 90

`unknown_bl11`
BL, DXF 90

`unknown_bl12`
BL, DXF 90

`unknown_bl13`
BL, DXF 90

`unknown_bl14`
BL, DXF 90

`unknown_bl15`
BL, DXF 90

`unknown_bl16`
BL, DXF 90

`unknown_bl17`
BL, DXF 90

`unknown_bl18`
BL, DXF 90

`unknown_bl19`
BL, DXF 90

`unknown_bl20`
BL, DXF 90

`unknown_bl21`
BL, DXF 90

`unknown_bl22`
BL, DXF 90

`unknown_bl23`
BL, DXF 90

`unknown_bl24`
BL, DXF 90

`unknown_bl25`
BL, DXF 90

`unknown_bl26`
BL, DXF 90

`unknown_bl27`
BL, DXF 90

`unknown_bl28`
BL, DXF 90

`unknown_bl29`
BL, DXF 90

`unknown_bl30`
BL, DXF 90

`unknown_bl31`
BL, DXF 90

`unknown_bl32`
BL, DXF 90

`unknown_bl33`
BL, DXF 90

`unknown_bl34`
BL, DXF 90

`unknown_bl35`
BL, DXF 90

`unknown_bl36`
BL, DXF 90

`unknown_b37`
B, DXF 290

ASSOCPLANESURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

`class_version`
BL, DXF 90

ASSOCPOINTREFACTIONPARAM

`parent` struct _dwg_object_object*

`is_r2013` BS, DXF 90

`aap_version`
BL, DXF 90

`name` T, DXF 1

`class_version`
BS, DXF 90

`bs1` BS, DXF 90

`num_params`
BL, DXF 90

`params` H*, DXF 360

`has_child_param`
B

`child_status`
BS, DXF 90

`child_id` BL, DXF 90

`child_param`
H, DXF 330

`h330_2` H, DXF 330

`bl2` BL, DXF 90

`h330_3` H, DXF 330

ASSOCRESTOREENTITYSTATEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`class_version`
BL, DXF 90

`entity` H, DXF 330

ASSOCREVOLVEDSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

`class_version`
BL, DXF 90

ASSOCROTATEDDIMACTIONBODY

`parent` struct _dwg_object_object*

`aaab_version`
BS, DXF 90

`assoc_dep`
H, DXF 330

`aab_version`
BS, DXF 90

`actionbody`
H, DXF 360

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`class_version`
BS, DXF 90

`r_node` H, DXF 330

`d_node` H, DXF 330

ASSOCSWEPTSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

`class_version`
BL, DXF 90

ASSOCTRIMSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`aab_version`
BL, DXF 90

`pab` Dwg_ASSOCPARAMBASEDACTIONBODY

`sab` Dwg_ASSOCSURFACEACTIONBODY

`pbsab_status`
BL, DXF 90

`class_version`
BL, DXF 90

`b1` B, DXF 290

`b2` B, DXF 290

`distance` BD, DXF 40

ASSOCVALUEDEPENDENCY

`parent` struct _dwg_object_object*

`assocdep` Dwg_Object_ASSOCDEPENDENCY

ASSOCVARIABLE

`parent` struct _dwg_object_object*

`av_class_version`
BS, DXF 90

`class_version`
BS, DXF 90

`geometry_status`
BL, DXF 90

`owningnetwork`
H, DXF 330

`actionbody`
H, DXF 360

`action_index`
BL, DXF 90

`max_assoc_dep_index`
BL, DXF 90

`num_deps` BL, DXF 90

`deps` Dwg_ASSOCACTION_Deps*

`num_owned_params`
BL

`owned_params`
H*

`num_values`
BL

`values` Dwg_VALUEPARAM*

`name` T, DXF 1

`t58` T, DXF 1

`evaluator`
T, DXF 1

`desc` T, DXF 1

`value` Dwg_EvalVariant

`has_t78` B, DXF 290

`t78` T, DXF 1

`b290` B, DXF 290

ASSOCVERTEXACTIONPARAM

`parent` struct _dwg_object_object*

`is_r2013` BS, DXF 90

`aap_version`
BL, DXF 90

`name` T, DXF 1

`asdap_class_version`
BL, DXF 90

`dep` H, DXF 330

`class_version`
BL, DXF 90

`pt` 3BD, DXF 10

BLKREFOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`rotation` BD, DXF 50

`ins_pt` 3BD, DXF 10

`scale_factor`
3BD_1, DXF 42

BLOCKALIGNEDCONSTRAINTPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
3BD

`endpt` 3BD

`dependency`
H, DXF 330

`expr_name`
T, DXF 305

`expr_description`
T, DXF 306

`value` BD, DXF 140

`value_set`
Dwg_BLOCKPARAMVALUESET

BLOCKALIGNMENTGRIP

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`bg_bl91` BL, DXF 91

`bg_bl92` BL, DXF 92

`bg_location`
3BD, DXF 1010

`bg_insert_cycling`
B, DXF 280

`bg_insert_cycling_weight`
BLd, DXF 93

`orientation`
3BD_1, DXF 140

BLOCKALIGNMENTPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
3BD

`endpt` 3BD

`align_perpendicular`
B, DXF 280

BLOCKANGULARCONSTRAINTPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
: 3BD

`endpt` 3BD

`dependency`
: H, DXF 330

`center_pt`
: 3BD, DXF 1011

`end_pt` 3BD, DXF 1012

`expr_name`
: T, DXF 305

`expr_description`
: T, DXF 306

`angle` BD, DXF 140

`orientation_on_both_grips`
: B, DXF 280

`value_set`
: Dwg_BLOCKPARAMVALUESET

BLOCKARRAYACTION

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`display_location`
: 3BD, DXF 1010

`num_actions`
: BL, DXF 70

`actions` BL*, DXF 91

`num_deps` BL, DXF 71

`deps` H*, DXF 330

`conn_pts` Dwg_BLOCKACTION_connectionpts

`column_offset`
: BD, DXF 140

`row_offset`
: BD, DXF 141

BLOCKBASEPOINTPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_pt` 3BD, DXF 1010

`num_propinfos`
BL, DXF 93

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`pt` 3BD, DXF 1011

`base_pt` 3BD, DXF 1012

BLOCKDIAMETRICCONSTRAINTPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
3BD

`endpt` 3BD

`dependency`
H, DXF 330

`expr_name`
T, DXF 305

`expr_description`
T, DXF 306

`distance` BD, DXF 140

`orientation_on_both_grips`
B

`value_set`
Dwg_BLOCKPARAMVALUESET

BLOCKFLIPACTION

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`display_location`
3BD, DXF 1010

`num_actions`
BL, DXF 70

`actions` BL*, DXF 91

`num_deps` BL, DXF 71

`deps` H*, DXF 330

`conn_pts` Dwg_BLOCKACTION_connectionpts

`action_offset_x`
BD

`action_offset_y`
BD

`angle_offset`
BD

BLOCKFLIPGRIP

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`bg_bl91` BL, DXF 91

`bg_bl92` BL, DXF 92

`bg_location`
3BD, DXF 1010

`bg_insert_cycling`
B, DXF 280

`bg_insert_cycling_weight`
BLd, DXF 93

`combined_state`
BL, DXF 93

`orientation`
3BD_1, DXF 140

`upd_state`
BS

`state` BS

BLOCKFLIPPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
3BD

`endpt` 3BD

`flip_label`
T, DXF 305

`flip_label_desc`
T, DXF 306

`base_state_label`
T, DXF 307

`flipped_state_label`
T, DXF 308

`def_label_pt`
3BD, DXF 1012

`bl96` BL, DXF 96

`tooltip` T, DXF 309

BLOCKGRIPLOCATIONCOMPONENT

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`grip_type`
BL, DXF 91

`grip_expr`
T, DXF 300

BLOCKHORIZONTALCONSTRAINTPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
3BD

`endpt` 3BD

`dependency`
: H, DXF 330

`expr_name`
: T, DXF 305

`expr_description`
: T, DXF 306

`value` BD, DXF 140

`value_set`
: Dwg_BLOCKPARAMVALUESET

BLOCKLINEARCONSTRAINTPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
: B, DXF 280

`chain_actions`
: B, DXF 281

`def_basept`
: 3BD, DXF 1010

`def_endpt`
: 3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
: BL*, DXF 91

`parameter_base_location`
: BS, DXF 177

`upd_basept`
: 3BD

`basept` 3BD

`upd_endpt`
: 3BD

`endpt` 3BD

`dependency`
H, DXF 330

`expr_name`
T, DXF 305

`expr_description`
T, DXF 306

`value` BD, DXF 140

`value_set`
Dwg_BLOCKPARAMVALUESET

BLOCKLINEARGRIP

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`bg_bl91` BL, DXF 91

`bg_bl92` BL, DXF 92

`bg_location`
3BD, DXF 1010

`bg_insert_cycling`
B, DXF 280

`bg_insert_cycling_weight`
BLd, DXF 93

`orientation`
3BD_1, DXF 140

BLOCKLINEARPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
3BD

`endpt` 3BD

`distance_name`
T, DXF 305

`distance_desc`
T, DXF 306

`distance` BD, DXF 140

`value_set`
Dwg_BLOCKPARAMVALUESET

BLOCKLOOKUPACTION

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`display_location`
3BD, DXF 1010

`num_actions`
BL, DXF 70

`actions` BL*, DXF 91

`num_deps` BL, DXF 71

`deps` H*, DXF 330

`numelems` BL

`numrows` BL, DXF 92

`numcols` BL, DXF 93

`lut` Dwg_BLOCKLOOKUPACTION_lut*

`exprs` TV*, DXF 302

`b280` B, DXF 280

BLOCKLOOKUPGRIP

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`bg_bl91` BL, DXF 91

`bg_bl92` BL, DXF 92

`bg_location`
3BD, DXF 1010

`bg_insert_cycling`
B, DXF 280

`bg_insert_cycling_weight`
BLd, DXF 93

BLOCKLOOKUPPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_pt` 3BD, DXF 1010

`num_propinfos`
BL, DXF 93

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`lookup_name`
T, DXF 303

`lookup_desc`
T, DXF 304

`index` BL, DXF 94

`unknown_t`
T

BLOCKMOVEACTION

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`display_location`
3BD, DXF 1010

`num_actions`
BL, DXF 70

`actions` BL*, DXF 91

`num_deps` BL, DXF 71

`deps` H*, DXF 330

`conn_pts` Dwg_BLOCKACTION_connectionpts

`action_offset_x`
BD, DXF 140

`action_offset_y`
BD, DXF 141

`angle_offset`
BD

BLOCKPARAMDEPENDENCYBODY

`parent` struct _dwg_object_object*

`adb_version`
BS, DXF 90

`dimbase_version`
BS, DXF 90

`name` T, DXF 1

`class_version`
BS, DXF 90

BLOCKPOINTPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_pt` 3BD, DXF 1010

`num_propinfos`
BL, DXF 93

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`position_name`
T, DXF 303

`position_desc`
T, DXF 304

`def_label_pt`
3BD, DXF 1011

BLOCKPOLARGRIP

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`bg_bl91` BL, DXF 91

`bg_bl92` BL, DXF 92

`bg_location`
3BD, DXF 1010

`bg_insert_cycling`
B, DXF 280

`bg_insert_cycling_weight`
BLd, DXF 93

BLOCKPOLARPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
3BD

`endpt` 3BD

`angle_name`
T, DXF 305

`angle_desc`
T, DXF 306

`distance_name`
T, DXF 305

`distance_desc`
T, DXF 306

`offset` BD, DXF 140

`angle_value_set`
Dwg_BLOCKPARAMVALUESET

`distance_value_set`
Dwg_BLOCKPARAMVALUESET

BLOCKPOLARSTRETCHACTION

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`display_location`
3BD, DXF 1010

`num_actions`
BL, DXF 70

`actions` BL*, DXF 91

`num_deps` BL, DXF 71

`deps` H*, DXF 330

`conn_pts` Dwg_BLOCKACTION_connectionpts

`num_pts` BL, DXF 72

`pts` 2RD*, DXF 10

`num_hdls` BL, DXF 73

`hdls` H*, DXF 331

`shorts` BS*, DXF 74

`num_codes`
BL, DXF 75

`codes` BL*, DXF 76

BLOCKPROPERTIESTABLE

`parent` struct _dwg_object_object*

BLOCKPROPERTIESTABLEGRIP

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`bg_bl91` BL, DXF 91

`bg_bl92` BL, DXF 92

`bg_location`
3BD, DXF 1010

`bg_insert_cycling`
B, DXF 280

`bg_insert_cycling_weight`
BLd, DXF 93

BLOCKRADIALCONSTRAINTPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
: 3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
: BL*, DXF 91

`parameter_base_location`
: BS, DXF 177

`upd_basept`
: 3BD

`basept` 3BD

`upd_endpt`
: 3BD

`endpt` 3BD

`dependency`
: H, DXF 330

`expr_name`
: T, DXF 305

`expr_description`
: T, DXF 306

`distance` BD, DXF 140

`value_set`
: Dwg_BLOCKPARAMVALUESET

BLOCKREPRESENTATION

`parent` struct _dwg_object_object*

`flag` BS, DXF 70

`block` H, DXF 340

BLOCKROTATEACTION

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`display_location`
3BD, DXF 1010

`num_actions`
BL, DXF 70

`actions` BL*, DXF 91

`num_deps` BL, DXF 71

`deps` H*, DXF 330

`offset` 3BD, DXF 1011

`conn_pts` Dwg_BLOCKACTION_connectionpts

`dependent`
B, DXF 280

`base_pt` 3BD, DXF 1012

BLOCKROTATIONGRIP

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`bg_bl91` BL, DXF 91

`bg_bl92` BL, DXF 92

`bg_location`
3BD, DXF 1010

`bg_insert_cycling`
B, DXF 280

`bg_insert_cycling_weight`
BLd, DXF 93

BLOCKROTATIONPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
3BD

`endpt` 3BD

`def_base_angle_pt`
3BD, DXF 1011

`angle_name`
T, DXF 305

`angle_desc`
T, DXF 306

`angle` BD, DXF 140

`angle_value_set`
Dwg_BLOCKPARAMVALUESET

BLOCKSCALEACTION

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`display_location`
3BD, DXF 1010

`num_actions`
BL, DXF 70

`actions` BL*, DXF 91

`num_deps` BL, DXF 71

`deps` H*, DXF 330

`offset` 3BD, DXF 1011

`conn_pts` Dwg_BLOCKACTION_connectionpts

`dependent`
B, DXF 280

`base_pt` 3BD, DXF 1012

BLOCKSTRETCHACTION

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`display_location`
3BD, DXF 1010

`num_actions`
BL, DXF 70

`actions` BL*, DXF 91

`num_deps` BL, DXF 71

`deps` H*, DXF 330

`conn_pts` Dwg_BLOCKACTION_connectionpts

`num_pts` BL, DXF 72

`pts` 2RD*, DXF 1011

`num_hdls` BL, DXF 73

`hdls` Dwg_BLOCKSTRETCHACTION_handles*

`num_codes`
BL, DXF 75

`codes` Dwg_BLOCKSTRETCHACTION_codes*

`action_offset_x`
BD, DXF 140

`action_offset_y`
BD, DXF 141

`angle_offset`
BD

BLOCKUSERPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_pt` 3BD, DXF 1010

`num_propinfos`
BL, DXF 93

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`flag` BS, DXF 90

`assocvariable`
H, DXF 330

`expr` T, DXF 301

`value` Dwg_EvalVariant

`type` BS, DXF 170

BLOCKVERTICALCONSTRAINTPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
3BD

`endpt` 3BD

`dependency`
H, DXF 330

`expr_name`
T, DXF 305

`expr_description`
T, DXF 306

`value` BD, DXF 140

`value_set`
Dwg_BLOCKPARAMVALUESET

BLOCKVISIBILITYGRIP

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`bg_bl91` BL, DXF 91

`bg_bl92` BL, DXF 92

`bg_location`
3BD, DXF 1010

`bg_insert_cycling`
B, DXF 280

`bg_insert_cycling_weight`
BLd, DXF 93

BLOCKVISIBILITYPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_pt` 3BD, DXF 1010

`num_propinfos`
BL, DXF 93

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`is_initialized`
B, DXF 281

`unknown_bool`
B, DXF 91

`blockvisi_name`
T, DXF 301

`blockvisi_desc`
T, DXF 302

`num_blocks`
BL, DXF 93

`blocks` H*, DXF 331

`num_states`
BL, DXF 92

`states` Dwg_BLOCKVISIBILITYPARAMETER_state*

BLOCKXYGRIP

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`bg_bl91` BL, DXF 91

`bg_bl92` BL, DXF 92

`bg_location`
3BD, DXF 1010

`bg_insert_cycling`
B, DXF 280

`bg_insert_cycling_weight`
BLd, DXF 93

BLOCKXYPARAMETER

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

`name` T, DXF 300

`be_major` BL, DXF 98

`be_minor` BL, DXF 99

`eed1071` BL, DXF 1071

`show_properties`
B, DXF 280

`chain_actions`
B, DXF 281

`def_basept`
3BD, DXF 1010

`def_endpt`
3BD, DXF 1011

`prop1` Dwg_BLOCKPARAMETER_PropInfo

`prop2` Dwg_BLOCKPARAMETER_PropInfo

`prop3` Dwg_BLOCKPARAMETER_PropInfo

`prop4` Dwg_BLOCKPARAMETER_PropInfo

`prop_states`
BL*, DXF 91

`parameter_base_location`
BS, DXF 177

`upd_basept`
3BD

`basept` 3BD

`upd_endpt`
3BD

`endpt` 3BD

`x_label` T, DXF 305

`x_label_desc`
T, DXF 306

`y_label` T, DXF 307

`y_label_desc`
T, DXF 308

`x_value` BD, DXF 142

`y_value` BD, DXF 141

`x_value_set`
Dwg_BLOCKPARAMVALUESET

`y_value_set`
Dwg_BLOCKPARAMVALUESET

BLOCK_CONTROL

BLOCK_CONTROL is a table_control object.

`parent` struct _dwg_object_object*

`num_entries`
BS, DXF 70

`entries` H*

`flags_r11`
RS

`model_space`
H

`paper_space`
H

BLOCK_HEADER

BLOCK_HEADER is a table object.

`parent` struct _dwg_object_object*

`flag` RC

`name` TV

`used` RSd

`is_xref_ref`
B

`is_xref_resolved`
BS

`is_xref_dep`
B

`xref` H

`__iterator`
BL

`anonymous`
B

`hasattrs` B

`blkisxref`
B

`xrefoverlaid`
B

`xref_loaded`
B

`num_owned`
BL

`base_pt` 3DPOINT, DXF 10

`xref_pname`
T, DXF 1

`num_inserts`
RL

`description`
T, DXF 4

`preview_size`
BL

`preview` TF, DXF 310

`insert_units`
BS, DXF 70

`explodable`
B, DXF 280

`block_scaling`
RC, DXF 281

`block_entity`
H

`first_entity`
H

`last_entity`
H

`entities` H*

`endblk_entity`
H

`inserts` H*, DXF 331

`layout` H, DXF 340

`flag2` RS

`unknown_r11`
RC

`block_offset_r11`
RL

BREAKDATA

`parent` struct _dwg_object_object*

`num_pointrefs`
BL, DXF 90

`pointrefs`
H*, DXF 330

`dimref` H, DXF 331

BREAKPOINTREF

`parent` struct _dwg_object_object*

CELLSTYLEMAP

`parent` struct _dwg_object_object*

`num_cells`
BL, DXF 90

`cells` Dwg_TABLESTYLE_CellStyle*

CONTEXTDATAMANAGER

`parent` struct _dwg_object_object*

`objectcontext`
H

`num_submgrs`
BL

`submgrs` Dwg_CONTEXTDATA_submgr*

CSACDOCUMENTOPTIONS

`parent` struct _dwg_object_object*

`class_version`
BS

CURVEPATH

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 90

`entity` H, DXF 340

DATALINK

`parent` struct _dwg_object_object*

`class_version`
BS

`data_adapter`
T, DXF 1

`description`
T, DXF 300

`tooltip` T, DXF 301

`connection_string`
T, DXF 302

`option` BL, DXF 90

`update_option`
BL, DXF 91

`bl92` BL, DXF 92

`year` BS, DXF 170

`month` BS, DXF 171

`day` BS, DXF 172

`hour` BS, DXF 173

`minute` BS, DXF 174

`seconds` BS, DXF 175

`msec` BS, DXF 176

`path_option`
BS, DXF 177

`bl93` BL, DXF 93

`update_status`
T, DXF 304

`num_customdata`
BL, DXF 94

`customdata`
Dwg_DATALINK_customdata*

`hardowner`
H, DXF 360

DATATABLE

`parent` struct _dwg_object_object*

`flags` BS, DXF 70

`num_cols` BL, DXF 90

`num_rows` BL, DXF 91

`table_name`
T, DXF 1

`cols` Dwg_DATATABLE_column*

DBCOLOR

`parent` struct _dwg_object_object*

`color` CMC, DXF 62

DETAILVIEWSTYLE

`parent` struct _dwg_object_object*

`mdoc_class_version`
BS, DXF 70

`desc` T, DXF 3

`is_modified_for_recompute`
B, DXF 290

`display_name`
T, DXF 300

`viewstyle_flags`
BL, DXF 90

`class_version`
BS, DXF 70

`flags` BL, DXF 90

`identifier_style`
H, DXF 340

`identifier_color`
CMC, DXF 62

`identifier_height`
BD, DXF 40

`identifier_exclude_characters`
T, DXF 300

`identifier_offset`
BD, DXF 40

`identifier_placement`
RC, DXF 280

`arrow_symbol`
H, DXF 340

`arrow_symbol_color`
CMC, DXF 62

`arrow_symbol_size`
BD, DXF 40

`boundary_ltype`
H, DXF 340

`boundary_linewt`
BLd, DXF 90

`boundary_line_color`
CMC, DXF 62

`viewlabel_text_style`
H, DXF 340

`viewlabel_text_color`
CMC, DXF 62

`viewlabel_text_height`
BD, DXF 40

`viewlabel_attachment`
BL, DXF 90

`viewlabel_offset`
BD, DXF 40

`viewlabel_alignment`
BL, DXF 90

`viewlabel_pattern`
T, DXF 300

`connection_ltype`
H, DXF 340

`connection_linewt`
BLd, DXF 90

`connection_line_color`
CMC, DXF 62

`borderline_ltype`
H, DXF 340

`borderline_linewt`
BLd, DXF 90

`borderline_color`
CMC, DXF 62

`model_edge`
RC, DXF 280

DICTIONARY

`parent` struct _dwg_object_object*

`numitems` BL

`is_hardowner`
RC, DXF 280

`cloning` BS, DXF 281

`texts` T*, DXF 3

`itemhandles`
H*, DXF 350

DICTIONARYVAR

`parent` struct _dwg_object_object*

`schema` RC, DXF 280

`strvalue` T, DXF 1

DICTIONARYWDFLT

`parent` struct _dwg_object_object*

`numitems` BL

`is_hardowner`
RC, DXF 280

`cloning` BS, DXF 281

`texts` T*, DXF 3

`itemhandles`
H*, DXF 350

`defaultid`
H, DXF 340

DIMASSOC

`parent` struct _dwg_object_object*

`dimensionobj`
H, DXF 330

`associativity`
BL, DXF 90

`trans_space_flag`
B, DXF 70

`rotated_type`
RC, DXF 71

`ref` Dwg_DIMASSOC_Ref*

DIMSTYLE

DIMSTYLE is a table object.

`parent` struct _dwg_object_object*

`flag` RC, DXF 70

`name` T

`used` RSd

`is_xref_ref`
B

`is_xref_resolved`
BS

`is_xref_dep`
B

`xref` H

`DIMTOL` B, DXF 71

`DIMLIM` B, DXF 72

`DIMTIH` B, DXF 73

`DIMTOH` B, DXF 74

`DIMSE1` B, DXF 75

`DIMSE2` B, DXF 76

DIMALT	B, DXF 170
DIMTOFL	B, DXF 172
DIMSAH	B, DXF 173
DIMTIX	B, DXF 174
DIMSOXD	B, DXF 175
DIMALTD	BS, DXF 171
DIMZIN	BS, DXF 78
DIMSD1	B, DXF 281
DIMSD2	B, DXF 282
DIMTOLJ	BS, DXF 283
DIMJUST	BS, DXF 280
DIMFIT	BS, DXF 287
DIMUPT	B, DXF 288
DIMTZIN	BS, DXF 284
DIMALTZ	BS, DXF 285
DIMALTTZ	BS, DXF 286
DIMTAD	BS, DXF 77
DIMUNIT	BS, DXF 270
DIMAUNIT	BS, DXF 275
DIMDEC	BS, DXF 271
DIMTDEC	BS, DXF 272
DIMALTU	BS, DXF 273
DIMALTTD	BS, DXF 274
DIMSCALE	BD, DXF 40
DIMASZ	BD, DXF 41
DIMEXO	BD, DXF 42
DIMDLI	BD, DXF 43
DIMEXE	BD, DXF 44
DIMRND	BD, DXF 45
DIMDLE	BD, DXF 46
DIMTP	BD, DXF 47
DIMTM	BD, DXF 48
DIMFXL	BD, DXF 49

`DIMJOGANG`
: BD, DXF 50

`DIMTFILL` BS, DXF 69

`DIMTFILLCLR`
: CMC, DXF 70

`DIMAZIN` BS, DXF 79

`DIMARCSYM`
: BS, DXF 90

`DIMTXT` BD, DXF 140

`DIMCEN` BD, DXF 141

`DIMTSZ` BD, DXF 142

`DIMALTF` BD, DXF 143

`DIMLFAC` BD, DXF 144

`DIMTVP` BD, DXF 145

`DIMTFAC` BD, DXF 146

`DIMGAP` BD, DXF 147

`DIMPOST` T, DXF 3

`DIMAPOST` T, DXF 4

`DIMBLK_T` TV, DXF 5

`DIMBLK1_T`
: TV, DXF 6

`DIMBLK2_T`
: TV, DXF 7

`DIMALTRND`
: BD, DXF 148

`DIMCLRD_N`
: RS, DXF 176

`DIMCLRE_N`
: RS, DXF 177

`DIMCLRT_N`
: RS, DXF 178

`DIMCLRD` CMC, DXF 176

`DIMCLRE` CMC, DXF 177

`DIMCLRT` CMC, DXF 178

`DIMADEC` BS, DXF 179

`DIMFRAC` BS, DXF 276

`DIMLUNIT` BS, DXF 277

`DIMDSEP` BS, DXF 278

`DIMTMOVE` BS, DXF 279

`DIMATFIT` BS, DXF 289

`DIMFXLON` B, DXF 290

`DIMTXTDIRECTION`
B, DXF 294

`DIMALTMZF`
BD

`DIMALTMZS`
T

`DIMMZF` BD

`DIMMZS` T

`DIMLWD` BSd, DXF 371

`DIMLWE` BSd, DXF 372

`flag0` B

`DIMTXSTY` H, DXF 340

`DIMLDRBLK`
H, DXF 341

`DIMBLK` H, DXF 342

`DIMBLK1` H, DXF 343

`DIMBLK2` H, DXF 344

`DIMLTYPE` H, DXF 345

`DIMLTEX1` H, DXF 346

`DIMLTEX2` H, DXF 347

DIMSTYLE_CONTROL

DIMSTYLE_CONTROL is a table_control object.

`parent` struct _dwg_object_object*

`num_entries`
BS, DXF 70

`entries` H*

`flags_r11`
RS

`num_morehandles`
RC, DXF 71

`morehandles`
H*, DXF 340

DMDIMOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`dimension`
Dwg_OCD_Dimension

`first_arc_pt`
3BD, DXF 11

`def_pt` 3BD, DXF 12

DUMMY

`parent` struct _dwg_object_object*

DYNAMICBLOCKPROXYNODE

`parent` struct _dwg_object_object*

`evalexpr` Dwg_EvalExpr

DYNAMICBLOCKPURGEPREVENTER

`parent` struct _dwg_object_object*

`flag` BS, DXF 70

`block` H

EVALUATION_GRAPH

`parent` struct _dwg_object_object*

`major` BL

`minor` BL

`first_nodeid`
BLd, DXF 96

`first_nodeid_copy`
BLd, DXF 97

`num_nodes`
BL

`nodes` Dwg_EVAL_Node*

`has_graph`
B

`num_edges`
BL

`edges` Dwg_EVAL_Edge*

FCFOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`location` 3BD, DXF 10

`horiz_dir`
3BD, DXF 11

FIELD

`parent` struct _dwg_object_object*

`id` T, DXF 1

`code` T, DXF 2

`num_childs`
BL, DXF 90

`childs` H*, DXF 360

`num_objects`
BL, DXF 97

`objects` H*, DXF 331

`format` TV, DXF 4

`evaluation_option`
BL, DXF 91

`filing_option`
BL, DXF 92

`field_state`
BL, DXF 94

`evaluation_status`
BL, DXF 95

`evaluation_error_code`
BL, DXF 96

`evaluation_error_msg`
T, DXF 300

`value` Dwg_TABLE_value

`value_string`
T, DXF 301

`value_string_length`
BL, DXF 98

`num_childval`
BL, DXF 93

`childval` Dwg_FIELD_ChildValue*

FIELDLIST

`parent` struct _dwg_object_object*

`num_fields`
BL, DXF 90

`unknown` B

`fields` H*, DXF 330

GEODATA

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`host_block`
H, DXF 330

`coord_type`
BS, DXF 70

`design_pt`
3BD, DXF 10

`ref_pt` 3BD_1, DXF 11

`obs_pt` 3BD, DXF 11

`scale_vec`
3BD_1, DXF 43

`unit_scale_horiz`
BD, DXF 40

`units_value_horiz`
BL, DXF 91

`unit_scale_vert`
BD, DXF 41

`units_value_vert`
: BL, DXF 92

`up_dir` 3BD, DXF 210

`north_dir`
: 2RD, DXF 12

`scale_est`
: BL, DXF 95

`user_scale_factor`
: BD, DXF 141

`do_sea_level_corr`
: B, DXF 294

`sea_level_elev`
: BD, DXF 142

`coord_proj_radius`
: BD, DXF 143

`coord_system_def`
: T, DXF 301

`geo_rss_tag`
: T, DXF 302

`coord_system_datum`
: T, DXF 303

`coord_system_wkt`
: T, DXF 304

`observation_from_tag`
: T, DXF 305

`observation_to_tag`
: T, DXF 306

`observation_coverage_tag`
: T, DXF 307

`num_geomesh_pts`
: BL, DXF 93

`geomesh_pts`
: Dwg_GEODATA_meshpt*

`num_geomesh_faces`
: BL, DXF 96

`geomesh_faces`
: Dwg_GEODATA_meshface*

`has_civil_data`
: B

`obsolete_false`
B, DXF 292

`ref_pt2d` 2RD, DXF 15

`zero1` 3BD, DXF 16

`unknown1` BL, DXF 93

`unknown2` BL, DXF 94

`unknown_b`
B, DXF 293

`north_dir_angle_deg`
BD, DXF 54

`north_dir_angle_rad`
BD, DXF 140

GEOMAPIMAGE

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`pt0` 3BD, DXF 10

`image_size`
2RD, DXF 13

`display_props`
BS, DXF 70

`clipping` B, DXF 280

`brightness`
RC, DXF 281

`contrast` RC, DXF 282

`fade` RC, DXF 283

`rotation` BD

`image_width`
BD

`image_height`
BD

`name` T

`image_file`
BD

`image_visibility`
BD

`transparency`
BS

`height` BD

`width` BD

`show_rotation`
B

`scale_factor`
BD

`geoimage_brightness`
BS

`geoimage_contrast`
BS

`geoimage_fade`
BS

`geoimage_position`
BS

`geoimage_width`
BS

`geoimage_height`
BS

GRADIENT_BACKGROUND

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`color_top`
BLx, DXF 90

`color_middle`
BLx, DXF 91

`color_bottom`
BLx, DXF 92

`horizon` BD, DXF 140

`height` BD, DXF 141

`rotation` BD, DXF 142

GROUND_PLANE_BACKGROUND

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`color_sky_zenith`
: BLx, DXF 90

`color_sky_horizon`
: BLx, DXF 91

`color_underground_horizon`
: BLx, DXF 92

`color_underground_azimuth`
: BLx, DXF 93

`color_near`
: BLx, DXF 94

`color_far`
: BLx, DXF 95

GROUP

`parent` struct _dwg_object_object*

`name` T, DXF 300

`unnamed` BS, DXF 70

`selectable`
: BS, DXF 71

`num_groups`
: BL

`groups` H*, DXF 340

IBL_BACKGROUND

`parent` struct _dwg_object_object*

`class_version`
: BL, DXF 90

`enable` B, DXF 290

`name` T, DXF 1

`rotation` BD, DXF 40

`display_image`
: B, DXF 290

`secondary_background`
: H, DXF 340

IDBUFFER

`parent` struct _dwg_object_object*

`unknown` RC

`num_obj_ids`
: BL

`obj_ids` H*, DXF 330

IMAGEDEF

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`image_size`
2RD, DXF 10

`file_path`
T, DXF 1

`is_loaded`
B, DXF 280

`resunits` RC, DXF 281

`pixel_size`
2RD, DXF 11

IMAGEDEF_REACTOR

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

IMAGE_BACKGROUND

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`filename` T, DXF 300

`fit_to_screen`
B, DXF 290

`maintain_aspect_ratio`
B, DXF 291

`use_tiling`
B, DXF 292

`offset` 2BD_1, DXF 140

`scale` 2BD_1, DXF 142

INDEX

`parent` struct _dwg_object_object*

`last_updated`
TIMEBLL, DXF 40

LAYER

LAYER is a table object.

`parent` struct _dwg_object_object*

`flag` BS

`name` T

`used` RSd

`is_xref_ref`
B

`is_xref_resolved`
BS

`is_xref_dep`
B

`xref` H

`flag0` BS

`frozen` B

`off` B

`frozen_in_new`
B

`locked` B

`plotflag` B, DXF 290

`linewt` RC, DXF 370

`color` CMC, DXF 62

`plotstyle`
H, DXF 390

`material` H, DXF 347

`ltype` H, DXF 6

`visualstyle`
H, DXF 348

LAYERFILTER

`parent` struct _dwg_object_object*

`num_names`
BL

`names` TV*, DXF 8

LAYER_CONTROL

LAYER_CONTROL is a table_control object.

`parent` struct _dwg_object_object*

`num_entries`
BS, DXF 70

`entries` H*

`flags_r11`
RS

LAYER_INDEX

`parent` struct _dwg_object_object*

`last_updated`
TIMEBLL, DXF 40

`num_entries`
BL

`entries` Dwg_LAYER_entry*

LAYOUT

`parent` struct _dwg_object_object*

`plotsettings`
Dwg_Object_PLOTSETTINGS

`layout_name`
T, DXF 1

`tab_order`
BS, DXF 71

`layout_flags`
BS, DXF 70

`INSBASE` 3DPOINT, DXF 12

`LIMMIN` 2DPOINT, DXF 10

`LIMMAX` 2DPOINT, DXF 11

`UCSORG` 3DPOINT, DXF 13

`UCSXDIR` 3DPOINT, DXF 16

`UCSYDIR` 3DPOINT, DXF 17

`ucs_elevation`
BD, DXF 146

`UCSORTHOVIEW`
BS, DXF 76

`EXTMIN` 3DPOINT, DXF 14

`EXTMAX` 3DPOINT, DXF 15

`block_header`
H, DXF 330

`active_viewport`
: H, DXF 331

`base_ucs` H, DXF 346

`named_ucs`
: H, DXF 345

`num_viewports`
: BL

`viewports`
: H*

LEADEROBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
: BS, DXF 70

`is_default`
: B, DXF 290

`scale` H, DXF 340

`num_points`
: BL, DXF 70

`points` 3DPOINT*, DXF 10

`b290` B, DXF 290

`x_direction`
: 3DPOINT, DXF 11

`inspt_offset`
: 3DPOINT, DXF 12

`endptproj`
: 3DPOINT, DXF 13

LIGHTLIST

`parent` struct _dwg_object_object*

`class_version`
: BL, DXF 90

`num_lights`
: BL, DXF 90

`lights` Dwg_LIGHTLIST_light*

LONG_TRANSACTION

`parent` struct _dwg_object_object*

LTYPE

LTYPE is a table object.

`parent` struct _dwg_object_object*

`flag` RC

`name` TV

`used` RSd

`is_xref_ref`
B

`is_xref_resolved`
BS

`is_xref_dep`
B

`xref` H

`description`
T, DXF 3

`pattern_len`
BD, DXF 40

`alignment`
RC, DXF 72

`numdashes`
RC, DXF 73

`dashes` Dwg_LTYPE_dash*

`dashes_r11`
RD, DXF 49

`has_strings_area`
B

`strings_area`
TF

`unknown_r11`
RC

LTYPE_CONTROL

LTYPE_CONTROL is a table_control object.

`parent` struct _dwg_object_object*

`num_entries`
BS, DXF 70

`entries` H*

`flags_r11`
: RS

`bylayer` H

`byblock` H

MATERIAL

`parent` struct _dwg_object_object*

`name` T, DXF 1

`description`
: T, DXF 2

`ambient_color`
: Dwg_MATERIAL_color

`diffuse_color`
: Dwg_MATERIAL_color

`diffusemap`
: Dwg_MATERIAL_mapper

`specular_gloss_factor`
: BD, DXF 44

`specular_color`
: Dwg_MATERIAL_color

`specularmap`
: Dwg_MATERIAL_mapper

`reflectionmap`
: Dwg_MATERIAL_mapper

`opacity_percent`
: BD, DXF 140

`opacitymap`
: Dwg_MATERIAL_mapper

`bumpmap` Dwg_MATERIAL_mapper

`refraction_index`
: BD, DXF 145

`refractionmap`
: Dwg_MATERIAL_mapper

`color_bleed_scale`
: BD, DXF 460

`indirect_bump_scale`
: BD, DXF 461

`reflectance_scale`
: BD, DXF 462

`transmittance_scale`
: BD, DXF 463

`two_sided_material`
: B, DXF 290

`luminance`
: BD, DXF 464

`luminance_mode`
: BS, DXF 270

`translucence`
: BD, DXF 148

`self_illumination`
: BD, DXF 149

`reflectivity`
: BD, DXF 468

`illumination_model`
: BL, DXF 93

`channel_flags`
: BL, DXF 94

`mode` BL, DXF 282

`genprocname`
: T

`genproctype`
: BS

`genprocvalbool`
: B

`genprocvalint`
: BS

`genprocvalreal`
: BD

`genprocvaltext`
: T

`genprocvalcolor`
: CMC

`genproctableend`
: B

`num_gentextures`
: BS

`gentextures`
: Dwg_MATERIAL_gentexture*

MENTALRAYRENDERSETTINGS

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`name` T, DXF 1

`fog_enabled`
B, DXF 290

`fog_background_enabled`
B, DXF 290

`backfaces_enabled`
B, DXF 290

`environ_image_enabled`
B, DXF 290

`environ_image_filename`
T, DXF 1

`description`
T, DXF 1

`display_index`
BL, DXF 90

`has_predefined`
B, DXF 290

`mr_version`
BL, DXF 90

`sampling1`
BL, DXF 90

`sampling2`
BL, DXF 90

`sampling_mr_filter`
BS, DXF 70

`sampling_filter1`
BD, DXF 40

`sampling_filter2`
BD, DXF 40

`sampling_contrast_color1`
BD, DXF 40

`sampling_contrast_color2`
BD, DXF 40

`sampling_contrast_color3`
BD, DXF 40

`sampling_contrast_color4`
BD, DXF 40

`shadow_mode`
BS, DXF 70

`shadow_maps_enabled`
B, DXF 290

`ray_tracing_enabled`
B, DXF 290

`ray_trace_depth1`
BL, DXF 90

`ray_trace_depth2`
BL, DXF 90

`ray_trace_depth3`
BL, DXF 90

`global_illumination_enabled`
B, DXF 290

`gi_sample_count`
BL, DXF 90

`gi_sample_radius_enabled`
B, DXF 290

`gi_sample_radius`
BD, DXF 40

`gi_photons_per_light`
BL, DXF 90

`photon_trace_depth1`
BL, DXF 90

`photon_trace_depth2`
BL, DXF 90

`photon_trace_depth3`
BL, DXF 90

`final_gathering_enabled`
B, DXF 290

`fg_ray_count`
BL, DXF 90

`fg_sample_radius_state1`
B, DXF 290

`fg_sample_radius_state2`
B, DXF 290

`fg_sample_radius_state3`
B, DXF 290

`fg_sample_radius1`
BD, DXF 40

`fg_sample_radius2`
BD, DXF 40

`light_luminance_scale`
BD, DXF 40

`diagnostics_mode`
BS, DXF 70

`diagnostics_grid_mode`
BS, DXF 70

`diagnostics_grid_float`
BD, DXF 40

`diagnostics_photon_mode`
BS, DXF 70

`diagnostics_bsp_mode`
BS, DXF 70

`export_mi_enabled`
B, DXF 290

`mr_description`
T, DXF 1

`tile_size`
BL, DXF 90

`tile_order`
BS, DXF 70

`memory_limit`
BL, DXF 90

`diagnostics_samples_mode`
B, DXF 290

`energy_multiplier`
BD, DXF 40

MLEADEROBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

MLEADERSTYLE

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 179

`content_type`
BS, DXF 170

`mleader_order`
BS, DXF 171

`leader_order`
BS, DXF 172

`max_points`
BL, DXF 90

`first_seg_angle`
BD, DXF 40

`second_seg_angle`
BD, DXF 41

`type` BS, DXF 173

`line_color`
CMC, DXF 91

`line_type`
H, DXF 340

`linewt` BLd, DXF 92

`has_landing`
B, DXF 290

`has_dogleg`
B, DXF 291

`landing_gap`
BD, DXF 42

`landing_dist`
BD, DXF 43

`description`
T, DXF 3

`arrow_head`
H, DXF 341

`arrow_head_size`
BD, DXF 44

`text_default`
T, DXF 300

`text_style`
H, DXF 342

`attach_left`
BS, DXF 174

`attach_right`
BS, DXF 178

`text_angle_type`
BS, DXF 175

`text_align_type`
BS, DXF 176

`text_color`
CMC, DXF 93

`text_height`
BD, DXF 45

`has_text_frame`
B, DXF 292

`text_always_left`
B, DXF 297

`align_space`
BD, DXF 46

`block` H, DXF 343

`block_color`
CMC, DXF 94

`block_scale`
3BD

`use_block_scale`
B, DXF 293

`block_rotation`
BD, DXF 141

`use_block_rotation`
B, DXF 294

`block_connection`
BS, DXF 177

`scale` BD, DXF 142

`is_changed`
B, DXF 295

`is_annotative`
: B, DXF 296

`break_size`
: BD, DXF 143

`attach_dir`
: BS, DXF 271

`attach_top`
: BS, DXF 273

`attach_bottom`
: BS, DXF 272

`text_extended`
: B, DXF 298

MLINESTYLE

`parent` struct _dwg_object_object*

`name` T, DXF 2

`description`
: T, DXF 3

`flag` BS, DXF 70

`fill_color`
: CMC, DXF 62

`start_angle`
: BD, DXF 51

`end_angle`
: BD, DXF 52

`num_lines`
: RC, DXF 71

`lines` Dwg_MLINESTYLE_line*

MOTIONPATH

`parent` struct _dwg_object_object*

`class_version`
: BS, DXF 90

`camera_path`
: H, DXF 340

`target_path`
: H, DXF 340

`viewtable`
: H, DXF 340

`frames` BS, DXF 90

`frame_rate`
BS, DXF 90

`corner_decel`
B, DXF 290

MTEXTATTRIBUTEOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`horizontal_mode`
BS, DXF 70

`rotation` BD, DXF 50

`ins_pt` 2RD, DXF 10

`alignment_pt`
2RD, DXF 11

`enable_context`
B, DXF 290

`context` Dwg_Object_SCALE

MTEXTOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`attachment`
BL, DXF 70

`ins_pt` 3BD, DXF 10

`x_axis_dir`
3BD, DXF 11

`rect_height`
BD, DXF 41

`rect_width`
BD, DXF 40

`extents_width`
BD, DXF 42

`extents_height`
BD, DXF 43

`column_type`
BL, DXF 71

`column_width`
BD, DXF 44

`gutter` BD, DXF 45

`auto_height`
B, DXF 73

`flow_reversed`
B, DXF 74

`num_column_heights`
BL, DXF 72

`column_heights`
BD*, DXF 46

NAVISWORKSMODELDEF

`parent` struct _dwg_object_object*

`flags` BS, DXF 70

`path` T, DXF 1

`status` B, DXF 290

`min_extent`
3BD, DXF 10

`max_extent`
3BD, DXF 11

`host_drawing_visibility`
B, DXF 290

OBJECT_PTR

`parent` struct _dwg_object_object*

ORDDIMOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`dimension`
: Dwg_OCD_Dimension

`feature_location_pt`
: 3BD, DXF 11

`leader_endpt`
: 3BD, DXF 12

PARTIAL_VIEWING_INDEX

`parent` struct _dwg_object_object*

`num_entries`
: BL

`has_entries`
: B

`entries` Dwg_PARTIAL_VIEWING_INDEX_Entry*

PERSUBENTMGR

`parent` struct _dwg_object_object*

`class_version`
: BL, DXF 90

`unknown_0`
: BL, DXF 90

`unknown_2`
: BL, DXF 90

`numassocsteps`
: BL, DXF 90

`numassocsubents`
: BL, DXF 90

`num_steps`
: BL, DXF 90

`steps` BL*, DXF 90

`num_subents`
: BL, DXF 90

`subents` BL*, DXF 90

PLACEHOLDER

`parent` struct _dwg_object_object*

PLOTSETTINGS

`parent` struct _dwg_object_object*

`printer_cfg_file`
: T, DXF 1

`paper_size`
: T, DXF 2

`canonical_media_name`
: T, DXF 4

`plot_flags`
: BS, DXF 70

`plotview` H, DXF 6

`plotview_name`
: T, DXF 6

`left_margin`
: BD, DXF 40

`bottom_margin`
: BD, DXF 41

`right_margin`
: BD, DXF 42

`top_margin`
: BD, DXF 43

`paper_width`
: BD, DXF 44

`paper_height`
: BD, DXF 45

`plot_origin`
: 2BD_1, DXF 46

`plot_window_ll`
: 2BD_1, DXF 48

`plot_window_ur`
: 2BD_1, DXF 140

`plot_paper_unit`
: BS, DXF 72

`plot_rotation_mode`
: BS, DXF 73

`plot_type`
: BS, DXF 74

`paper_units`
: BD, DXF 142

`drawing_units`
: BD, DXF 143

`stylesheet`
: T, DXF 7

`std_scale_type`
BS, DXF 75

`std_scale_factor`
BD, DXF 147

`paper_image_origin`
2BD_1, DXF 148

`shadeplot_type`
BS, DXF 76

`shadeplot_reslevel`
BS, DXF 77

`shadeplot_customdpi`
BS, DXF 78

`shadeplot`
H, DXF 333

POINTCLOUDCOLORMAP

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`def_intensity_colorscheme`
T, DXF 1

`def_elevation_colorscheme`
T, DXF 1

`def_classification_colorscheme`
T, DXF 1

`num_colorramps`
BL, DXF 90

`colorramps`
Dwg_POINTCLOUDCOLORMAP_Ramp*

`num_classification_colorramps`
BL, DXF 90

`classification_colorramps`
Dwg_POINTCLOUDCOLORMAP_Ramp*

POINTCLOUDDEF

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`source_filename`
T, DXF 1

`is_loaded`
B, DXF 280

`numpoints`
RLL, DXF 160

`extents_min`
3BD, DXF 10

`extents_max`
3BD, DXF 11

POINTCLOUDDEFEX

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`source_filename`
T, DXF 1

`is_loaded`
B, DXF 280

`numpoints`
RLL, DXF 160

`extents_min`
3BD, DXF 10

`extents_max`
3BD, DXF 11

POINTCLOUDDEF_REACTOR

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

POINTCLOUDDEF_REACTOR_EX

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

POINTPATH

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 90

`point` 3BD, DXF 10

PROXY_OBJECT

`parent` struct _dwg_object_object*

`proxy_id` BL, DXF 90

`class_id` BL, DXF 91

`dwg_versions`
BL, DXF 95

`maint_version`
BL

`dwg_version`
BL

`from_dxf` B, DXF 70

`data_numbits`
BL, DXF 93

`data_size`
BL

`data` TF, DXF 310

`num_objids`
BL

`objids` H*

RADIMLGOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`dimension`
Dwg_OCD_Dimension

`ovr_center`
3BD, DXF 12

`jog_point`
3BD, DXF 13

RADIMOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`dimension`
: Dwg_OCD_Dimension

`first_arc_pt`
: 3BD, DXF 11

RAPIDRTRENDERSETTINGS

`parent` struct _dwg_object_object*

`class_version`
: BL, DXF 90

`name` T, DXF 1

`fog_enabled`
: B, DXF 290

`fog_background_enabled`
: B, DXF 290

`backfaces_enabled`
: B, DXF 290

`environ_image_enabled`
: B, DXF 290

`environ_image_filename`
: T, DXF 1

`description`
: T, DXF 1

`display_index`
: BL, DXF 90

`has_predefined`
: B, DXF 290

`rapidrt_version`
: BL, DXF 90

`render_target`
: BL, DXF 70

`render_level`
: BL, DXF 90

`render_time`
: BL, DXF 90

`lighting_model`
: BL, DXF 70

`filter_type`
: BL, DXF 70

`filter_width`
: BD, DXF 40

`filter_height`
BD, DXF 40

RASTERVARIABLES

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`image_frame`
BS, DXF 70

`image_quality`
BS, DXF 71

`units` BS, DXF 72

RENDERENTRY

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`image_file_name`
T, DXF 1

`preset_name`
T, DXF 1

`view_name`
T, DXF 1

`dimension_x`
BL, DXF 90

`dimension_y`
BL, DXF 90

`start_year`
BS, DXF 70

`start_month`
BS, DXF 70

`start_day`
BS, DXF 70

`start_minute`
BS, DXF 70

`start_second`
BS, DXF 70

`start_msec`
BS, DXF 70

`render_time`
BD, DXF 40

`memory_amount`
BL, DXF 90

`material_count`
BL, DXF 90

`light_count`
BL, DXF 90

`triangle_count`
BL, DXF 90

`display_index`
BL, DXF 90

RENDERENVIRONMENT

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`fog_enabled`
B, DXF 290

`fog_background_enabled`
B, DXF 290

`fog_color_r`
RC, DXF 280

`fog_color_g`
RC, DXF 280

`fog_color_b`
RC, DXF 280

`fog_density_near`
BD, DXF 40

`fog_density_far`
BD, DXF 40

`fog_distance_near`
BD, DXF 40

`fog_distance_far`
BD, DXF 40

`environ_image_enabled`
B, DXF 290

`environ_image_filename`
T, DXF 1

RENDERGLOBAL

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`procedure`
BL, DXF 90

`destination`
BL, DXF 90

`save_enabled`
B, DXF 290

`save_filename`
T, DXF 1

`image_width`
BL, DXF 90

`image_height`
BL, DXF 90

`predef_presets_first`
B, DXF 290

`highlevel_info`
B, DXF 290

RENDERSETTINGS

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`name` T, DXF 1

`fog_enabled`
B, DXF 290

`fog_background_enabled`
B, DXF 290

`backfaces_enabled`
B, DXF 290

`environ_image_enabled`
B, DXF 290

`environ_image_filename`
T, DXF 1

`description`
T, DXF 1

`display_index`
BL, DXF 90

`has_predefined`
B, DXF 290

SCALE

`parent` struct _dwg_object_object*

`flag` BS, DXF 70

`name` T, DXF 300

`paper_units`
BD, DXF 140

`drawing_units`
BD, DXF 141

`is_unit_scale`
B, DXF 290

SECTIONVIEWSTYLE

`parent` struct _dwg_object_object*

`mdoc_class_version`
BS, DXF 70

`desc` T, DXF 3

`is_modified_for_recompute`
B, DXF 290

`display_name`
T, DXF 300

`viewstyle_flags`
BL, DXF 90

`class_version`
BS, DXF 70

`flags` BL, DXF 90

`identifier_style`
H, DXF 340

`identifier_color`
CMC, DXF 62

`identifier_height`
BD, DXF 40

`arrow_start_symbol`
H, DXF 340

`arrow_end_symbol`
H, DXF 340

`arrow_symbol_color`
: CMC, DXF 62

`arrow_symbol_size`
: BD, DXF 40

`identifier_exclude_characters`
: T, DXF 300

`identifier_position`
: BLd, DXF 90

`identifier_offset`
: BD, DXF 40

`arrow_position`
: BLd, DXF 90

`arrow_symbol_extension_length`
: BD, DXF 40

`plane_ltype`
: H, DXF 340

`plane_linewt`
: BLd, DXF 90

`plane_line_color`
: CMC, DXF 62

`bend_ltype`
: H, DXF 340

`bend_linewt`
: BLd, DXF 90

`bend_line_color`
: CMC, DXF 62

`bend_line_length`
: BD, DXF 40

`end_line_overshoot`
: BD, DXF 40

`end_line_length`
: BD, DXF 40

`viewlabel_text_style`
: H, DXF 340

`viewlabel_text_color`
: CMC, DXF 62

`viewlabel_text_height`
: BD, DXF 40

`viewlabel_attachment`
BL, DXF 90

`viewlabel_offset`
BD, DXF 40

`viewlabel_alignment`
BL, DXF 90

`viewlabel_pattern`
T, DXF 300

`hatch_color`
CMC, DXF 62

`hatch_bg_color`
CMC, DXF 62

`hatch_pattern`
T, DXF 300

`hatch_scale`
BD, DXF 40

`hatch_transparency`
BLd, DXF 90

`unknown_b1`
B, DXF 290

`unknown_b2`
B, DXF 290

`num_hatch_angles`
BL, DXF 90

`hatch_angles`
BD*, DXF 40

SECTION_MANAGER

`parent` struct _dwg_object_object*

`is_live` B, DXF 70

`num_sections`
BS, DXF 90

`sections` H*, DXF 330

SECTION_SETTINGS

`parent` struct _dwg_object_object*

`curr_type`
BS, DXF 90

`num_types`
BL, DXF 91

`types` Dwg_SECTION_typesettings*

SKYLIGHT_BACKGROUND

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`sunid` H, DXF 340

SOLID_BACKGROUND

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`color` BLx, DXF 90

SORTENTSTABLE

`parent` struct _dwg_object_object*

`num_ents` BL

`sort_ents`
H*, DXF 5

`block_owner`
H

`ents` H*, DXF 331

SPATIAL_FILTER

`parent` struct _dwg_object_object*

`num_clip_verts`
BS, DXF 70

`clip_verts`
2RD*, DXF 10

`extrusion`
BE, DXF 210

`origin` 3BD, DXF 11

`display_boundary_on`
BS, DXF 71

`front_clip_on`
BS, DXF 72

`front_clip_z`
BD, DXF 40

`back_clip_on`
BS, DXF 73

`back_clip_z`
: BD, DXF 41

`inverse_transform`
: BD*, DXF 40

`transform`
: BD*, DXF 40

SPATIAL_INDEX

`parent` struct _dwg_object_object*

`last_updated`
: TIMEBLL, DXF 40

`num1` BD, DXF 40

`num_hdls` BL, DXF 90

`hdls` H*, DXF 330

`bindata_size`
: BL, DXF 90

`bindata` TF, DXF 310

STYLE

STYLE is a table object.

`parent` struct _dwg_object_object*

`flag` RC

`name` TV

`used` RSd

`is_xref_ref`
: B

`is_xref_resolved`
: BS

`is_xref_dep`
: B

`xref` H

`is_shape` B

`is_vertical`
: B

`text_size`
: BD, DXF 40

`width_factor`
: BD, DXF 41

`oblique_angle`
BD, DXF 50

`generation`
RC, DXF 71

`last_height`
BD, DXF 42

`font_file`
T, DXF 3

`bigfont_file`
T, DXF 4

STYLE_CONTROL

STYLE_CONTROL is a table_control object.

`parent` struct _dwg_object_object*

`num_entries`
BS, DXF 70

`entries` H*

`flags_r11`
RS

SUN

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`is_on` B, DXF 290

`color` CMC, DXF 63

`intensity`
BD, DXF 40

`has_shadow`
B, DXF 291

`julian_day`
BL, DXF 91

`msecs` BL, DXF 92

`is_dst` B, DXF 292

`shadow_type`
BL, DXF 70

`shadow_mapsize`
BS, DXF 71

`shadow_softness`
RC, DXF 280

SUNSTUDY

`parent` struct _dwg_object_object*

`class_version`
BL, DXF 90

`setup_name`
T, DXF 1

`description`
T, DXF 2

`output_type`
BL, DXF 70

`sheet_set_name`
T, DXF 3

`use_subset`
B, DXF 290

`sheet_subset_name`
T, DXF 4

`select_dates_from_calendar`
B, DXF 291

`num_dates`
BL, DXF 91

`dates` Dwg_SUNSTUDY_Dates*

`select_range_of_dates`
B, DXF 292

`start_time`
BL, DXF 93

`end_time` BL, DXF 94

`interval` BL, DXF 95

`num_hours`
BL, DXF 91

`hours` B*, DXF 290

`shade_plot_type`
BL, DXF 74

`numvports`
BL, DXF 75

`numrows` BL, DXF 76

`numcols` BL, DXF 77

`spacing` BD, DXF 40

`lock_viewports`
B, DXF 293

`label_viewports`
B, DXF 294

`page_setup_wizard`
H, DXF 340

`view` H, DXF 341

`visualstyle`
H, DXF 342

`text_style`
H, DXF 343

TABLECONTENT

`parent` struct _dwg_object_object*

`ldata` Dwg_LinkedData

`tdata` Dwg_LinkedTableData

`fdata` Dwg_FormattedTableData

`tablestyle`
H, DXF 340

TABLEGEOMETRY

`parent` struct _dwg_object_object*

`numrows` BL, DXF 90

`numcols` BL, DXF 91

`num_cells`
BL, DXF 92

`cells` Dwg_TABLEGEOMETRY_Cell*

TABLESTYLE

`parent` struct _dwg_object_object*

`class_version`
BS

`name` T, DXF 3

`flags` BS, DXF 71

`flow_direction`
BS, DXF 70

`horiz_cell_margin`
BD, DXF 40

`vert_cell_margin`
BD, DXF 41

`is_title_suppressed`
B, DXF 280

`is_header_suppressed`
B, DXF 281

`unknown_rc`
RC, DXF 70

`unknown_bl1`
BL

`unknown_bl2`
BL

`cellstyle`
H

`sty` Dwg_TABLESTYLE_CellStyle

`numoverrides`
BL

`unknown_bl3`
BL

`ovr` Dwg_TABLESTYLE_CellStyle

`num_rowstyles`
BL

`rowstyles`
Dwg_TABLESTYLE_rowstyles*

TEXTOBJECTCONTEXTDATA

`parent` struct _dwg_object_object*

`class_version`
BS, DXF 70

`is_default`
B, DXF 290

`scale` H, DXF 340

`horizontal_mode`
BS, DXF 70

`rotation` BD, DXF 50

`ins_pt` 2RD, DXF 10

`alignment_pt`
2RD, DXF 11

TVDEVICEPROPERTIES

`parent` struct _dwg_object_object*

`flags` BL

`max_regen_threads`
BS

`use_lut_palette`
BL

`alt_hlt` BLL

`alt_hltcolor`
BLL

`geom_shader_usage`
BLL

`blending_mode`
BL

`antialiasing_level`
BD

`bd2` BD

UCS

UCS is a table object.

`parent` struct _dwg_object_object*

`flag` RC

`name` TV

`used` RSd

`is_xref_ref`
B

`is_xref_resolved`
BS

`is_xref_dep`
B

`xref` H

`ucsorg` 3BD, DXF 10

`ucsxdir` 3BD, DXF 11

`ucsydir` 3BD, DXF 12

`ucs_elevation`
BD, DXF 146

`UCSORTHOVIEW`
: BS, DXF 79

`base_ucs` H, DXF 346

`named_ucs`
: H

`num_orthopts`
: BS

`orthopts` Dwg_UCS_orthopts*

UCS_CONTROL

UCS_CONTROL is a table_control object.

`parent` struct _dwg_object_object*

`num_entries`
: BS, DXF 70

`entries` H*

`flags_r11`
: RS

UNKNOWN_OBJ

`parent` struct _dwg_object_object*

VBA_PROJECT

`parent` struct _dwg_object_object*

`data_size`
: BL, DXF 90

`data` TF, DXF 310

VIEW

VIEW is a table object.

`parent` struct _dwg_object_object*

`flag` RC

`name` TV

`used` RSd

`is_xref_ref`
: B

`is_xref_resolved`
: BS

`is_xref_dep`
: B

`xref` H

`VIEWSIZE` BD, DXF 40

`view_width`
BD, DXF 41

`aspect_ratio`
BD

`VIEWCTR` 2RD, DXF 10

`view_target`
3BD, DXF 12

`VIEWDIR` 3BD, DXF 11

`twist_angle`
BD, DXF 50

`lens_length`
BD, DXF 42

`front_clip_z`
BD, DXF 43

`back_clip_z`
BD, DXF 44

`VIEWMODE` 4BITS, DXF 71

`render_mode`
RC, DXF 281

`use_default_lights`
B, DXF 292

`default_lightning_type`
RC, DXF 282

`brightness`
BD, DXF 141

`contrast` BD, DXF 142

`ambient_color`
CMC, DXF 63

`is_pspace`
B

`associated_ucs`
B, DXF 72

`ucsorg` 3BD, DXF 110

`ucsxdir` 3BD, DXF 111

`ucsydir` 3BD, DXF 112

`ucs_elevation`
BD, DXF 146

`UCSORTHOVIEW`
BS, DXF 79

`is_camera_plottable`
B, DXF 73

`background`
H, DXF 332

`visualstyle`
H, DXF 348

`sun` H, DXF 361

`base_ucs` H, DXF 346

`named_ucs`
H, DXF 345

`livesection`
H, DXF 334

`flag_3d` RS

`unknown_r2`
RC

`unknown_r11`
RC

VIEW_CONTROL

VIEW_CONTROL is a table_control object.

`parent` struct _dwg_object_object*

`num_entries`
BS, DXF 70

`entries` H*

`flags_r11`
RS

VISUALSTYLE

`parent` struct _dwg_object_object*

`description`
T, DXF 2

`style_type`
BL, DXF 70

`ext_lighting_model`
BS, DXF 177

`internal_only`
B, DXF 291

`face_lighting_model`
: BL, DXF 71

`face_lighting_model_int`
: BS, DXF 176

`face_lighting_quality`
: BL, DXF 72

`face_lighting_quality_int`
: BS, DXF 176

`face_color_mode`
: BL, DXF 73

`face_color_mode_int`
: BS, DXF 176

`face_opacity`
: BD, DXF 40

`face_opacity_int`
: BS, DXF 176

`face_specular`
: BD, DXF 41

`face_specular_int`
: BS, DXF 176

`face_modifier`
: BL, DXF 90

`face_modifier_int`
: BS, DXF 176

`face_mono_color`
: CMC, DXF 63

`face_mono_color_int`
: BS, DXF 176

`edge_model`
: BS, DXF 74

`edge_model_int`
: BS, DXF 176

`edge_style`
: BL, DXF 91

`edge_style_int`
: BS, DXF 176

`edge_intersection_color`
: CMC, DXF 64

`edge_intersection_color_int`
: BS, DXF 176

`edge_obscured_color`
: CMC, DXF 65

`edge_obscured_color_int`
: BS, DXF 176

`edge_obscured_ltype`
: BL, DXF 75

`edge_obscured_ltype_int`
: BS, DXF 176

`edge_intersection_ltype`
: BL, DXF 175

`edge_intersection_ltype_int`
: BS, DXF 176

`edge_crease_angle`
: BD, DXF 42

`edge_crease_angle_int`
: BS, DXF 176

`edge_modifier`
: BL, DXF 92

`edge_modifier_int`
: BS, DXF 176

`edge_color`
: CMC, DXF 66

`edge_color_int`
: BS, DXF 176

`edge_opacity`
: BD, DXF 43

`edge_opacity_int`
: BS, DXF 176

`edge_width`
: BL, DXF 76

`edge_width_int`
: BS, DXF 176

`edge_overhang`
: BL, DXF 77

`edge_overhang_int`
: BS, DXF 176

`edge_jitter`
: BL, DXF 78

`edge_jitter_int`
: BS, DXF 176

`edge_silhouette_color`
: CMC, DXF 67

`edge_silhouette_color_int`
: BS, DXF 176

`edge_silhouette_width`
: BL, DXF 79

`edge_silhouette_width_int`
: BS, DXF 176

`edge_halo_gap`
: BL, DXF 170

`edge_halo_gap_int`
: BS, DXF 176

`edge_isolines`
: BL, DXF 171

`edge_isolines_int`
: BS, DXF 176

`edge_do_hide_precision`
: B, DXF 290

`edge_do_hide_precision_int`
: BS, DXF 176

`edge_style_apply`
: BL, DXF 174

`edge_style_apply_int`
: BS

`display_settings`
: BL, DXF 93

`display_settings_int`
: BS, DXF 176

`display_brightness_bl`
: BLd, DXF 44

`display_brightness`
: BD, DXF 44

`display_brightness_int`
: BS, DXF 176

`display_shadow_type`
: BL, DXF 173

`display_shadow_type_int`
: BS, DXF 176

`bd2007_45`
: BD, DXF 45

`num_props`
: BS, DXF 70

`b_prop1c` B, DXF 290

`b_prop1c_int`
: BS, DXF 176

`b_prop1d` B, DXF 290

`b_prop1d_int`
: BS, DXF 176

`b_prop1e` B, DXF 290

`b_prop1e_int`
: BS, DXF 176

`b_prop1f` B, DXF 290

`b_prop1f_int`
: BS, DXF 176

`b_prop20` B, DXF 290

`b_prop20_int`
: BS, DXF 176

`b_prop21` B, DXF 290

`b_prop21_int`
: BS, DXF 176

`b_prop22` B, DXF 290

`b_prop22_int`
: BS, DXF 176

`b_prop23` B, DXF 290

`b_prop23_int`
: BS, DXF 176

`b_prop24` B, DXF 290

`b_prop24_int`
: BS, DXF 176

`bl_prop25`
: BL, DXF 90

`bl_prop25_int`
: BS, DXF 176

`bd_prop26`
: BD, DXF 40

`bd_prop26_int`
: BS, DXF 176

`bd_prop27`
: BD, DXF 40

`bd_prop27_int`
: BS, DXF 176

`bl_prop28`
: BL, DXF 90

`bl_prop28_int`
: BS, DXF 176

`c_prop29` CMC, DXF 62

`c_prop29_int`
: BS, DXF 176

`bl_prop2a`
: BL, DXF 90

`bl_prop2a_int`
: BS, DXF 176

`bl_prop2b`
: BL, DXF 90

`bl_prop2b_int`
: BS, DXF 176

`c_prop2c` CMC, DXF 62

`c_prop2c_int`
: BS, DXF 176

`b_prop2d` B, DXF 290

`b_prop2d_int`
: BS, DXF 176

`bl_prop2e`
: BL, DXF 90

`bl_prop2e_int`
: BS, DXF 176

`bl_prop2f`
: BL, DXF 90

`bl_prop2f_int`
: BS, DXF 176

`bl_prop30`
: BL, DXF 90

`bl_prop30_int`
: BS, DXF 176

`b_prop31` B, DXF 290

`b_prop31_int`
: BS, DXF 176

`bl_prop32`
: BL, DXF 90

`bl_prop32_int`
: BS, DXF 176

`c_prop33` CMC, DXF 62

`c_prop33_int`
: BS, DXF 176

`bd_prop34`
: BD, DXF 40

`bd_prop34_int`
: BS, DXF 176

`edge_wiggle`
: BL, DXF 90

`edge_wiggle_int`
: BS, DXF 176

`strokes` T, DXF 1

`strokes_int`
: BS, DXF 176

`b_prop37` B, DXF 290

`b_prop37_int`
: BS, DXF 176

`bd_prop38`
: BD, DXF 40

`bd_prop38_int`
: BS, DXF 176

`bd_prop39`
: BD, DXF 40

`bd_prop39_int`
: BS, DXF 176

VPORT

VPORT is a table object.

`parent` struct _dwg_object_object*

`flag` RC

`name` TV

`used` RSd

`is_xref_ref`
B

`is_xref_resolved`
BS

`is_xref_dep`
B

`xref` H

`VIEWSIZE` BD, DXF 40

`view_width`
BD

`aspect_ratio`
BD, DXF 41

`VIEWCTR` 2RD, DXF 12

`view_target`
3BD, DXF 17

`VIEWDIR` 3BD, DXF 16

`view_twist`
BD, DXF 51

`lens_length`
BD, DXF 42

`front_clip_z`
BD, DXF 43

`back_clip_z`
BD, DXF 44

`VIEWMODE` 4BITS, DXF 71

`render_mode`
RC, DXF 281

`use_default_lights`
B, DXF 292

`default_lightning_type`
RC, DXF 282

`brightness`
BD, DXF 141

`contrast` BD, DXF 142

`ambient_color`
: CMC, DXF 63

`lower_left`
: 2RD, DXF 10

`upper_right`
: 2RD, DXF 11

`UCSFOLLOW`
: B, DXF 71

`circle_zoom`
: BS, DXF 72

`FASTZOOM` B, DXF 73

`UCSICON` RC, DXF 74

`GRIDMODE` B, DXF 76

`GRIDUNIT` 2RD, DXF 15

`SNAPMODE` B, DXF 75

`SNAPSTYLE`
: B, DXF 77

`SNAPISOPAIR`
: BS, DXF 78

`SNAPANG` BD, DXF 50

`SNAPBASE` 2RD, DXF 13

`SNAPUNIT` 2RD, DXF 14

`ucs_at_origin`
: B

`UCSVP` B, DXF 71

`ucsorg` 3BD, DXF 110

`ucsxdir` 3BD, DXF 111

`ucsydir` 3BD, DXF 112

`ucs_elevation`
: BD, DXF 146

`UCSORTHOVIEW`
: BS, DXF 79

`grid_flags`
: BS, DXF 60

`grid_major`
: BS, DXF 61

`background`
: H, DXF 332

`visualstyle`
: H, DXF 348

`sun`
: H, DXF 361

`named_ucs`
: H, DXF 345

`base_ucs`
: H, DXF 346

VPORT_CONTROL

VPORT_CONTROL is a table_control object.

`parent`
: struct _dwg_object_object*

`num_entries`
: BS, DXF 70

`entries`
: H*

`flags_r11`
: RS

VX_CONTROL

VX_CONTROL is a table_control object.

`parent`
: struct _dwg_object_object*

`num_entries`
: BS, DXF 70

`entries`
: H*

`flags_r11`
: RS

VX_TABLE_RECORD

VX_TABLE_RECORD is a table object.

`parent`
: struct _dwg_object_object*

`flag`
: RC

`name`
: TV

`used`
: RSd

`is_xref_ref`
: B

`is_xref_resolved`
: BS

`is_xref_dep`
: B

`xref` H

`is_on` B, DXF 290

`viewport` H, DXF 338

`prev_entry`
H, DXF 340

`vport_entity_address`
RS

`r11_viewport_index`
RSd

`r11_prev_entry_index`
RSd

WIPEOUTVARIABLES

`parent` struct _dwg_object_object*

`display_frame`
BS, DXF 70

XRECORD

`parent` struct _dwg_object_object*

`cloning` BS, DXF 280

`xdata_size`
BL

`num_xdata`
BL

`xdata` Dwg_Resbuf*

`num_objid_handles`
BL

`objid_handles`
H*, DXF 340

PDFDEFINITION

`parent` struct _dwg_object_object*

`filename` T, DXF 1

`name` T, DXF 2

DGNDEFINITION

See [UNDERLAYDEFINITION], page 219,

DWFDEFINITION

See [UNDERLAYDEFINITION], page 219,

ASSOCARRAYMODIFYPARAMETERS

`parent` struct _dwg_object_object*

`aap_version`
BL

`num_items`
BL

`classname`
TV

`items` Dwg_ASSOCARRAYITEM*

`numitems` BL

`numrows` BL

`numlevels`
BL

ASSOCARRAYPATHPARAMETERS
See [ASSOCARRAYPARAMETERS], page 219,

ASSOCARRAYPOLARPARAMETERS
See [ASSOCARRAYPARAMETERS], page 219,

ASSOCARRAYRECTANGULARPARAMETERS
See [ASSOCARRAYPARAMETERS], page 219,

Dwg_3DSOLID_material

`parent` struct _dwg_entity_3DSOLID*

`array_index`
BL

`mat_absref`
BL

`material_handle`
H

Dwg_3DSOLID_silhouette

`parent` struct _dwg_entity_3DSOLID*

`vp_id` BL

`vp_target`
3BD

`vp_dir_from_target`
3BD

`vp_up_dir`
3BD

`vp_perspective`
B

`has_wires`
B

`num_wires`
BL

`wires` Dwg_3DSOLID_wire*

Dwg_3DSOLID_wire

`parent` struct _dwg_entity_3DSOLID*

`type` RC

`selection_marker`
BLd

`color` BL

`acis_index`
BLd

`num_points`
BL

`points` 3BD*

`transform_present`
B

`axis_x` 3BD

`axis_y` 3BD

`axis_z` 3BD

`translation`
3BD

`scale` 3BD

`has_rotation`
B

`has_reflection`
B

`has_shear`
B

Dwg_ACSH_HistoryNode

`major` BL

`minor` BL

`trans` BD*, DXF 40

`color` CMC

`step_id` BL

`material` H

Dwg_ACSH_SubentColor

`major` BL

`minor` BL

`transparency` BL

`bl93` BL

`is_face_variable` B

Dwg_ACSH_SubentMaterial

`major` BL

`minor` BL

`reflectance` BL

`displacement` BL

Dwg_ACTIONBODY

`parent` struct _dwg_object_ASSOCNETWORK*

`evaluatorid` T

`expression` T

`value` BL

Dwg_ARRAYITEMLOCATOR

`parent` struct _dwg_object_ASSOCARRAYMODIFYACTIONBODY*

`itemloc` BL

Dwg_ASSOCACTIONBODY_action

`parent` struct _dwg_object_ASSOCMLEADERACTIONBODY*

`depid` BL

`dep` H, DXF 330

Dwg_ASSOCACTION_Deps

`parent` struct _dwg_object_ASSOCACTION*

`is_owned` B

`dep` H

Dwg_ASSOCARRAYITEM

`parent` struct _dwg_abstractobject_ASSOCARRAYPARAMETERS*

`class_version`
: BL, DXF 90

`itemloc` BL

`flags` BL

`is_default_transmatrix`
: int

`x_dir` 3BD

`transmatrix`
: BD*

`rel_transform`
: BD*

`has_h1` int

`h1` H

`h2` H

Dwg_ASSOCPARAMBASEDACTIONBODY

`parent` struct _dwg_object_object*

`version` BL

`minor` BL

`num_deps` BL, DXF 90

`deps` H*

`l4` BL

`l5` BL

`assocdep` H

`num_values`
: BL

`values` Dwg_VALUEPARAM*

Dwg_ASSOCSURFACEACTIONBODY

`parent` struct _dwg_object_object*

`version` BL

`is_semi_assoc`
: B

`l2` BL

`is_semi_ovr`
: B

`grip_status`
BS

`assocdep` H

Dwg_AcDbMTextObjectEmbedded

`attachment`
BL

`ins_pt` 3BD

`x_axis_dir`
3BD

`rect_height`
BD

`rect_width`
BD

`extents_width`
BD

`extents_height`
BD

`column_type`
BL

`column_width`
BD

`gutter` BD

`auto_height`
B

`flow_reversed`
B

`num_column_heights`
BL

`column_heights`
BD*

Dwg_AcDs

`file_signature`
RL

`file_header_size`
RL

`unknown_1`
RL

`version` RL

`unknown_2`
RL

`ds_version`
RL

`segidx_offset`
RL

`segidx_unknown`
RL

`num_segidx`
RL

`schidx_segidx`
RL

`datidx_segidx`
RL

`search_segidx`
RL

`prvsav_segidx`
RL

`file_size`
RL

`total_segments`
BL

`segidx` Dwg_AcDs_SegmentIndex*

`datidx` Dwg_AcDs_DataIndex

`data` Dwg_AcDs_Data*

`blob01` Dwg_AcDs_DataBlob

`schidx` Dwg_AcDs_SchemaIndex

`schdat` Dwg_AcDs_SchemaData

`search` Dwg_AcDs_Search

`segments` Dwg_AcDs_Segment*

Dwg_AcDs_Data

`record_hdrs`
Dwg_AcDs_Data_RecordHdr*

`records` Dwg_AcDs_Data_Record*

Dwg_AcDs_DataBlob

`data_size`
RLL

`page_count`
RL

`record_size`
RL

`page_size`
RL

`unknown_1`
RL

`unknown_2`
RL

`ref` Dwg_AcDs_DataBlobRef*

Dwg_AcDs_DataBlob01

`total_data_size`
RLL

`page_start_offset`
RLL

`page_index`
int32_t

`page_count`
int32_t

`page_data_size`
RLL

`page_data`
RC*

Dwg_AcDs_DataBlobRef

`total_data_size`
RLL

`num_pages`
RL

`record_size`
RL

`page_size`
RL

`unknown_1`
RL

`unknown_2`
RL

`pages` Dwg_AcDs_DataBlobRef_Page*

Dwg_AcDs_DataBlobRef_Page

`segidx` RL

`size` RL

Dwg_AcDs_DataIndex

`num_entries`
RL

`di_unknown`
RL

`entries` Dwg_AcDs_DataIndex_Entry*

Dwg_AcDs_DataIndex_Entry

`segidx` RL

`offset` RL

`schidx` RL

Dwg_AcDs_Data_Record

`data_size`
RL

`blob` RC*

Dwg_AcDs_Data_RecordHdr

`entry_size`
RL

`unknown` RL

`handle` RLL

`offset` RL

Dwg_AcDs_Schema

`num_index`
RS

`index` RLL*

`num_props`
RS

`props` Dwg_AcDs_Schema_Prop*

Dwg_AcDs_SchemaData

`num_uprops`
RL

`uprops` Dwg_AcDs_SchemaData_UProp*

`num_schemas`
RL

`schemas` Dwg_AcDs_Schema*

`num_propnames`
RL

`propnames`
TV*

Dwg_AcDs_SchemaData_UProp

`size` RL

`flags` RL

Dwg_AcDs_SchemaIndex

`num_props`
RL

`si_unknown_1`
RL

`props` Dwg_AcDs_SchemaIndex_Prop*

`si_tag` RLL

`num_prop_entries`
RL

`si_unknown_2`
RL

`prop_entries`
Dwg_AcDs_SchemaIndex_Prop*

Dwg_AcDs_SchemaIndex_Prop

`index` RL

`segidx` RL

`offset` RL

Dwg_AcDs_Schema_Prop

`flags` RL

`namidx` RL

`type` RL

`type_size`
RL

`unknown_1`
RL

`unknown_2`
RL

`num_values`
RS

`values` RC*

Dwg_AcDs_Search

`num_search`
RL

`search` Dwg_AcDs_Search_Data*

Dwg_AcDs_Search_Data

`schema_namidx`
RL

`num_sortedidx`
RL

`sortedidx`
RLLd*

`num_ididxs`
RL

`unknown` RL

`ididxs` Dwg_AcDs_Search_IdIdxs*

Dwg_AcDs_Search_IdIdx

`handle` RLL

`num_ididx`
RL

`ididx` RLL*

Dwg_AcDs_Search_IdIdxs

`num_ididx`
RL

`ididx` Dwg_AcDs_Search_IdIdx*

Dwg_AcDs_Segment

`signature`
RS

`name` RC

`type` RCd

`segment_idx`
RL

`is_blob01`
: RL

`segsize` RL

`unknown_2`
: RL

`ds_version`
: RL

`unknown_3`
: RL

`data_algn_offset`
: RL

`objdata_algn_offset`
: RL

`padding` RC

Dwg_AcDs_SegmentIndex

`offset` RLL

`size` RL

Dwg_BLOCKACTION_connectionpts

`code` BL

`name` TV

Dwg_BLOCKLOOKUPACTION_lut

`parent` struct _dwg_object_BLOCKLOOKUPACTION*

`conn_pts` Dwg_BLOCKACTION_connectionpts

`b282` B, DXF 282

`b281` B, DXF 281

Dwg_BLOCKPARAMETER_PropInfo

`num_connections`
: BL

`connections`
: Dwg_BLOCKPARAMETER_connection*

Dwg_BLOCKPARAMETER_connection

`code` BL

`name` T

Dwg_BLOCKPARAMVALUESET

`desc` TV

`flags` BL

`minimum` BD

`maximum` BD

`increment`
BD

`num_valuelist`
BS

`valuelist`
BD*

Dwg_BLOCKSTRETCHACTION_codes

`parent` struct _dwg_object_BLOCKSTRETCHACTION*

`bl95` BL, DXF 95

`num_indexes`
BS, DXF 76

`indexes` BL*, DXF 94

Dwg_BLOCKSTRETCHACTION_handles

`parent` struct _dwg_object_BLOCKSTRETCHACTION*

`hdl` H, DXF 331

`num_indexes`
BS, DXF 74

`indexes` BL*, DXF 94

Dwg_BLOCKVISIBILITYPARAMETER_state

`parent` struct _dwg_object_BLOCKVISIBILITYPARAMETER*

`name` T, DXF 303

`num_blocks`
BL, DXF 94

`blocks` H*, DXF 332

`num_params`
BL, DXF 95

`params` H*, DXF 333

Dwg_COMPOUNDOBJECTID

`parent` struct _dwg_object_object*

`has_object`
B

`name` T

`object` H

Dwg_CONSTRAINTGROUPNODE

`parent` struct _dwg_object_ASSOC2DCONSTRAINTGROUP*

`nodeid` BL

`status` RC

`num_connections`
BL

`connections`
BL*

Dwg_CONTEXTDATA_dict

`parent` struct _dwg_CONTEXTDATA_submgr*

`text` T, DXF 3

`itemhandle`
H, DXF 350

Dwg_CONTEXTDATA_submgr

`parent` struct _dwg_object_CONTEXTDATAMANAGER*

`handle` H

`num_entries`
BL, DXF 90

`entries` Dwg_CONTEXTDATA_dict*

Dwg_CellContentGeometry

`dist_top_left`
3BD, DXF 10

`dist_center`
3BD, DXF 11

`content_width`
BD, DXF 43

`content_height`
BD, DXF 44

`width` BD, DXF 45

`height` BD, DXF 46

`unknown` BL, DXF 95

`cell_parent`
struct _dwg_TableCell*

`geom_parent`
struct _dwg_TABLEGEOMETRY_Cell*

Dwg_CellStyle

`type` BL, DXF 90

`data_flags`
BS, DXF 170

`property_override_flags`
BL, DXF 91

`merge_flags`
BL, DXF 92

`bg_color` CMC, DXF 62

`content_layout`
BL, DXF 93

`content_format`
Dwg_ContentFormat

`margin_override_flags`
BS, DXF 171

`vert_margin`
BD, DXF 40

`horiz_margin`
BD, DXF 40

`bottom_margin`
BD, DXF 40

`right_margin`
BD, DXF 40

`margin_horiz_spacing`
BD, DXF 40

`margin_vert_spacing`
BD, DXF 40

`num_borders`
BL, DXF 94

`borders` Dwg_GridFormat*

`tablerow_parent`
struct _dwg_TableRow*

`tabledatacolumn_parent`
struct _dwg_TableDataColumn*

Dwg_ColorRamp

`parent` struct _dwg_POINTCLOUDCOLORMAP_Ramp*

`colorscheme`
T, DXF 1

`unknown_bl`
: BL, DXF 91

`unknown_b`
: B, DXF 290

Dwg_ContentFormat

`property_override_flags`
: BL, DXF 90

`property_flags`
: BL, DXF 91

`value_data_type`
: BL, DXF 92

`value_unit_type`
: BL, DXF 93

`value_format_string`
: T, DXF 300

`rotation` BD, DXF 40

`block_scale`
: BD, DXF 140

`cell_alignment`
: BL, DXF 94

`content_color`
: CMC, DXF 62

`text_style`
: H

`text_height`
: BD, DXF 144

Dwg_DATALINK_customdata

`parent` struct _dwg_object_DATALINK*

`target` H

`text` T, DXF 304

Dwg_DATATABLE_column

`parent` struct _dwg_object_DATATABLE*

`type` BL, DXF 92

`text` T, DXF 2

`rows` Dwg_DATATABLE_row*

Dwg_DATATABLE_row

`parent` struct _dwg_DATATABLE_column*

`value` Dwg_TABLE_value

Dwg_DIMASSOC_Ref

`parent` struct _dwg_object_DIMASSOC*

`classname`
T, DXF 1

`osnap_type`
RC, DXF 72

`osnap_dist`
BD, DXF 40

`osnap_pt` 3BD, DXF 10

`num_xrefs`
BS

`xrefs` H*, DXF 331

`main_subent_type`
BS, DXF 73

`main_gsmarker`
BL, DXF 91

`num_xrefpaths`
BS

`xrefpaths`
TV*, DXF 301

`has_lastpt_ref`
B, DXF 75

`lastpt_ref`
3BD

`num_intsectobj`
BL, DXF 74

`intsectobj`
H*, DXF 332

Dwg_DIMENSION_common

`parent` struct _dwg_object_entity*

`class_version`
RC, DXF 280

`extrusion`
BE, DXF 210

`def_pt` 3BD, DXF 10

`text_midpt`
2RD, DXF 11

`elevation`
BD, DXF 31

`flag` RC, DXF 70

`flag1` RC

`user_text`
TV, DXF 1

`text_rotation`
BD, DXF 53

`horiz_dir`
BD, DXF 51

`ins_scale`
3BD

`ins_rotation`
BD, DXF 54

`attachment`
BS, DXF 71

`lspace_style`
BS, DXF 72

`lspace_factor`
BD, DXF 41

`act_measurement`
BD, DXF 42

`unknown` B, DXF 73

`flip_arrow1`
B, DXF 74

`flip_arrow2`
B, DXF 75

`clone_ins_pt`
2RD, DXF 12

`dimstyle` H, DXF 3

`block` H, DXF 2

Dwg_EVAL_Edge

`parent` struct _dwg_object_EVALUATION_GRAPH*

`id` BL, DXF 92

`nextid` BLd, DXF 93

`e1` BLd, DXF 94

`e2` BLd, DXF 91

`e3` BLd, DXF 91

`out_edge` BLd

Dwg_EVAL_Node

`parent` struct _dwg_object_EVALUATION_GRAPH*

`id` BL, DXF 91

`edge_flags`
BL, DXF 93

`nextid` BLd, DXF 95

`evalexpr` H, DXF 360

`node` BLd

`active_cycles`
B

Dwg_EvalExpr

`parentid` BLd

`major` BL

`minor` BL

`value_code`
BSd

`value.num40`
BD

`value.pt2d`
2RD

`value.pt3d`
3BD

`value.text1`
TV

`value.long90`
BL

`value.handle91`
H

`value.short70`
BS

`nodeid` BL

Dwg_EvalVariant

`code` BS

`u.bd` BD

`u.bl` BL

`u.bs` BS

`u.rc` RC

`u.text` TV

`u.handle` H

Dwg_FIELD_ChildValue

`parent` struct _dwg_object_FIELD*

`key` TV, DXF 6

`value` Dwg_TABLE_value

Dwg_FileDepList_Files

`filename` T32

`filepath` T32

`fingerprint`
T32

`version` T32

`feature_index`
RL

`timestamp`
RL

`filesize` RL

`affects_graphics`
RS

`refcount` RL

Dwg_FormattedTableData

`parent` struct _dwg_object_TABLECONTENT*

`cellstyle`
Dwg_CellStyle

`num_merged_cells`
BL, DXF 90

`merged_cells`
Dwg_FormattedTableMerged*

Dwg_FormattedTableMerged

`parent` struct _dwg_FormattedTableData*

`top_row` BL, DXF 91

`left_col` BL, DXF 92

`bottom_row`
BL, DXF 93

`right_col`
BL, DXF 94

Dwg_GEODATA_meshface

`face1` BL

`face2` BL

`face3` BL

Dwg_GEODATA_meshpt

`source_pt`
2RD

`dest_pt` 2RD

Dwg_GridFormat

`parent` struct _dwg_CellStyle*

`index_mask`
BL, DXF 95

`border_overrides`
BL, DXF 90

`border_type`
BL, DXF 91

`color` CMC, DXF 62

`linewt` BLd, DXF 92

`ltype` H, DXF 340

`visible` B, DXF 93

`double_line_spacing`
BD, DXF 40

Dwg_HATCH_Color

`parent` struct _dwg_entity_HATCH*

`shift_value`
BD, DXF 463

`color` CMC, DXF 63

Dwg_HATCH_ControlPoint

`parent` struct _dwg_HATCH_PathSeg*

`point` 2RD, DXF 10

`weight` BD, DXF 40

Dwg_HATCH_DefLine

`parent` struct _dwg_entity_HATCH*

`angle` BD, DXF 53

`pt0` 2BD, DXF 43

`offset` 2BD, DXF 45

`num_dashes`
BS, DXF 79

`dashes` BD*, DXF 49

Dwg_HATCH_Path

`parent` struct _dwg_entity_HATCH*

`flag` BL, DXF 92

`num_segs_or_paths`
BL, DXF 93

`segs` Dwg_HATCH_PathSeg*

`bulges_present`
B, DXF 72

`closed` B, DXF 73

`polyline_paths`
Dwg_HATCH_PolylinePath*

`num_boundary_handles`
BL, DXF 97

`boundary_handles`
H*, DXF 330

Dwg_HATCH_PathSeg

`parent` struct _dwg_HATCH_Path*

`curve_type`
RC, DXF 72

`first_endpoint`
2RD, DXF 10

`second_endpoint`
2RD, DXF 11

`center` 2RD, DXF 10

`radius` BD, DXF 40

`start_angle`
BD, DXF 50

`end_angle`
BD, DXF 51

`is_ccw` B, DXF 73

`endpoint` 2RD, DXF 11

`minor_major_ratio`
BD, DXF 40

`degree` BL, DXF 94

`is_rational`
B, DXF 73

`is_periodic`
B, DXF 74

`num_knots`
BL, DXF 95

`num_control_points`
BL, DXF 96

`knots` BD*

`control_points`
Dwg_HATCH_ControlPoint*

`num_fitpts`
BL, DXF 97

`fitpts` 2RD*

`start_tangent`
2RD, DXF 12

`end_tangent`
2RD, DXF 13

Dwg_HATCH_PolylinePath

`parent` struct _dwg_HATCH_Path*

`point` 2RD, DXF 10

`bulge` BD, DXF 42

Dwg_LAYER_entry

`parent` struct _dwg_object_LAYER_INDEX*

`numlayers`
BL, DXF 90

`name` T, DXF 8

`handle` H, DXF 360

Dwg_LEADER_ArrowHead

`parent` struct _dwg_entity_MULTILEADER*

`is_default`
B, DXF 94

`arrowhead`
H, DXF 345

Dwg_LEADER_BlockLabel

`parent` struct _dwg_entity_MULTILEADER*

`attdef` H, DXF 330

`label_text`
TV, DXF 302

`ui_index` BS, DXF 177

`width` BD, DXF 44

Dwg_LEADER_Break

`parent` struct _dwg_LEADER_Line*

`start` 3BD, DXF 11

`end` 3BD, DXF 12

Dwg_LEADER_Line

`parent` struct _dwg_LEADER_Node*

`num_points`
BL

`points` 3DPOINT*

`num_breaks`
BL

`breaks` Dwg_LEADER_Break*

`line_index`
BL, DXF 91

`type` BS, DXF 170

`color` CMC, DXF 92

`ltype` H, DXF 340

`linewt` BLd, DXF 171

`arrow_size`
BD, DXF 40

`arrow_handle`
H, DXF 341

`flags` BL, DXF 93

Dwg_LEADER_Node

`parent` struct _dwg_entity_MULTILEADER*

`has_lastleaderlinepoint`
B, DXF 290

`has_dogleg`
B, DXF 291

`lastleaderlinepoint`
3BD, DXF 10

`dogleg_vector`
3BD, DXF 11

`branch_index`
BL, DXF 90

`dogleg_length`
BD, DXF 40

`num_lines`
BL

`lines` Dwg_LEADER_Line*

`num_breaks`
BL

`breaks` Dwg_LEADER_Break*

`attach_dir`
BS, DXF 271

Dwg_LIGHTLIST_light

`parent` struct _dwg_object_LIGHTLIST*

`name` T, DXF 1

`handle` H, DXF 5

Dwg_LTYPE_dash

`parent` struct _dwg_object_LTYPE*

`length` BD, DXF 49

`complex_shapecode`
BS, DXF 75

`style` H, DXF 340

`x_offset` RD, DXF 44

`y_offset` RD, DXF 45

`scale` BD, DXF 46

`rotation` BD, DXF 50

`shape_flag`
BS, DXF 74

`text` T, DXF 9

Dwg_LWPOLYLINE_width

`start` BD, DXF 40

`end` BD, DXF 41

Dwg_LinkedData

`name` T, DXF 1

`description`
T, DXF 300

Dwg_LinkedTableData

`num_cols` BL, DXF 90

`cols` Dwg_TableDataColumn*

`num_rows` BL, DXF 90

`rows` Dwg_TableRow*

`num_field_refs`
BL

`field_refs`
H*

Dwg_MATERIAL_color

`parent` struct _dwg_object_object*

`flag` RC

`factor` BD

`rgb` BL

Dwg_MATERIAL_gentexture

`parent` struct _dwg_object_MATERIAL*

`genprocname`
T

`material` struct _dwg_object_MATERIAL*

Dwg_MATERIAL_mapper

`parent` struct _dwg_object_object*

`blendfactor`
BD

`transmatrix`
BD*

`filename` T

`color1` Dwg_MATERIAL_color

`color2` Dwg_MATERIAL_color

`source` RC

`projection`
RC

`tiling` RC

`autotransform`
RC

`texturemode`
BS

Dwg_MESH_edge

`parent` struct _dwg_entity_MESH*

`idxfrom` BL, DXF 90

`idxto` BL, DXF 90

Dwg_MLEADER_AnnotContext

`num_leaders`
BL

`leaders` Dwg_LEADER_Node*

`attach_dir`
BS

`scale_factor`
BD, DXF 40

`content_base`
3BD, DXF 10

`text_height`
BD, DXF 41

`arrow_size`
BD, DXF 140

`landing_gap`
BD, DXF 145

`text_left`
BS, DXF 174

`text_right`
BS, DXF 175

`text_angletype`
BS, DXF 176

`text_alignment`
BS, DXF 177

`has_content_txt`
B, DXF 290

`has_content_blk`
B, DXF 296

`content` Dwg_MLEADER_Content

`base` 3BD, DXF 110

`base_dir` 3BD, DXF 111

`base_vert`
3BD, DXF 112

`is_normal_reversed`
B, DXF 297

`text_top` BS, DXF 273

`text_bottom`
BS, DXF 272

Dwg_MLEADER_Content_Block

`type` RC

`normal` 3BD

`location` 3BD

`rotation` BD

`block_table`
H

`scale` 3BD

`color` CMC

`transform`
BD*

Dwg_MLEADER_Content_MText

`type` RC

`normal` 3BD

`location` 3BD

`rotation` BD

`style` H

`direction`
3BD

`color` CMC

`width` BD

`height` BD

`line_spacing_factor`
BD

`default_text`
T

`line_spacing_style`
BS

`alignment`
BS

`flow` BS

`bg_color` CMC

`bg_scale` BD

`bg_transparency`
BL

`is_bg_fill`
B

`is_bg_mask_fill`
B

`col_type` BS

`is_height_auto`
B

`col_width`
BD

`col_gutter`
BD

`is_col_flow_reversed`
B

`num_col_sizes`
BL

`col_sizes`
BD*

`word_break`
B

`unknown` B

Dwg_MLINESTYLE_line

`parent` struct _dwg_object_MLINESTYLE*

`offset` BD, DXF 49

`color` CMC, DXF 62

`lt.index` BSd, DXF 6

`lt.ltype` H, DXF 6

Dwg_MLINE_line

`parent` struct _dwg_MLINE_vertex*

`num_segparms`
BS, DXF 74

`segparms` BD*

`num_areafillparms`
BS, DXF 75

`areafillparms`
BD*

Dwg_MLINE_vertex

`parent` struct _dwg_entity_MLINE*

`vertex` 3BD, DXF 11

`vertex_direction`
3BD, DXF 12

`miter_direction`
3BD, DXF 13

`num_lines`
RC

`lines` Dwg_MLINE_line*

Dwg_OCD_Dimension

`b293` B, DXF 293

`def_pt` 2RD, DXF 10

`is_def_textloc`
B, DXF 294

`text_rotation`
BD, DXF 140

`block` H, DXF 2

`dimtofl` B, DXF 298

`dimosxd` B, DXF 291

`dimatfit` B, DXF 70

`dimtix` B, DXF 292

`dimtmove` B, DXF 71

`override_code`
RC, DXF 280

`has_arrow2`
B, DXF 295

`flip_arrow2`
B, DXF 296

`flip_arrow1`
B, DXF 297

Dwg_PARTIAL_VIEWING_INDEX_Entry

`parent` struct _dwg_object_PARTIAL_VIEWING_INDEX*

`extents_min`
3BD

`extents_max`
3BD

`object` H

Dwg_POINTCLOUDCOLORMAP_Ramp

`parent` struct _dwg_object_POINTCLOUDCOLORMAP*

`class_version`
BS, DXF 70

`num_ramps`
BL, DXF 90

`ramps` Dwg_ColorRamp*

Dwg_POINTCLOUDEX_Croppings

`parent` struct _dwg_entity_POINTCLOUDEX*

`type` BS, DXF 280

`is_inside`
B, DXF 290

`is_inverted`
B, DXF 290

`crop_plane`
3BD, DXF 13

`crop_x_dir`
3BD, DXF 213

`crop_y_dir`
3BD, DXF 213

`num_pts` BL, DXF 93

`pts` 3BD*

Dwg_POINTCLOUD_Clippings

`parent` struct _dwg_entity_POINTCLOUD*

`is_inverted`
B

`type` BS

`num_vertices`
BL

`vertices` 2RD*

`z_min` BD

`z_max` BD

Dwg_POINTCLOUD_IntensityStyle

`parent` struct _dwg_entity_POINTCLOUD*

`min_intensity`
BD

`max_intensity`
BD

`intensity_low_treshold`
BD

`intensity_high_treshold`
BD

Dwg_PROXY_LWPOLYLINE

`parent` struct _dwg_entity_PROXY_ENTITY*

`data_size`
RL

`flags` BS

`const_width`
BD

`elevation`
BD

`thickness`
BD

`extrusion`
BE

`num_points`
BL

`points` 2RD*

`num_bulges`
BL

`bulges` BD*

`num_widths`
BL

`widths` Dwg_LWPOLYLINE_width*

`unknown_1`
RC

`unknown_2`
RC

`unknown_3`
RC

Dwg_R2004_Header

`file_ID_string`
RC

`header_address`
RLx

`header_size`
RL

`x04` RL

`root_tree_node_gap`
RLd

`lowermost_left_tree_node_gap`
RLd

`lowermost_right_tree_node_gap`
RLd

`unknown_long`
RL

`last_section_id`
RL

`last_section_address`
RLL

`secondheader_address`
RLL

`numgaps` RL

`numsections`
RL

`x20` RL

`x80` RL

`x40` RL

`section_map_id`
RL

`section_map_address`
RLL

`section_info_id`
RLd

`section_array_size`
RLd

`gap_array_size`
RL

`crc32` RLx

`padding` RC

`section_type`
RL

`decomp_data_size`
RL

`comp_data_size`
RL

`compression_type`
RL

`checksum` RLx

Dwg_SECTION_geometrysettings

`parent` struct _dwg_SECTION_typesettings*

`num_geoms`
BL, DXF 90

`hexindex` BL, DXF 91

`flags` BL, DXF 92

`color` CMC, DXF 62

`layer` T, DXF 8

`ltype` T, DXF 6

`ltype_scale`
BD, DXF 40

`plotstyle`
T, DXF 1

`linewt` BLd, DXF 370

`face_transparency`
BS, DXF 70

`edge_transparency`
BS, DXF 71

`hatch_type`
BS, DXF 72

`hatch_pattern`
T, DXF 2

`hatch_angle`
BD, DXF 41

`hatch_spacing`
BD, DXF 42

`hatch_scale`
BD, DXF 43

Dwg_SECTION_typesettings

`parent` struct _dwg_object_SECTION_SETTINGS*

`type` BS, DXF 90

`generation`
BS, DXF 91

`num_sources`
BL, DXF 92

`sources` H*, DXF 330

`destblock`
H, DXF 331

`destfile` T, DXF 1

`num_geom` BL, DXF 93

`geom` Dwg_SECTION_geometrysettings*

Dwg_SPLINE_control_point

`parent` struct _dwg_entity_SPLINE*

`x` BD

`y` BD

`z` BD

`w` BD, DXF 41

Dwg_SUNSTUDY_Dates

`julian_day`
BL, DXF 90

`msecs` BL, DXF 90

Dwg_SummaryInfo_Property

`tag` T16

`value` T16

Dwg_TABLEGEOMETRY_Cell

`parent` struct _dwg_object_TABLEGEOMETRY*

`geom_data_flag`
BL, DXF 93

`width_w_gap`
BD, DXF 40

`height_w_gap`
BD, DXF 41

`tablegeometry`
H, DXF 330

`num_geometry`
BL, DXF 94

`geometry` Dwg_CellContentGeometry*

Dwg_TABLESTYLE_CellStyle

`parent` struct _dwg_object_TABLESTYLE*

`id` BL, DXF 90

`type` BL, DXF 91

`name` T, DXF 300

`cellstyle.type`
BL

`cellstyle.data_flags`
BS

`cellstyle.property_override_flags`
BL

`cellstyle.merge_flags`
BL

`cellstyle.bg_color`
CMC

`cellstyle.content_layout`
BL

`cellstyle.content_format`
Dwg_ContentFormat

`cellstyle.margin_override_flags`
BS

`cellstyle.vert_margin`
BD

`cellstyle.horiz_margin`
BD

`cellstyle.bottom_margin`
BD

`cellstyle.right_margin`
BD

`cellstyle.margin_horiz_spacing`
BD

`cellstyle.margin_vert_spacing`
BD

`cellstyle.num_borders`
BL

`cellstyle.borders`
Dwg_GridFormat*

`cellstyle.tablerow_parent`
struct _dwg_TableRow*

`cellstyle.tabledatacolumn_parent`
struct _dwg_TableDataColumn*

`cellstyle`
struct _dwg_CellStyle

Dwg_TABLESTYLE_border

`linewt` BSd

`visible` B

`color` CMC

Dwg_TABLESTYLE_rowstyles

`parent` struct _dwg_object_TABLESTYLE*

`text_style`
H, DXF 7

`text_height`
BD, DXF 140

`text_alignment`
BS, DXF 170

`text_color`
CMC, DXF 62

`fill_color`
: CMC, DXF 63

`has_bgcolor`
: B, DXF 283

`num_borders`
: BL

`borders` Dwg_TABLESTYLE_border*

`data_type`
: BL

`unit_type`
: BL

`format_string`
: TU

Dwg_TABLE_AttrDef

`parent` struct _dwg_TABLE_Cell*

`attdef` H, DXF 331

`index` BS, DXF 179

`text` T, DXF 300

Dwg_TABLE_BreakHeight

`parent` struct _dwg_entity_TABLE*

`position` 3BD

`height` BD

`flag` BL

Dwg_TABLE_BreakRow

`parent` struct _dwg_entity_TABLE*

`position` 3BD

`start` BL

`end` BL

Dwg_TABLE_Cell

`parent` struct _dwg_entity_TABLE*

`type` BS, DXF 171

`flags` RC, DXF 172

`is_merged_value`
: B, DXF 173

`is_autofit_flag`
B, DXF 174

`merged_width_flag`
BL, DXF 175

`merged_height_flag`
BL, DXF 176

`rotation` BD, DXF 145

`text_value`
T, DXF 1

`text_style`
H, DXF 7

`block_handle`
H, DXF 340

`block_scale`
BD, DXF 144

`additional_data_flag`
B

`cell_flag_override`
BL, DXF 177

`virtual_edge_flag`
RC, DXF 178

`cell_alignment`
RS, DXF 170

`bg_fill_none`
B, DXF 283

`bg_color` CMC, DXF 63

`content_color`
CMC, DXF 64

`text_height`
BD, DXF 140

`top_grid_color`
CMC, DXF 69

`top_grid_linewt`
BS, DXF 279

`top_visibility`
BS, DXF 289

`right_grid_color`
CMC, DXF 65

`right_grid_linewt`
BS, DXF 275

`right_visibility`
BS, DXF 285

`bottom_grid_color`
CMC, DXF 66

`bottom_grid_linewt`
BS, DXF 276

`bottom_visibility`
BS, DXF 286

`left_grid_color`
CMC, DXF 68

`left_grid_linewt`
BS, DXF 278

`left_visibility`
BS, DXF 288

`unknown` BL

`value` Dwg_TABLE_value

`num_attr_defs`
BL

`attr_defs`
Dwg_TABLE_AttrDef*

Dwg_TABLE_CustomDataItem

`name` T, DXF 300

`value` Dwg_TABLE_value

`cell_parent`
struct _dwg_TableCell*

`row_parent`
struct _dwg_TableRow*

Dwg_TABLE_value

`flags` BL

`format_flags`
BL

`data_type`
BL, DXF 90

`data_size`
BL, DXF 92

`data_long`
BL, DXF 91

`data_double`
BD, DXF 140

`data_string`
T, DXF 1

`data_date`
TF

`data_point`
2RD, DXF 11

`data_3dpoint`
3RD, DXF 11

`data_handle`
H

`unit_type`
BL, DXF 94

`format_string`
T, DXF 300

`value_string`
T, DXF 302

Dwg_TableCell

`flag` BL, DXF 90

`tooltip` TV, DXF 300

`customdata`
BL, DXF 91

`num_customdata_items`
BL, DXF 90

`customdata_items`
Dwg_TABLE_CustomDataItem*

`has_linked_data`
BL

`data_link`
H

`num_rows` BL

`num_cols` BL

`unknown` BL

`num_cell_contents`
BL

`cell_contents`
: Dwg_TableCellContent*

`style_id` BL

`has_geom_data`
: BL

`geom_data_flag`
: BL

`width_w_gap`
: BD

`height_w_gap`
: BD

`tablegeometry`
: H

`num_geometry`
: BL

`geometry` Dwg_CellContentGeometry*

`style_parent`
: struct _dwg_CellStyle*

`row_parent`
: struct _dwg_TableRow*

Dwg_TableCellContent

`parent` struct _dwg_TableCell*

`type` BL, DXF 90

`value` Dwg_TABLE_value

`handle` H, DXF 340

`num_attrs`
: BL, DXF 91

`attrs` Dwg_TableCellContent_Attr*

`has_content_format_overrides`
: BS

`content_format`
: Dwg_ContentFormat

Dwg_TableCellContent_Attr

`parent` struct _dwg_TableCellContent*

`attdef` H, DXF 330

`value` TV, DXF 301

`index` BL, DXF 92

Dwg_TableDataColumn

`parent` struct _dwg_LinkedTableData*

`name` T, DXF 300

`custom_data`
BL, DXF 91

`cellstyle`
Dwg_CellStyle

`cellstyle_id`
BL

`width` BL

Dwg_TableRow

`parent` struct _dwg_LinkedTableData*

`num_cells`
BL

`cells` Dwg_TableCell*

`custom_data`
BL

`num_customdata_items`
BL

`customdata_items`
Dwg_TABLE_CustomDataItem*

`cellstyle`
Dwg_CellStyle

`style_id` BL

`height` BL

Dwg_UCS_orthopts

`parent` struct _dwg_object_UCS*

`type` BS, DXF 71

`pt` 3BD, DXF 13

Dwg_VALUEPARAM

`parent` struct _dwg_object_object*

`class_version`
BL

`name` T

`unit_type`
: BL

`num_vars` BL

`vars` Dwg_VALUEPARAM_vars*

`controlled_objdep`
: H

Dwg_VALUEPARAM_vars

`value` Dwg_EvalVariant

`handle` H

Dwg_MLEADER_Content

`txt` Dwg_MLEADER_Content_MText

`blk` Dwg_MLEADER_Content_Block

Common Entity fields

`__iterator`
: BL

`color` CMC, DXF 62

`color_r11`
: RCd, DXF 62

`dwg` struct _dwg_struct*

`edge_visualstyle`
: H, DXF 348

`eed` Dwg_Eed*

`elevation_r11`
: RD, DXF 38

`entmode` BB, DXF 67

`extra_r11`
: RS

`face_visualstyle`
: H, DXF 348

`flag_r11` RC

`full_visualstyle`
: H, DXF 348

`has_ds_data`
: B

`has_edge_visualstyle`
: B

`has_face_visualstyle`
: B

`has_full_visualstyle`
: B

`invisible`
: BS, DXF 60

`is_xdic_missing`
: B

`isbylayerlt`
: B

`layer` H, DXF 8

`linewt` RC, DXF 370

`ltype` H, DXF 6

`ltype_flags`
: BB

`ltype_scale`
: BD, DXF 48

`material` H, DXF 347

`material_flags`
: BB

`next_entity`
: H

`nolinks` B

`num_eed` BL

`num_reactors`
: BL

`objid` BL

`opts_r11` RS

`ownerhandle`
: H, DXF 330

`plotstyle`
: H, DXF 390

`plotstyle_flags`
: BB

`prev_entity`
: H

`preview` TF, DXF 310

`preview_exists`
: B

`preview_is_proxy`
: B

`preview_size`
: BLL, DXF 160

`reactors` H*, DXF 330

`shadow` H

`shadow_flags`
: RC, DXF 284

`thickness_r11`
: RD, DXF 39

`viewport` H

`xdicobjhandle`
: H, DXF 360

Common Object fields

`dwg` struct _dwg_struct*

`eed` Dwg_Eed*

`handleref`
: Dwg_Handle*

`has_ds_data`
: B

`is_xdic_missing`
: B

`num_eed` BL

`num_reactors`
: BL

`objid` BL

`ownerhandle`
: H, DXF 330

`reactors` H*, DXF 330

`xdicobjhandle`
: H, DXF 360

5 Sections

The r2000 format (used for r13-r2000) knows the following 6 sections:

HEADER CLASSES HANDLES OBJFREESPACE + 2NDHEADER MEASUREMENT/TEMPLATE AUXHEADER (only r2000)

The r2004 and r2007 format (used for r2004-r2018) knows the following sections:

R2004_Header UNKNOWN SUMMARYINFO PREVIEW VBAPROJECT APPINFO APPINFOHISTORY FILEDEPLIST ACDS REVHISTORY SECURITY OBJECTS OBJFREESPACE TEMPLATE HANDLES CLASSES AUXHEADER HEADER SIGNATURE INFO SYSTEM_MAP

The old pre-R13 formats (from r1.1 to r11) have those sections, with the tables interleaved into the HEADER.

HEADER ENTITIES BLOCKS entities EXTRAS entities

With (from r1.1 to r10) those 5 Section 5.22 [Tables], page 268:

BLOCKS LAYER STYLE LTYPE VIEW

With r11 came the additional tables:

UCS VPORT APPID DIMSTYLE VX

But we convert them internally to r2004 table record and table control objects. See Section 5.22 [Tables], page 268.

5.1 HEADER Section

See Chapter 4 [Objects], page 8.

5.2 OBJECTS Section

The OBJECTS Section is usually split up into multiple pages (separate sections of type AcDbObjects) and contains all entities and objects. It is indexed by Section 5.4 [HANDLES], page 266.

See Chapter 4 [Objects], page 8.

5.3 CLASSES Section

The **Classes** Section contains the basic info for all dynamically loaded types for entities and objects. Its types start with 500, and are variable. An entity which has no class loaded is displayed as proxy.

LibreDWG contains support for many classes, but not all. See `src/classes.inc` and `src/classes.c`. We define a stability for each class, one of stable, unstable, debugging and unhandled.

Objects in **stable** classes are treated as the fixed-type objects with full support. Changes are treated as API breaking.

Objects in **unstable** classes are sometimes written to DXF or JSON, but not to DWG. Changes are not treated as API breaking. Usually such objects are converted to UNKNOWN_OBJ or UNKNOWN_ENT objects, and when written to DWG converted to PLACEHOLDER, DUMMY or POINT objects with EED pointing to the original class

and content. Only when rewriting from-to the very same version with the full known unknown_bits blob (e.g. dwgrewrite or json) such classes can persist as such.

Objects in **debugging** classes are only handled with the developer `configure --enable-debug` flag, otherwise ignored. See unstable above.

Objects in **undhandled** classes are always ignored. There are no fields known, only its type.

5.4 HANDLES Section

The Handles section contains a sorted list of all object handles and its position in the Objects stream. All values are stored relatively, as offsets. Handles only increase and can contain holews when an object is deleted, offsets can jump back also.

5.5 R2004_Header

The R2004_Header section at fixed position `0x100` in the DWG contains some meta-data for r2004 sections to find the two important sections INFO and SYSTEM_MAP.

5.6 UNKNOWN Section

The content of the UNKNOWN section with type 0 is unknown and does not always exist.

5.7 SummaryInfo

All Section SummaryInfo fields:

`TITLE` TU16, DXF 1

`SUBJECT` TU16, DXF 1

`AUTHOR` TU16, DXF 1

`KEYWORDS` TU16, DXF 1

`COMMENTS` TU16, DXF 1

`LASTSAVEDBY`
TU16, DXF 1

`REVISIONNUMBER`
TU16, DXF 1

`HYPERLINKBASE`
TU16, DXF 1

`TDINDWG` TIMERLL

`TDCREATE` TIMERLL

`TDUPDATE` TIMERLL

`num_props`
RS

`props` Dwg_SummaryInfo_Property*

`unknown1` RL

`unknown2` RL

See [Dwg_SummaryInfo_Property], page 254,

5.8 Preview

The optional Preview section contains the thumbnail stream of BMP, WMF or PNG data of the drawing. Note that blocks or proxy objects can also contain its own preview fields. The program **dwgbmp** can extract the preview image from this section.

5.9 VBAProject

5.10 AppInfo

Which product and version exactly created that DWG.

5.11 AppInfoHistory

5.12 FileDepList

Features and File Dependencies. Image files, fonts, xrefs, plotconfigs.

5.13 AcDS

The AcDsPrototype_1b DataStorage, used mostly for binary ACIS blobs, embedded fonts, ...

5.14 RevHistory

Revision History

5.15 Security

Password Info

5.16 ObjFreeSpace

Some Objects meta-data

5.17 Template

Contains one Measurement Header variable.

5.18 AuxHeader

In case the original Header gets lost.

5.19 Signature

5.20 INFO

The info of all used sections.

5.21 SYSTEM_MAP

The map of all used sections and its chunked pages.

5.22 Tables

The old pre-R13 formats (from r1.1 to r11) have no objects and no sections, just tables, which we store in the sections indexed by the enum **Dwg_Section_Type_r11**.

BLOCKS LAYER STYLE LTYPE VIEW

With r11 came the additional tables:

UCS VPORT APPID DIMSTYLE VX

Since r13 all those tables are stored as table control objects and tablerecord objects.

From pre-r13 DWG's these tables are imported as old r11 sections and as new CONTROL objects, so that all entities are accessible via the single BLOCK_CONTROL.model_space -> BLOCK_HEADER.entities iterator, all layers via the LAYER_CONTROL.entries -> LAYER objects, and so on. All blocks are accessed via all other model_space BLOCK_HEADER's, `get_first_owned_block (BLOCK_HEADER)`. Each CONTROL object holds a list of all table records, i.e. entries. Each table record entry has a name and other common table fields.

The VX table (also vport_entity table) is a special table until r2004 for VIEWPORT entities, which are VPORT extensions, esp. for Paper Space, but have no old-style table records. So VX_CONTROL contains a list of matching VX_TABLE_RECORD's, which are pointed to by VIEWPORT.vport_entity_header handles. Likewise the VX_TABLE_RECORD object links in its viewport field back to the VIEWPORT. The VX table is not represented in DXF, on DXF import we need to create them with the VIEWPORT entities.

5.23 EXTRAS entities section

Before R13 we had no objects, just the 5-10 tables, and the entities divided into 3 sections. The entities, the blocks and the extras. Blocks just contains the entities from each BLOCK to the ENDBLK entity.

Extras contain entities which had no room in the original section, e.g. when closing a polyline, which needs one additional byte. Thus the original entity is replaced by an undocumented JUMP entity, which gives the offset into the EXTRAS section, until a JUMP in the EXTRAS jumps back to the next original entity.

See e.g. `r10/entities.dwg`

The JUMP [31] replaces the POLYLINE_2D entity which got later closed. An open POLYLINE_2D needs size 8, but closed 9 bytes. Hence they added the new closed replacement to the extras section at offset 0, index [66], and added another JUMP [67] back to the next original entity. Here it jumps back to the VERTEX_2D [32] at offset 0x84f.

```
type: 18 [RCd]
Add entity JUMP [31] Decode entity JUMP
```

```
==========================
Entity number: 31, Type: 18, Addr: 847
flag_r11: 0x0 [RC 0]
size: 8 [RS]
jump_address_raw: 0x80000000 [RLx 0]
jump_entity_section: DWG_EXTRA_SECTION
jump_address: 0x0

type: 20 [RCd]
Add entity VERTEX_2D [32] Decode entity VERTEX_2D
==========================
Entity number: 32, Type: 20, Addr: 84f
flag_r11: 0x0 [RC 0]
size: 24 [RS]
layer: 0 [H(RSd) 8]
opts_r11: 0x0 [RSx 0]
point: (5.500000, 7.500000) [2RD 10]

....

extras entities: (0x10c0-0x10d1 (0), size 17)
========================================
type: 19 [RCd]
Add entity POLYLINE_2D [66] Decode entity POLYLINE_2D
==========================
Entity number: 66, Type: 19, Addr: 10c0
flag_r11: 0x80 [RC 0]
          HAS_ATTRIBS(0x80)
size: 9 [RS]
layer: 0 [H(RSd) 8]
opts_r11: 0x1 [RSx 0]
          HAS_FLAG(0x1)
flag: 0x1 [RC 70]
      CLOSED(0x1)

type: 18 [RCd]
Add entity JUMP [67] Decode entity JUMP
==========================
Entity number: 67, Type: 18, Addr: 10c9
flag_r11: 0x0 [RC 0]
size: 8 [RS]
jump_address_raw: 0x84f [RLx 0]
jump_entity_section: DWG_ENTITY_SECTION
jump_address: 0x84f
========================================
extras entities: end
```

The entity iterator knows about these jumps. The DXF structure resolves those jumps, and inserts the replaced entities.

6 Structures

6.1 EED

"Extended Entity Data" (EED) may be optionally attached to each object. They consist of a handle to the registered APPID, and a list of typed data. Each block is preceded with a size, the processing stops with size 0.

Internally libredwg stores each eed line as an array of num_eed structs. If the size > 0, then new block starts with a handle, an optional raw string (when reading from a DWG), and a number of typed data entries. Only the first eed struct of each block has a size, all subsequent eed structs have size 0.

Example:

```
EED[0] size: 109 [BS]
EED[0] handle: 5.2.762
EED[0] code: 70 [RC] short: 2 [RS]
EED[1] code: 70 [RC] short: 0 [RS]
EED[2] code: 70 [RC] short: 0 [RS]
EED[3] code: 11 [RC] 3dpoint: (0.000000, 0.000000, 0.000000) [3RD]
EED[4] code: 11 [RC] 3dpoint: (1.000000, 0.000000, 0.000000) [3RD]
EED[5] code: 11 [RC] 3dpoint: (0.000000, 1.000000, 0.000000) [3RD]
EED[6] code: 11 [RC] 3dpoint: (0.000000, 0.000000, 1.000000) [3RD]
EED[7] size: 6 [BS]
EED[7] handle: 5.2.763
EED[7] code: 70 [RC] short: 0 [RS]
EED[8] code: 70 [RC] short: 0 [RS]
EED[9] size: 23 [BS]
EED[9] handle: 5.1.12
EED[9] code: 0 [RC] string: "RTMaterial" len=10 cp=30
EED[10] code: 5 [RC] entity: 0x6507000000000000 [RLL]
- size: 0 [BS]
```

These 10 num_eed structs consist of 3 blocks with 3 size and handle entries. EED[0] starts with size 109, the handle pointing to object 762, 3 shorts and 4 points. The next block at EED[7] has size 6, the handle pointing to object 763 and 2 shorts. The last block at EED[9] has size 9, the handle pointing to object 12 (the APPID.ACAD application) and a string and an entity reference. The size is calculated by the needed room for all data code + values, without the handle. E.g. EED[7] size: 6 is 1 + 2 for EED[7] RC + RS, and 1 + 2 for EED[8] RC + RS.

Each data block consists of a RC code, and a variable value. A string may be a an old pre-r2007 ASCII string with a RC length (max 255 chars), a codepage and the string. Or a r2007+ wide string with a RS length (max 32767 chars) and a UCS-2 wide string.

decode stores both, the raw data, and the structured data. in_dxf just the data. encode prefers raw over the data.

6.2 XDATA

XRECORD XDATA are very similar to the EED array, but internally it is a single linked-list, consisting of something like the EED data code + value pairs. There's only one size, xdata_size, and only one handle to the APPID, which handles this XRECORD XDATA.

7 Functions

You can use LibreDWG immediately upon loading, without any particular initialization. Only when using some see Section 7.4 [dynapi], page 277, functions you might need to initialize the version via `dwg_api_init_version(&dwg)`, when you need other formats than r2000 and you call an API function which does not store the version internally. Most do. This limitation will soon be fixed.

You usually use one set of functions - either decoding or encoding - at a time. All functions use the common data types (see Chapter 3 [Types], page 6). All functions return an error code, and the high-level functions for multiple objects add the error bitmask, which is sorted by severity. When the error exceeds DWG_ERR_CRITICAL, processing is stopped.

The new see Section 7.4 [dynapi], page 277, has dynamic get and set functions for all objects and its fields. You can get and set a property value from any object pointer by the object name and the field name.

7.1 Decoding

The highest level function for decoding a file is `dwg_read_file`.

`int dwg_read_file (char *filename, Dwg_Data *dwg)` [Function]
Open *filename* and decode it, saving information into *dwg*. Return 0 if successful.

You can then iterate over the entities in model space or paper space via two ways:

1. by using the `dwg.h` data structures. Via `dwg->object[0]`, which is of type `Dwg_Object_BLOCK_CONTROL`, and a custom `void process_BLOCK_HEADER(Dwg_Object_Ref* ref)`:

```
Dwg_Object_BLOCK_CONTROL* block_control = dwg->block_control;
// first all entities in the model space
process_BLOCK_HEADER (dwg->header_vars.BLOCK_RECORD_MSPACE);
// then all entities in the blocks
for (i=0; i < block_control->num_entries; i++)
  {
    process_BLOCK_HEADER (block_control->entries[i]);
  }
// and last all entities in the paper space
process_BLOCK_HEADER (dwg->header_vars.BLOCK_RECORD_PSPACE);
```

or 2. by using the API functions from `dwg_api.h`:

```
Dwg_Object_BLOCK_CONTROL* block_control = dwg_block_control(dwg);
process_BLOCK_HEADER (dwg_model_space_ref(dwg));
for (i=0; i < block_control->num_entries; i++)
  {
    process_BLOCK_HEADER (block_control->entries[i]);
  }
process_BLOCK_HEADER (dwg_paper_space_ref(dwg));
```

and inside the `process_BLOCK_HEADER` function, you iterate over the entities from the block_header via:

```
Dwg_Object* obj = get_first_owned_entity(ref->obj);
while (obj)
  {
    process_object(obj);
    obj = get_next_owned_entity(ref->obj, obj);
  }
```

where `process_object` checks the type of each entity under the *Dwg_Object* obj*.

For each entity or object type (i.e. a non-graphical dwg object, also tables) there also exist the simple and expensive `dwg_getall_ENTITY` and `dwg_getall_OBJECT` functions:

`int dwg_getall_ENTITY (Dwg_Object_Ref *block_header_ref)` [Function]
Return a malloc'ed NULL-terminated array of all such entities for Model Space, Paper Space or an individual block.

`int dwg_getall_OBJECT (Dwg_Data *dwg)` [Function]
Return a malloc'ed NULL-terminated array of all such DWG objects.

The decoder is driven by the fields definition in the `src/dwg.spec`, which adds each field to the object. This is done in the `src/decode.c` or `src/decode_r2007.c`.

`int dwg_decode_OBJECT (Bit_Chain *dat, Dwg_Object *obj)` [Function]
Sets the fields for the object from the DWG bitstream.

Note: Pre-R13 DWG's do contain all deleted entities, which e.g. where moved into a BLOCK. Those entities do have a type > 127. You need to filter them out by yourself, when processing the DWG.

7.2 Encoding

Encoding DWG files, i.e. DWG write support, can be disabled via `./configure --disable-write`. The default formats and only useful ones are currently r1.1 - r2000. Experimentally work is ongoing for the r2004 format, which is also used for r2010, r2013, and r2018. The r2007 format version is not covered yet. The pre-r13 formats are much simpler and can be written, but need some hand-holding and manual conversions when converting from newer formats still.

See `src/in_dxf.c` for a high-level usage example. The default codepage is Latin-1, 30.

The highest level function for encoding a bitstream to a file is `dwg_write_file`, which dumps the dwg to a file.

`int dwg_write_file (char *filename, Dwg_Data *dwg)` [Function]
Open *filename* and write the *dwg* to it. Return 0 if successful.

See Section 7.3 [add api], page 275, for:

`Dwg_Data* dwg_add_Document (const Dwg_Version_Type version,` [Function]
const int *imperial*, const int *loglevel*) Creates an initial template dwg structure in memory, suitable to be written to a DWG or DXF file, without any additional table

records or entities. Creates ModelSpace, PaperSpace and most Tables and basic Dictionaries.

and how to add entities and objects from scratch.

Low level-functions:

`int dwg_add_object (Dwg_Data *dwg)` [Function]
Adds a new uninitialized object to the *dwg*->*object*[] array. Return 0 or -1 if successful, otherwise DWG_ERR_OUTOFMEM. -1 is the array was re-allocated.

Then for each object or entity type there is a

`int dwg_setup_<OBJECT> (Dwg_Object *obj)` [Function]
Initializes an object for the given OBJECT or ENTITY type, with all fields being zero'ed. This does not initialize the *obj* size, type, address, handlestream_size, bitsize fields.

The encoder is driven by the fields definition in the `src/dwg.spec` and the generated `src/dynapi.c`, which adds each field to the object. This is done by `src/encode.c` or any `src/in_*.c` import module.

`int dwg_encode_<OBJECT> (Bit_Chain *dat, Dwg_Object *obj)` [Function]
Encodes the DWG bitstream from the fields of the object.

The iterator is similar to above, but you want to encode all data structures, not just the entities. But note that you need many helper functions, such as the Section 7.4 [dynapi], page 277, to create all needed sections to store a DWG if you didn't read a DWG into the right a *Dwg_Data** *dwg* struct already. This is especially important when importing from DXF or from an earlier or later DWG version.

7.3 add api

The add api functions are useful for CAD programs which want to write DWG. All the other API's are mostly to convert from and to DWG, so the main structures and links already do exist. With the add api you can easily create an empty DWG from scratch, add table entries (into fixed Tables or variables Dictionaries), and add entities. To set more entity fields use the Section 7.4 [dynapi], page 277.

For each almost each entity and table exists a function at to add it, with arguments to initialize some fields as in the VBA object model. The other objects are either created automatically, or handled separately.

All BITCODE_T strings are encoded as UTF-8, as with the dynapi. See Section 7.5 [strings], page 278. Most names are copied, since most names are considered to be constant. If not, you need to free them by yourself. Exceptions are dxfname (there exists a separate dxfname_u variant), the VX name, which does not exists anymore since r2000.

A very simple example using the add API is the example program See [dwgadd], page 284.

`Dwg_Data dwg_add_Document (const Dwg_Version_Type version, const int imperial, const int loglevel))` [Function]
Creates an initial template dwg structure in memory, suitable to be written to a DWG or DXF file, without any additional table records or entities. Creates ModelSpace, PaperSpace and most Tables and basic Dictionaries.

When writing DWG, a *version* of R_2000 is recommended, only R_1_2 - R_2000 are supported yet. For DXF you can try all versions R_13 - R_2018.

For each OBJECT and ENTITY type there exists a specific `dwg_add_<OBJECT>` function, which takes the owner and some default arguments. Entities are normally added to a block header, like modelspace, paperspace or any block. Objects are normally added to the dwg, or to some other object or entity. E.g.

`Dwg_Entity_LINE *line = dwg_add_LINE` [Function]
`(Dwg_Object_BLOCK_HEADER *modelspace,`
dwg_point_3d *start_pt, dwg_point_3d *end_pt)

`Dwg_Entity_TEXT* dwg_add_TEXT (Dwg_Object_BLOCK_HEADER` [Function]
`*restrict blkhdr, const char* restrict text_value, const dwg_point_3d *restrict ins_pt, const double height)`
Adds a TEXT entity to the ModelSpace, PaperSpace or a Block. Entity specific arguments are here the text, the point (as pointer to the struct of 3 doubles), and the text height.

`Dwg_Object_LAYER *layer = dwg_add_LAYER (Dwg_Data *dwg,` [Function]
`const char *name)`
Adds a new layer the Layer Table, i.e. creates the new LAYER object, and adds it to LAYER_CONTROL object, the list of layers.

Names and strings are encoded as UTF-8 and will be translated to type BITCODE_T (i.e. versions specific TU or TV types, either UCS-2 unicode or single-byte codepage) internally, as with the **dynapi**. Only internally you will have to deal with 2 different DWG text representations: UCS-2 since r2007, single-byte before. see Section 7.5 [strings], page 278.

To understand the object model for the add API see some VBA Object model documentation, such as e.g. `http://entercad.ru/acadauto.en/`.

The new add API mostly handles the direct `Dwg_Entity_ENTITY` structs, not all the generic `Dwg_Object` structs. Thus you can access the object specific fields directly, the common fields, not so easily.

The DWG Document consists of 3 basic entity containers `ModelSpace`, `PaperSpace` and `Blocks`, plus `Tables` (`Layers`, `Linetypes`, ...), `Dictionaries` as generic replacements of `Tables` with a root Dictionary, the `NOD` ("Named Object Dictionary"), and more support objects and complex entity groups.

Helper functions:

`dwg_add_u8_input (Dwg_Data *restrict dwg, const char` [Function]
`*restrict u8str)`
Convert UTF-8 strings to BITCODE_T fields. Returns a copy of the string. All external API's only deal with UTF-8 strings.

7.4 dynapi

The new dynapi replaced the old dwg_api functions to access each object field. The old dwg_api functions were deprecated, and need to be re-enabled by defining `CFLAGS="-DUSE_DEPRECATED_API"`. See see Chapter 4 [Objects], page 8, for an description of each object and its fields..

For each of header, entity, common or subclass there is a function to get and set the value of any type, or converted utf8 string.

`bool dwg_dynapi_entity_value (void *entity, const char *dxfname, const char *fieldname, void *out, Dwg_DYNAPI_field *fp)` [Function]

entity is of type `dwg_ent_generic`, that is the pointer to the object specific struct. *dxfname* is the dxfname of the object, *fieldname* is the field or property name of the field to be read from, **out* the result pointer and the optional **fp* is filled by the information for this field.

`bool dwg_dynapi_common_value (void *entity, const char *fieldname, void *out, Dwg_DYNAPI_field *fp)` [Function]

This accesses the common `Dwg_Object_Object*` or `Dwg_Object_Entity*` fields.

`bool dwg_dynapi_header_value (void *dwg, const char *fieldname, void *out, Dwg_DYNAPI_field *fp)` [Function]

This accesses the Header (or sometimes also called Database) fields.

`bool dwg_dynapi_subclass_value (void *ptr, const char *subclass, const char *fieldname, void *out, Dwg_DYNAPI_field *fp)` [Function]

This accesses a subclass, a structure within the object.

The utf8text functions convert version-specific text strings to UTF-8 strings. Internally the dwg stores strings as TU (unicode) or TV (single-byte codepage). The API treats all strings as UTF-8, as with JSON, DXF or the add API.

`bool dwg_dynapi_entity_utf8text (void *entity, const char *dxfname, const char *fieldname, char *textp, int *isnewp, Dwg_DYNAPI_field *fp)` [Function]

isnewp is set to 1 f the string is a fresh copy, for unicode strings.

`bool dwg_dynapi_common_utf8text (void *entity, const char *fieldname, char *textp, int *isnewp, Dwg_DYNAPI_field *fp)` [Function]

`bool dwg_dynapi_header_utf8text (void *dwg, const char *fieldname, char *textp, int *isnewp, Dwg_DYNAPI_field *fp)` [Function]

This accesses the Header (or sometimes also called Database) fields.

`bool dwg_dynapi_subclass_utf8text (void *ptr, const char *subclass, const char *fieldname, char *textp, int *isnewp, Dwg_DYNAPI_field *fp)` [Function]

This accesses a subclass, a structure within the object.

The setters don't differentiate between common values and strings.

`bool dwg_dynapi_entity_set_value (dwg_ent_generic *_obj, const char *fieldname, const void *value, const bool is_utf8)` [Function]

Sets the ENTITY.fieldname to a value. A malloc'ed struct is passed by ptr, not by the content. A non-malloc'ed struct is set by content. Arrays or strings must be malloced before. We just set the new pointer, the old value will be freed. If is_utf8 is set, the given value is a UTF-8 string, and will be converted to TV or TU

`bool dwg_dynapi_header_set_value (Dwg_Data *dwg, const char *fieldname, const void *value, const bool is_utf8)` [Function]

`bool dwg_dynapi_common_set_value (dwg_ent_generic *_obj, const char *fieldname, const void *value, const bool is_utf8)` [Function]

See the sourcecode of the importers or programs for the usage of the API's.

7.5 strings

Internally the DWG consists of multiple different string formats, see Chapter 3 [Types], page 6. The most important are `BITCODE_TV` (i.e. `char*`) encoded according to `dwg->header.codepage`, and `BITCODE_TU` (i.e. `wchar_t` on Windows, UCS-2).

Externally most functions get and set strings as UTF-8, as in DXF or JSON.

Before r2007 DWG's the TV and T strings are encoded in its codepage, and converted from and to their proper codepage to UTF-8 or \U+XXXX.

To encode unicode characters special \U+XXXX sequences are used, and pre-r2007 DXF MIF \M+nXXXX sequences, where n is one of the asian wide-character codepages 932 (Japanese), 950 (trad. Chinese), 949 (Korean Wansung), 1361 (Johab), and 936 (simplified Chinese).

On DWG's r2007 and later most strings (T and TU) are encoded in the Microsoft specific two-byte UCS-2 Unicode encoding, without proper support for surrogate pairs and the upper planes (i.e. emojis).

Fixed TF strings are not encoded and have a length stored also. Normal strings are all zero-delimited. EED and XDATA strings do have a length though, but have length limitations.

Strings in DXF and JSON also have quoting rules for special characters, like \r, \n, \" and so on.

Transformations:

DWG to DWG: decode reads the T and TU strings in its natural format into the field. encode translate it to TV or TU. encode needs `header.from_version` and how it was read, from DWG or from an importer (in_dxf or in_json) or the Section 7.3 [add api], page 275, (DWG_OPTS_IN).

DXF/JSON to DWG: in_dxf/json keeps the T and TU strings as TV. encode to <r2007 keeps it as TV, r2007+ translates it to TU. Unicode is encoded as \U+XXXX. It sets DWG_OPTS_IN.

DWG to DXF/JSON: decode keeps the T and TU strings as TV or TU. out_dxf/json translate them to TV or UTF-8 and quotes them via \U+XXXX.

add api to DWG/DXF: add reads strings as UTF-8, and encodes it from UTF-8 to TV or TU. (TU not yet, as we don't encode r2004+ yet). add sets DWG_OPTS_IN.

7.6 Other Formats

7.6.1 DXF

We can write ASCII DXF files in various versions, with much more data than other free DWG libraries, but not as stable as the unfree Teigha library yet. AutoCAD ® fails to import some of our files still (~10% failure rate).

Options: `--minimal` (see `dwg2dxf` or `dwgread`) creates only a short header with a ACADVER and HANDSEED element, and the entities, without any subclass markers, reactors or handles.

Support for the different r1.x DXF format is not planned.

Reading DXF works for most objects. Converting a minimal DXF to DWG may fail, and needs some fixes.

7.6.2 DXFB

We can almost write Binary DXF files in various versions.

Reading DXFB is under construction and un-tested.

7.6.3 JSON

We write to and read back from our own JSON format, which is a readable 1:1 mapping of the DWG structures, and carries much more information than the DXF format. The idea is to dump a DWG to JSON and filter/query or postprocess it with more powerful JSON query tools such as **jq** (`https://stedolan.github.io/jq/`), and optionally import it back in. See Chapter 9 [Programs: dwgfilter], page 282. JSON is much better structured than DXF.

The current first level objects are all the section names, like "HEADER", "CLASSES", "OBJECTS". For more see the specs.

Note that for the versions before R13 we convert all tables to table control and record objects internally, so that we can use our iterators needed for DXF support. This does not reflect the internal DWG structure. See Section 5.22 [Tables], page 268.

7.6.4 GeoJSON

`dwgread` supports writing to the GeoJSON format as specified at `http://geojson.org/geojson-spec.html`. See `dwgread` with the `--fmt GeoJSON` option.

We write in the RFC7946 format, the new GeoJSON format since 2016, which means smaller, less precision, and normalized polygons with proper right-hande rule orientation.

We write all coordinates as [x, y], not [y, x]. z-coordinates are optional, and only written if not 0.0. Colors are either written as palette index as integer if not 256 (ByLayer), or as TrueColor RGB hex string values for all r2004+ DWG's.

Missing entities: No 3D entities, HATCH by definition. ELLIPSE, polyline bulges would need segmention into line segments. MLINE, SPLINE, MINSERT, SOLID, TRACE, RAY(?), XLINE(?)

Due to implementation quirks with ending commas in JSON, we mostly add an empty dummy feature at the very end, with null properties and null geometry.

8 Errors

LibreDWG is mostly a library, and as such collects error codes from the highest level function down to the lowest level functions. The error codes are sorted by severity, and only if the error exceeds *DWG_ERR_CRITICAL*, i.e. `DWG_ERR_CLASSESNOTFOUND`, processing is stopped.

All error bitmasks are collected during read or write and returned at the end.

`DWG_ERR_WRONGCRC`
: 1

`DWG_ERR_NOTYETSUPPORTED`
: 2

`DWG_ERR_UNHANDLEDCLASS`
: 4

`DWG_ERR_INVALIDTYPE`
: 8

`DWG_ERR_INVALIDHANDLE`
: 16

`DWG_ERR_INVALIDEED`
: 32

`DWG_ERR_VALUEOUTOFBOUNDS`
: 64

`DWG_ERR_CLASSESNOTFOUND`
: 128 = `DWG_ERR_CRITICAL`

`DWG_ERR_SECTIONNOTFOUND`
: 256

`DWG_ERR_PAGENOTFOUND`
: 512

`DWG_ERR_INTERNALERROR`
: 1024

`DWG_ERR_INVALIDDWG`
: 2048

`DWG_ERR_IOERROR`
: 4096

`DWG_ERR_OUTOFMEM`
: 8192

Additionally, verbose warning and error messages are printed to stderr.

Unhandled class and Invalid type errors of objects are not severe. A DWG format can store a serialization of many third party classes and objects, and thus we will never be able read all possible types. Unknown types are just stored as binary blob without any DXF codes.

9 Programs

LibreDWG installs some binary programs to read or write DWG files.

`dwgread`

This reads a DWG file, and optionally converts its content to some output formats: JSON, Minimal JSON, GeoJSON, DXF, DXFB (i.e. Binary DXF), SVG.

`dwgread [OPTION]... DWGFILE`

Options:

-v[0-9], –verbose [0-9] verbosity

-O fmt, –format fmt fmt: JSON, DXF, DXFB, minJSON, GeoJSON.

More planned formats: YAML, XML/OGR, GPX, SVG, PS.

-o outfile, –file outfile also defines the output fmt. Default: stdout

–help display this help and exit

–version output version information and exit

minJSON is minimized JSON, without whitespace.

`dwgwrite`

Create a DWG from a given input file (see Section 7.6.1 [DXF], page 279, see Section 7.6.2 [DXFB], page 279, see Section 7.6.3 [JSON], page 279), optionally via `--as=rNNNN` as another version. For now can only create r1.2-r2000 DWG files.

`dxfwrite`

Create a DXF from a given input file (`DWG`, see Section 7.6.1 [DXF], page 279, see Section 7.6.2 [DXFB], page 279, see Section 7.6.3 [JSON], page 279), optionally via `--as=rNNNN` as another version. Experimental. Supports the same options as `dwg2dxf`.

`dwg2dxf`

Converts DWG files to DXF, optionally via `--as=rNNNN` as another version, an earlier or later version, or via `-m` or `--minimal` as a minimal DXF version, skipping most headers vars, classes, tables and objects.

`--binary` as a binary DXF file, with full precision, under construction.

The DXF files are created in the current directory and not overwritten, unless the option `--overwrite` or `-y` is given.

`dxf2dwg`

Converts DXF (or Binary DXF) files to DWG, optionally via `--as=rVER` as another version, an earlier or later version.

The DWG files are created in the current directory and not overwritten, unless the option `--overwrite` or `-y` is given.

This program is experimental and AutoCAD ® may fail to import it. For now can only create r1.2-r2000 DWG. The default is writing as r2000.

`dwgrewrite`

Read and write the DWG, optionally via `--as=rNNNN` as another version, an earlier or later version. The default is writing as r2000. For now can only create r1.2-r2000 DWG.

`dwglayers`

Prints all layers in a DWG. With `-x` or `--extnames` prints the extended displayed layer name with spaces, not the internally stored old-style name with `_` instead. Only relevant with old r13 and r14 DWGs, after that layers are always stored in the extended format. With `-f` or `--flags` also the status of frown, on/off and locked. With `--on` only the visible layers, which are on and not frozen.

You can get the same effect via this json filter:

```
dwgfilter '.OBJECTS[]' example.dwg | \
  grep -A22 '"object": "LAYER"' | grep name
```

`dwggrep`

Search regex pattern in all text values in a list of DWGs. dwggrep uses PCRE.

With `-i` searches case-insensitive. With `-c` prints only the count of found texts. With `-h` or `--no-filename` does not print the filename. With `--type NAME` search only NAME entities or objects. With `--dxf NUM` search only in DXF group NUM fields. With `--text` searches only TEXT-like entities: TEXT, MTEXT, ATTRIB, ATTDEF.

`dwgfilter`

Search and modify a single DWG file via `jq`, using the powerful JQ query expression language on a temporary json file. See `man jq`.

With `-i` replaces the DWG in-place. This only makes sense with an JQ expression which changes values.

`dwg2SVG`

Convert a DWG to a limited SVG. All paperspace or modelspace enties of type: TEXT, LINE, CIRCLE, ARC, POLYLINE_2D, LWPOLYLINE, INSERT, ELLIPSE (unrotated), SOLID, 3DFACE, RAY, XLINE.

With `-m` or `--mspace` all paper-space entities are ignored, and only model-space is printed. The default is to print all paper-space entities. But if there are none, print all model-space entities instead.

Limitations: Many other graphical entities and some properties are still missing.

`dwg2ps`

Convert a DWG to a very limited Postscript file. All paperspace and modelspace entities of type LINE, POLYLINE_2D, LWPOLYLINE, ARC and CIRCLE.

This requires installation of pslib `http://pslib.sourceforge.net/doc/pslib.html`.

Note that the graphical representation for PS and SVG output is severely lacking, block references (insert entities) are not yet exploded, UCS and paper space transformations per entity are not yet done.

Planned is **dwgplot**, via GNU Plotutils `https://www.gnu.org/software/plotutils/`, to replace `dwg2SVG` and `dwg2ps`. This supports much more bitmap and vector formats.

There are also some more examples in the source distribution:

`load_dwg` loads a DWG and adds some entities.

`dwg2svg2` converts a DWG to SVG similar to `dwg2SVG`, but via the `dwg_api.h` only. The graphical representation for PS and SVG output is severely lacking, block references (insert entities) are not yet exploded , UCS and paper space transformations per entity are not yet done.

`unknown` lists the not yet reverse-engineered blobs from our examples files, and is the framework to guess the field layout for these. It is optionally using picat (`http://picat-lang.org/`) to solve some of the field-packing problems.

`dwgfuzz` afl++ fuzzing frontend, to test and debug various fast shared-memory options for afl-clang-fast, with the following runtime options: `-indxf`, `-injson`, `-rw`, `-dwg`, `-dxf`, `-dxfb`, `-json`, `-geojson`. All other output formats, like BMP, SVG, PS need to be fuzzed via their programs, which is the recommended way. The now default and fastest method INMEM does not need the 2nd file argument `@@`, the 2nd method STDIN neither.

See also `https://github.com/LibreDWG/libredwg-fuzz` for our fuzzing setup to test new fuzzing campaigns automatically and find regressions.

`dwgadd` is the easiest way to create DWG's (or DXF, JSON) from scratch or add entities to an existing DWG. It accepts a very simple file with commands to create entities or objects and set its properties. See `man 1 dwgadd` and `man 5 dwgadd`.

10 Bindings

LibreDWG generates library bindings to python and perl5 via swig. These can be quite huge, and it is recommended to use `ccache`. You can easily add bindings to other swig-supported languages, like Go, C#, ruby, php, D, lua, tcl, common lisp, ocaml, or others by yourself. Patches accepted.

Bindings for gambas (which looks very close to VBA) are at GitHub (`https://github.com/LibreDWG/gambas3-bindings`) and will soon be added to gambas3 as gb.dwg component. This is in development and about 80% finished.

11 Reference API

See the separate refman (`https://www.gnu.org/software/libredwg/refman/`) manual (in pdf or html format, the pdf has ~1800 pages) for a detailed API description, or see the relevant `dwg.h`, `dwg_api.h` or the `*.spec` files.

For reference you might also want to check the public DXF reference manuals, the VBA object model and the ODA `OpenDesign_Specification_for_dwg_files.pdf`.

12 Reporting bugs

To report bugs or suggest enhancements for GNU LibreDWG, please "submit a bug" at Savannah (`http://savannah.gnu.org/projects/libredwg`) or send electronic mail to `libredwg@gnu.org`. (If you use the web interface, you don't need to also send email, since that is done automatically.) Issues and pull requests at the github mirror (`https://github.com/LibreDWG/libredwg`) are also accepted.

For bug reports, please include enough information for the maintainers to reproduce the problem. Generally speaking, that means:

- The version numbers of LibreDWG and any other program(s) or manual(s) involved.
- Hardware and operating system names and versions.
- The contents of any input files necessary to reproduce the bug.
- The expected behavior and/or output.
- A description of the problem and samples of any erroneous output.
- Options you gave to `configure` other than specifying installation directories.
- Anything else that you think would be helpful. Usually that's the failing part of the object processed with `dwgread -v5`, but only the failing part, not the whole output.

When in doubt whether something is needed or not, include it. It's better to include too much than to leave out something important.

Patches are welcome; if possible, please make them with '`git format-patch`' and include `ChangeLog` entries (see Section "Change Log" in *The GNU Emacs Manual*). Please follow the existing GNU coding conventions. For patches longer than 15 lines we need your copyright assignment to the FSF clerk. See `CONTRIBUTING` in the source distribution.

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